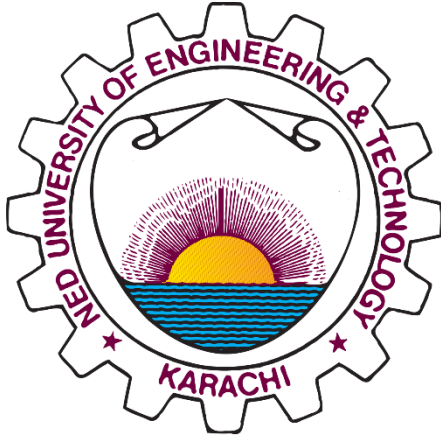


Engineering Drawing (ME-118)



Name: _____

Year: _____

Batch: _____

Roll No.: _____

Department: _____

Department of Mechanical Engineering
NED University of Engineering & Technology
Karachi 75270, Pakistan

Practical Workbook Engineering Drawing (ME-118)

Prepared by:

Dr. Tariq Jamil
(Associate Professor)

Dr. Muhammad Anique Azam
(Lecturer)

Mr. Abdul Samad Khan
(Lecturer)

This is certified that this Practical Workbook contains 92 pages



Fall 2025
Chairperson

Department of Mechanical Engineering
NED University of Engineering & Technology
Karachi – 75270, Pakistan

Certificate

of Laboratory Work Completion

This is to certify that Mr./Miss. _____ s/o, d/o
_____ bearing roll number _____, student of Spring /
Fall semester _____ has successfully completed the lab work for the
course of _____ (course code _____) and the
assessment done for OBE based practical in this workbook comply with the set OBE
rules.

Verified By: Course Coordinator

Name: _____

Signature: _____

Date: _____

Course learning outcome of Engineering Drawing

No.	CLO	PLO	Taxonomy Level
1	Illustrate 2D/3D drawings for effective visual communication	PLO – 9	Cognitive Level 3*
2	Practice acquired skills to develop drawings using relevant software	PLO – 5	Psychomotor Level 3*
3	Demonstrate the engineering drawing skills to illustrate the standard views of geometries and components used in the mechanical systems	PLO – 11	Psychomotor Level 4*

CLO Mapping on Practical

CLO	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
2					✓	✓	✓	✓	✓	✓	✓	✓
3	✓	✓	✓	✓								

Summary of Marks Obtained

Sr #	Components	Max. Marks	Marks Obtained
1	P3 (Psychomotor Domain) software use rubric-based evaluation	10	
2	P4 (Psychomotor Domain) rubric-based evaluation	05	
3	Hand drafting Tasks	05	
4	AutoCAD Tasks	05	
5	Miscellaneous (Attendance)	05	
Total Marks		30	

Marks Awarding Scheme:

- Marks for (Psychomotor Domain) OBE based practical =

$$\frac{\text{Sum of marks obtained from rubric form}}{\text{Max. marks from rubric form}} \times 10 \text{ (P3)}$$

$$\frac{\text{Sum of marks obtained from rubric form}}{\text{Max. marks from rubric form}} \times 5 \text{ (P4)}$$

Table of Contents

S. No.	Experiment	Page	Initial
1	To draw single-view drawings of various mechanical parts according to the dimensions using the concepts of geometrical construction	13-19	
2	To draw orthographic projections of a 3D representation of a part	20-29	
3	To construct isometric views from Multiview drawings	30-37	
4	To construct an assembly drawing from an exploded pictorial view	38-43	
5	Introduction to AutoCAD Graphical User Interface (GUI)	44-51	
6	To create 2D drawings using basic geometrical construction commands in AutoCAD	52-56	
7	To develop 2D drawings using Array function in AutoCAD	57-59	
8	Applications of Engineering Curves (Involute and Cycloids) in 2D Drawing with AutoCAD	60-66	
9	To develop and visualize parametric 3D models using basic geometrical construction commands in AutoCAD	67-74	
10	Modifying dimensions and units in AutoCAD on 2D & 3D drawings	75-80	
11	To create Orthographic 2D Views from a 3D Model using AutoCAD	81-83	
12	To create Piping & Instrumentation Diagram (P&ID) using AutoCAD Plant 3D	84-92	

Introduction to Drafting Instruments and Drawing Sheet Setup

Objective:

To get acquainted with engineering drawing, sheet format, drafting techniques and tool usage.

Instruments:

Drawing sheet (recommended ISO A3, imperial sizes can be used), masking tape/thumb pins, pencils (Grade 2H, HB, 2B), eraser, sharpener, fine-grit sandpaper for removable-lead pencils, T-square, set-squares, protractor, compass, and flexible curve/French curves.

Exercise 1 (Setting up): Preparation of the drawing sheet

You will have a half imperial sheet size which measures approximately 22×15 inches. For drawing on the sheet, you will have to affix the sheet in landscape orientation to your drawing board/drawing table. Use small pieces of masking tapes or thumb pins to fix the sheet at the four corners. This will secure your sheet from movement while drawing.

Before starting to draw, we need to specify the drawing area and create borders and margins. The thumb pins and any adhesive tape will cover the sheet in the margins of the sheet and will not interfere with the drawing region.

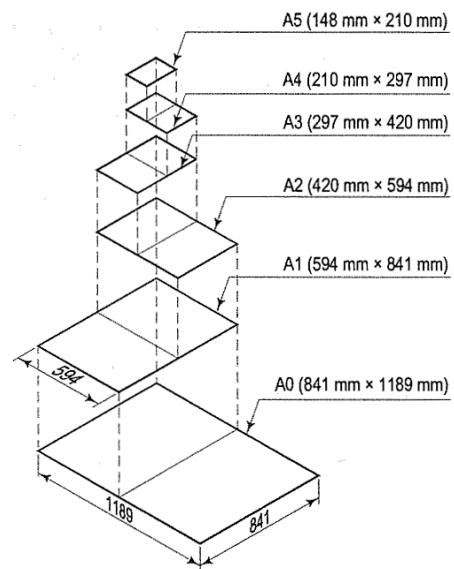


Fig. I

Align one long side of the sheet with the edge/rim of the drawing table. This will ensure straight fixing of the sheet. Make sure that your drawing table is levelled with a bubble-gauge, with perpendicular sides and right-angled corners. An example of drawing board is shown in Fig. II which you would use for out of campus tasks.

Also, ensure that your drawing surface is free from debris and dust (such as eraser debris, graphite shavings etc.) Always start with a clean surface and clean instruments.

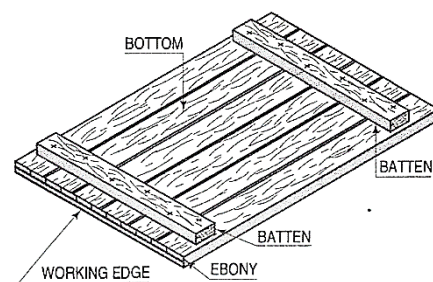


Fig. II

Exercise 2 (Practice): Freehand Sketching

Sketching freehand is an important skill to master. Freehand drawings are the first step between the ideation and the technical drawing with dimensions. On your drawing sheet, sketch by freehand a series of hatches in various directions. An example is given in Fig. VI. Afterwards, draw a freehand circle of radius 30 mm by using guides as shown in Fig. VII.

A standard border can be drawn as shown in the Fig. III (dimensions in mm). Use the grids at the bottom to denote your credentials as follows:

- (1) Name of department
- (2) Title of drawing
- (3) Name and roll number of the student

To draw straight lines, you would need to use a T-square (Fig. IV). T-square takes support from the working edge of the drawing table/board, thereby necessitating a clear, straight edge of the table/board.

Orient the stock of the T-square as shown in Fig. V to be able to slide the instrument's working edge along the edge of the table. Use the T-square to define the boundaries of your sheet.

Your basic drawing space is ready for you to start practicing your drawings. Plan your sheet regions in advance. For basic drawings and getting started, it would be ideal to divide your sheet into imaginary quarters and dedicate one quadrant to each specific exercise.

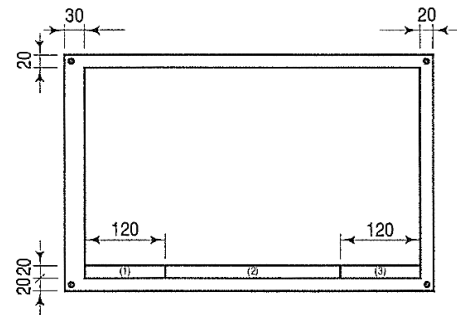


Fig. III

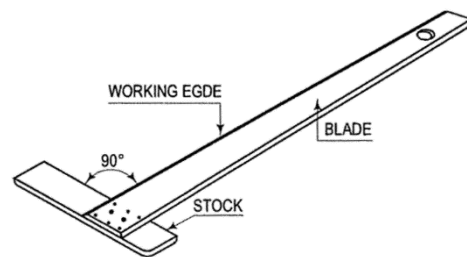


Fig. IV

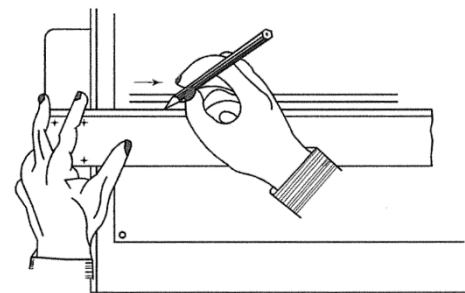
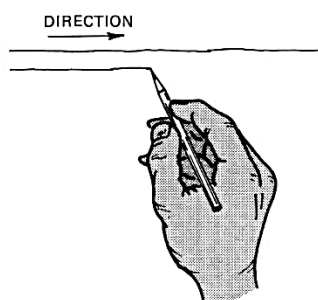
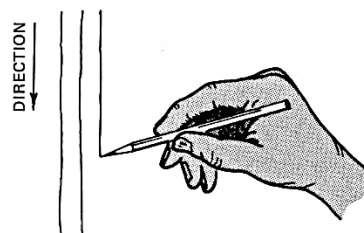


Fig. V



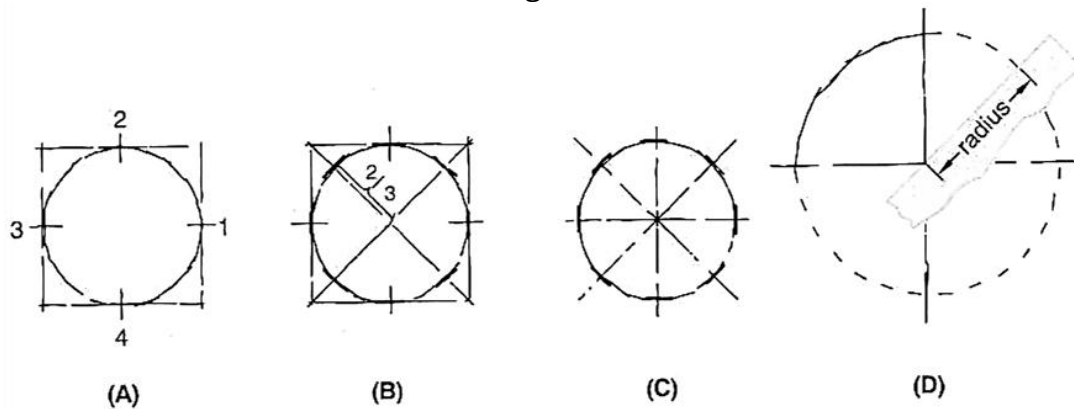
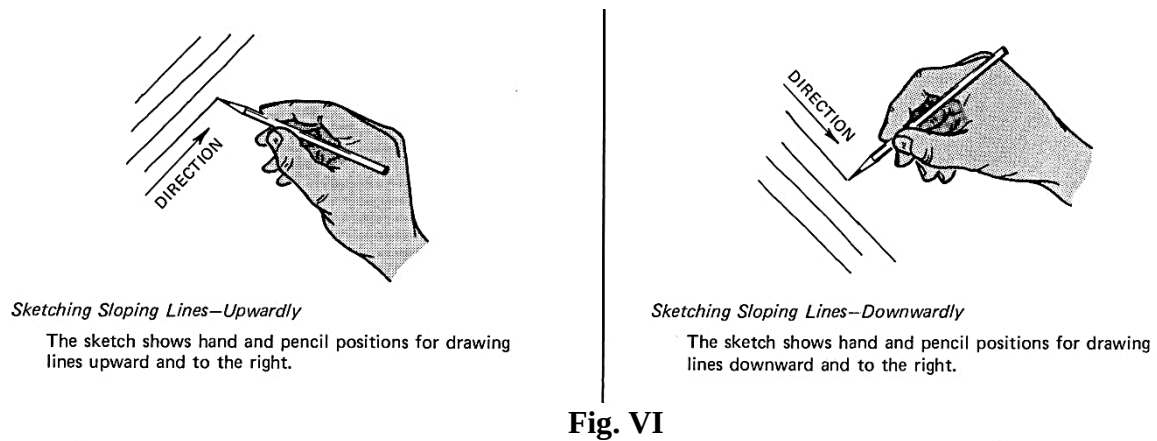
Sketching Horizontal Lines

Segments approximately 3 inches long are lightly drawn by using a sliding action of the forearm. Final lines are strengthened by applying pressure.



Sketching Vertical Lines

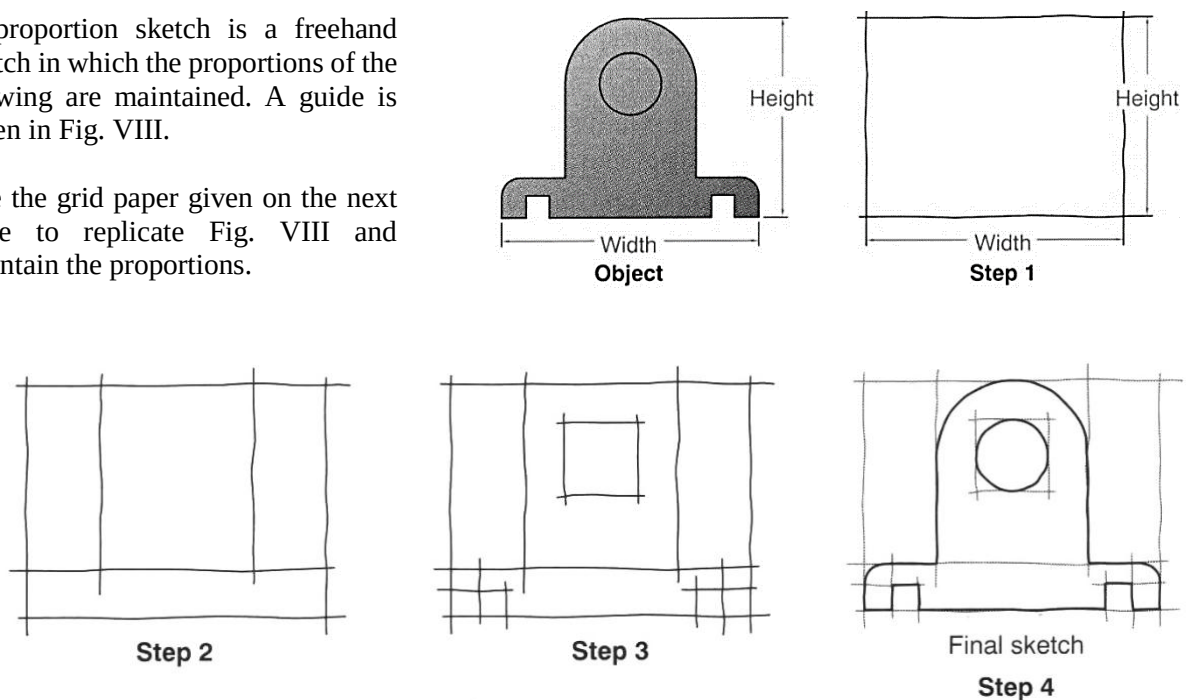
The pencil point extends about 2 inches beyond the fingers. The forearm and elbow should rest on the drawing surface.

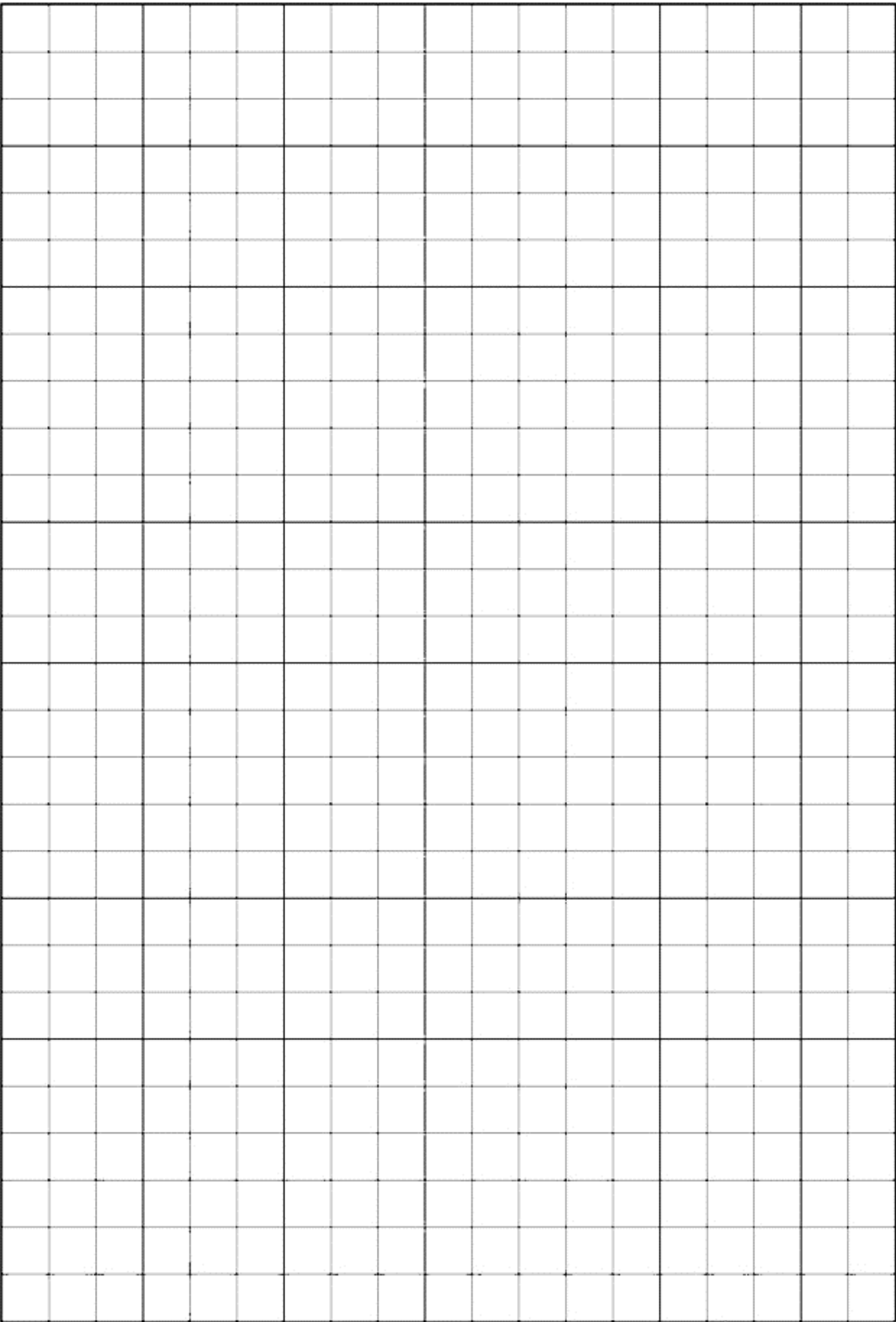


Exercise 3 (Practice): Proportion sketch with grid support

A proportion sketch is a freehand sketch in which the proportions of the drawing are maintained. A guide is given in Fig. VIII.

Use the grid paper given on the next page to replicate Fig. VIII and maintain the proportions.





Exercise 4 (Practice): Proportion sketch without grids

Use your freehand skills to replicate the drawings (Fig. IX and X) as shown below, in the blank space beside it.

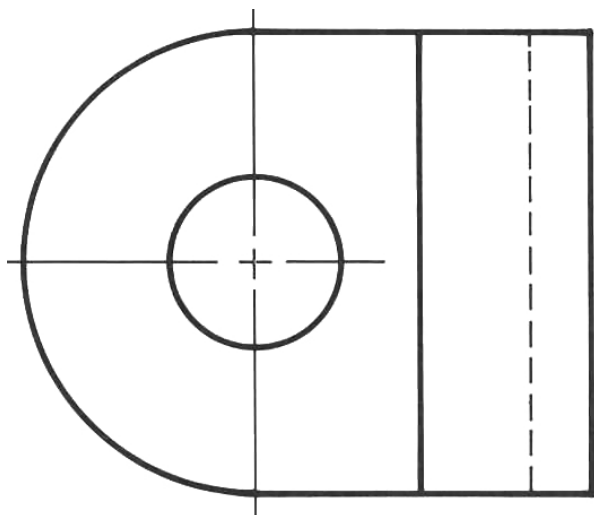


Fig. IX

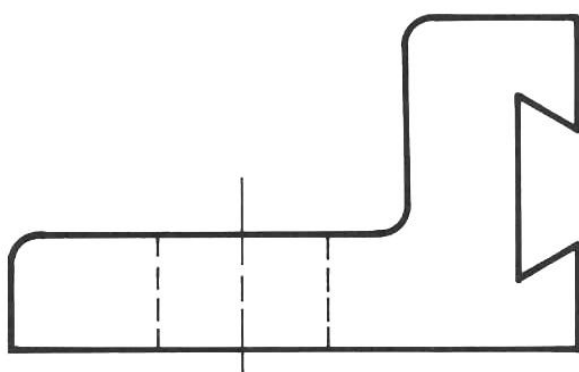


Fig. X

Exercise 5: Drawing various lines

Attach a new sheet following Exercise 1 of the current practical.

Use your existing knowledge of various types of lines used in engineering drawing and replicate them. Use space of the fourth quadrant of your sheet.

Equally space each line vertically by a distance of 20mm, between two horizontal bounds separated by a distance of 80mm.

Append an arrow for the dimension line and write down the dimensions as per practice.

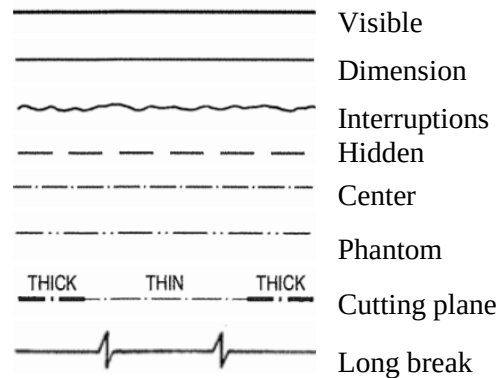


Fig. XI

Exercise 6: Drawing angular lines, grids, and straight-edge shapes

Use four equally spaced 100mm square boxes and hatch each box with equally spaced lines as follows:

- (1) Perpendicular grid (3 visible lines)
- (2) Perpendicular grid rotated by 45° (3 visible lines)
- (3) Grid with 60° angle (2 visible lines)
- (4) Grid with 15° angle (3 visible lines)

Use dimension lines to dimension each box, the inter-line spacing, and the angles. Use third quadrant of your sheet for the drawing.

Hint: Use set squares for the desired angles.

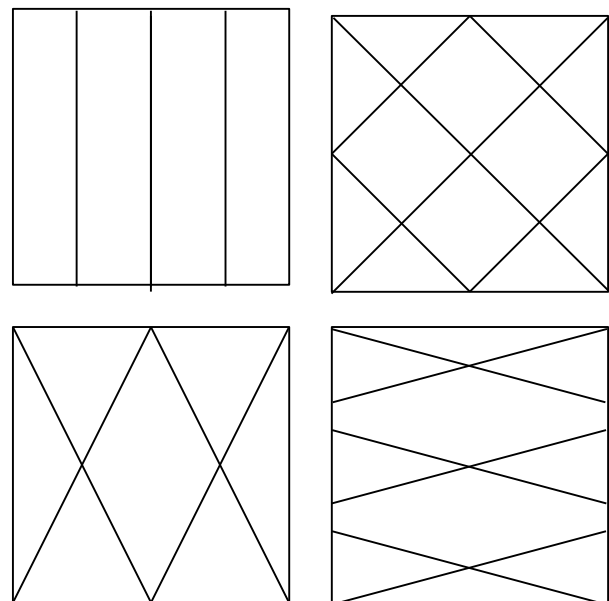


Fig. XII

Exercise 7: Drawing circle/circular arcs

Draw 2 concentric circles starting with a diameter of 50 mm with outer circle of 100 mm diameter. Draw a center-line at the middle of the two concentric circles and where required. Use the dimensioning line to denote the diameter and radius of the inner and outer circle respectively. Draw the object in the second quadrant.

Hint: Use compass for the construction. Diameter is denoted by a prefix “Ø” and radius is denoted by a prefix “R”.

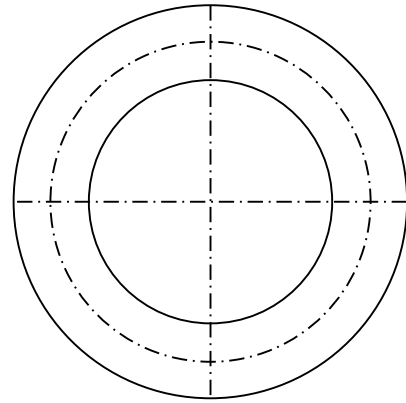


Fig. XIII

Exercise 8: Drawing curves/splines

Draw a spiral curve using French curves/flexible curve, or by freehand, such that the curve increases by a distance of 20mm every 45° beginning at center from horizontal east line clockwise and ending at 180°. Use first quadrant to make the drawing.

Hint: Use protractor/compass/set-square to draw the construction lines.

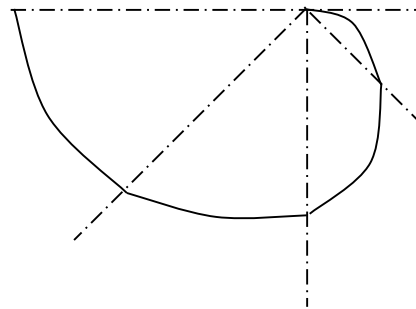


Fig. XIV

Practical No. 1

Objective:

To draw single-view drawings of various mechanical parts according to the dimensions using the concepts of geometrical construction.

Instruments:

Drawing sheet (recommended ISO A3, imperial sizes can be used), masking tape/thumb pins, pencils (Grade 2H, HB, 2B), eraser, sharpener, fine-grit sandpaper for removable-lead pencils, T-square, set-squares, protractor, compass, and flexible curve/French curves.

Exercise 1: Fixing and parting the sheet

Start by affixing your sheet according to the introduction, and divide the sheet into 4 quadrants to manage the space for the exercises in this practical. Use each quadrant for each drawing exercise.

Exercise 2: Extruded part with tangent arcs - Plate Cam

Extruded parts have same cross-section across their third axis; therefore, such parts can be effectively shown by a single view drawing.

Use your existing knowledge of drawing tangent arcs, construct the *Plate Cam* as shown in Fig. 1.1.

Deconstruct the geometry into two circles of radii R20 and R40. Locate the center of the tangent arcs of R55 and R150 by appropriate methodology.

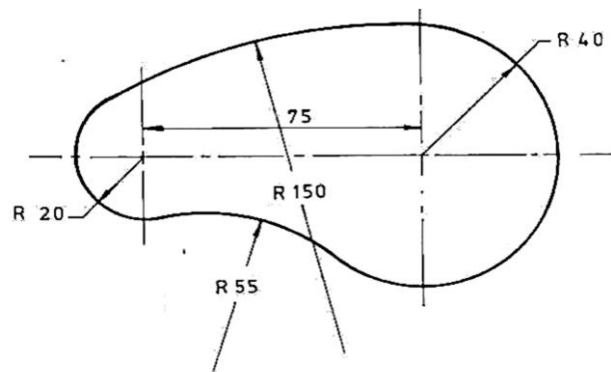


Fig. 1.1

Exercise 3: Revolved part with tangent arcs – File Handle

Parts by revolution are manufactured by revolving a certain profile about an axis. Such parts can also be effectively shown by a single view drawing. Using the knowledge of drawing tangent arcs, construct the *File Handle* as shown in Fig. 1.2.

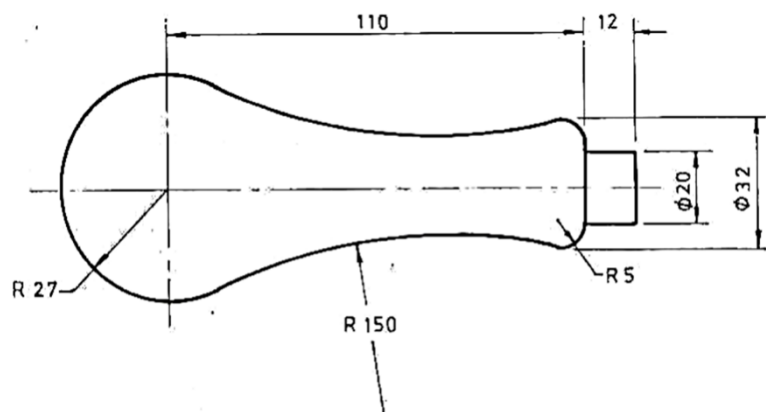


Fig. 1.2

Exercise 4: Sheet metal part with polygon - Trip catch

Like extruded parts, sheet metal parts also feature same cross-section, but their third dimension is negligible.

Using existing knowledge of tangent arcs, construct the *Trip catch* as shown in Fig. 1.3. The part also features a hexagonal through hole with each side of 12mm.

A regular hexagon is a combination of 6 equilateral triangles therefore the circumscribed circle of the hexagon will be of the radius equal to the length of one side. Using compass mark 6 equidistant points on the circumference to obtain the vertices of the hexagon.

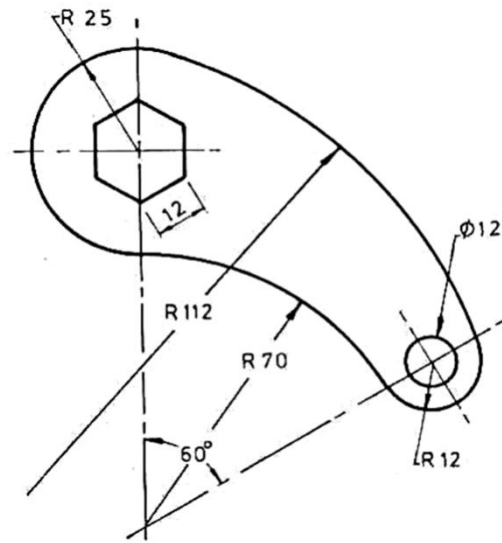


Fig. 1.3

Exercise 5: Parts made by casting – Spanner

While extrusion may carry a different meaning when it comes to production processes, some extruded parts are also fabricated by casting. A pattern is used to create a cast cavity and molten material is poured into the cavity to attain the shape. Heavy tools are an example.

Using the rules of geometrical construction, draw the *spanner* as shown in Fig. 1.4.

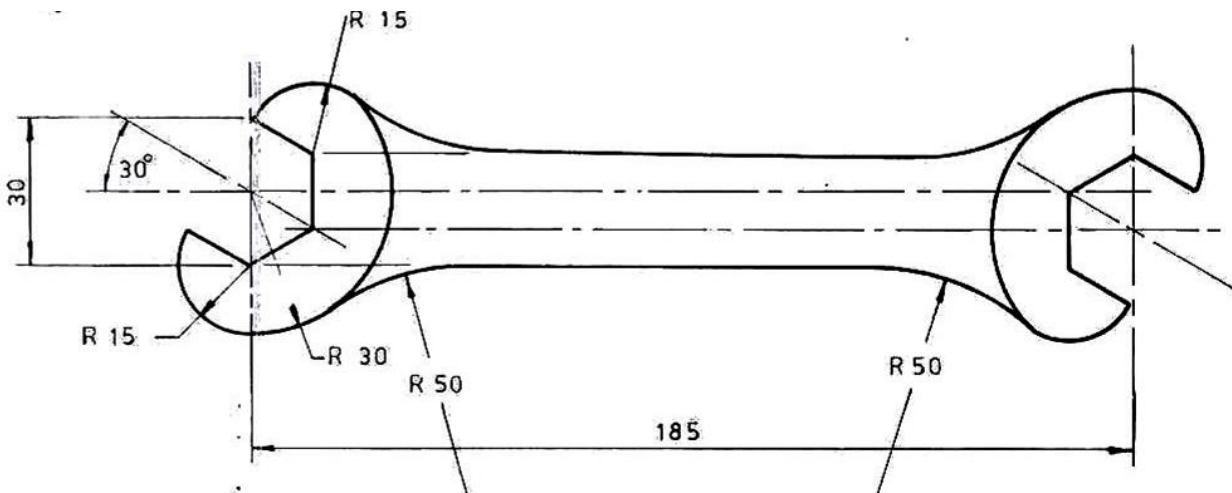


Fig. 1.4

Exercise 6: Optimizing material at drawing stage – Locking plate

With the existing knowledge of geometrical construction, draw the *Locking Plate* as shown in Fig. 1.5.

Could you identify the excess material if the tangent arcs of R40 are replaced by tangent lines?

In your drawing shown the saving in material by hatching the area in which the material is saved from.

Can you further save some material from the existing drawing by utilizing a tangent line?

Hint: Replace arc of R125 with a tangent line

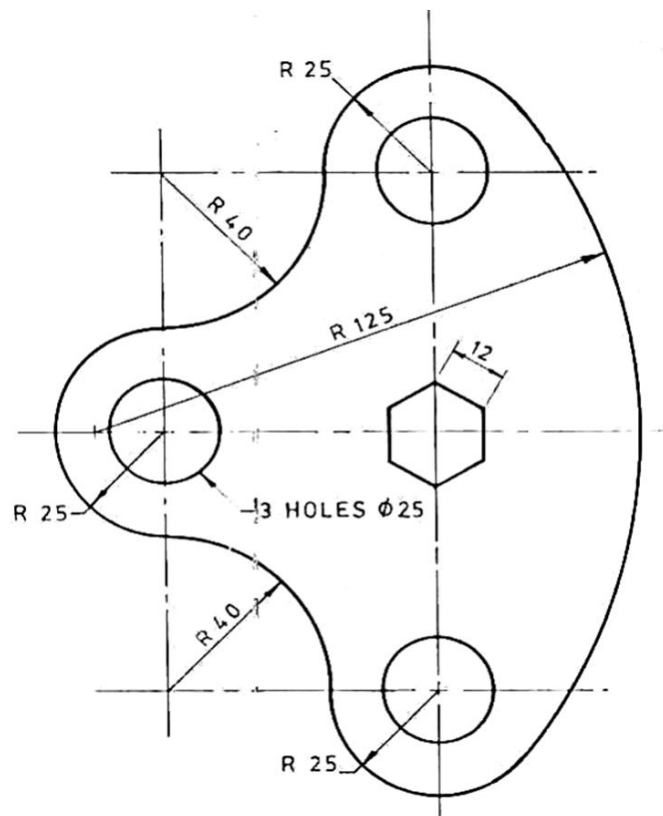


Fig. 1.5

Exercise 7 (Reading): Working sheets

Read the working sheet on the next page and answer the questions on the subsequent pages.



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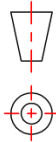
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THIRD ANGLE PROJECTION



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>50-100	±0.8	±0.50	±0.25
>100-200	±1.5	±0.80	±0.40
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>500	±5.0	±3.00	±1.50

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UNSPECIFIED RADII 0.25mm
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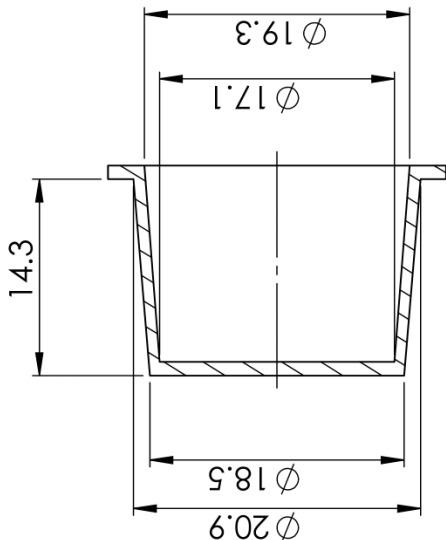
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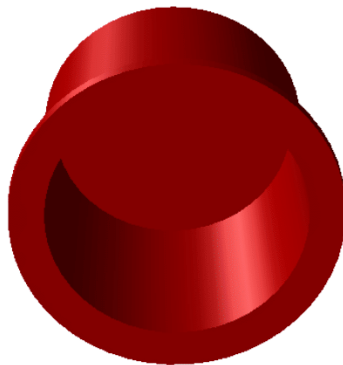
1 of 2

ISSUE

1



SECTION Y-Y





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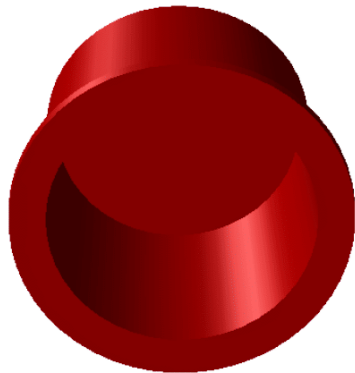
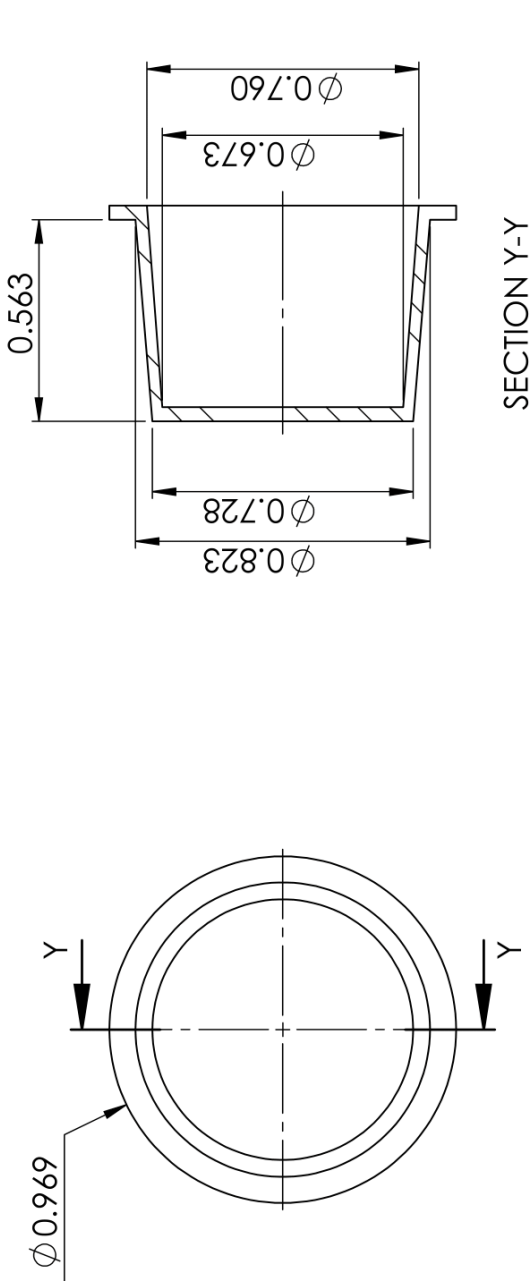
APPROVED DRAWING

UNLESS OTHERWISE STATED

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>1.181 - 1.969	±0.020 ±0.012 ±0.008		
>1.969 - 3.937	±0.031 ±0.020 ±0.010		
>3.937 - 7.874	±0.059 ±0.031 ±0.016		
>7.874 - 11.811	±0.079 ±0.047 ±0.031		
>11.811 - 19.685	±0.118 ±0.079 ±0.039		
>19.685	±0.197 ±0.118 ±0.059	ANGULAR TOLERANCE UNLESS SPECIFIED NOT 0.010 inch DIMENSIONS IN BRACKETS [] ARE IN INCHES FOR REFERENCE ONLY	

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			2 of 2	1



Answer the questions by reading the working sheets. Write your answer in the same box as that of the question.

Identify the manufacturer of the component from the above set of drawing sheets?

What is the difference between the above two drawing sheets?

What tolerances should be expected when ordering the part shown on the first drawing sheet?

Identify the material of the part. Are they different for the two drawing sheets?

What is the status of drawing? What is the level of control exercised on this working sheet?

NED University of Engineering & Technology
Department of Mechanical Engineering
Course Code and Title: **ME-118 Engineering Drawing**

Psychomotor Domain Assessment Rubric-Level P4					
Skill Sets	Extent of Achievement				
	0	1	2	3	4
<u>Equipment Identification</u> Sensory skill to <i>identify</i> equipment and/or its component for a lab work.	Not able to identify the equipment.	--	--	--	Able to identify equipment as well as its components.
<u>Equipment Use</u> Sensory skills to <i>demonstrate</i> the use of the equipment for the lab work.	Doesn't demonstrate the use of equipment.	Slightly demonstrates the use of equipment.	Somewhat demonstrates the use of equipment.	Moderately demonstrates the use of equipment.	Fully demonstrates the use of equipment.
<u>Procedural Skills</u> <i>Displays</i> skills to act upon sequence of steps in lab work.	Not able to either learn or perform lab work procedure.	Able to slightly understand lab work procedure and perform lab work.	Able to somewhat understand lab work procedure and perform lab work.	Able to moderately understand lab work procedure and perform lab work.	Able to fully understand lab work procedure and perform lab work.
<u>Response</u> Ability to <i>imitate</i> the lab work on his/her own.	Not able to imitate the lab work.	Able to slightly imitate the lab work.	Able to somewhat imitate the lab work.	Able to moderately imitate the lab work.	Able to fully imitate the lab work.
<u>Observation's Use</u> <i>Displays</i> skills to use the observations from lab work for experimental verifications and illustrations.	Not able to use the observations from lab work for experimental verifications and illustrations.	Slightly able to use the observations from lab work for experimental verifications and illustrations.	Somewhat able to use the observations from lab work for experimental verifications and illustrations.	Moderately able to use the observations from lab work for experimental verifications and illustrations.	Fully able to use the observations from lab work for experimental verifications and illustrations.
<u>Confidence in Completion</u> Ability to <i>complete with confidence</i> the drawing task	Cannot complete the task	Completes with frequent hesitation	Completes with some confidence	Completes with growing confidence	Completes confidently and independently
<u>Work Efficiency</u> Ability to <i>improve efficiency</i> while completing tasks	Shows no efficiency	Efficiency very low	Efficiency slowly improving	Efficiency generally good	Efficiency high and consistent
<u>Safety Adherence</u> Adherence to <i>safety</i> procedures.	Doesn't adhere to safety procedures.	Slightly adheres to safety procedures.	Somewhat adheres to safety procedures.	Moderately adheres to safety procedures.	Fully adheres to safety procedures.
<u>Equipment Handling</u> <i>Equipment care</i> during the use.	Doesn't handle equipment with required care.	Rarely handles equipment with required care.	Occasionally handles equipment with required care.	Often handles equipment with required care.	Handles equipment with required care.

Laboratory Session No. _____

Date: _____

Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	

Practical No. 2

Objective:

To draw orthographic projections of a 3D representation of a part.

Instruments:

Drawing sheet (recommended ISO A3, imperial sizes can be used), masking tape/thumb pins, pencils (Grade 2H, HB, 2B), eraser, sharpener, fine-grit sandpaper for removable-lead pencils, T-square, set-squares, protractor, compass, divider, and flexible curve/French curves.

Exercise 1 (Practice): Matching the views

Recall your understanding of Multiview drawings and projection theory. This practical will encompass drawing a part in both first angle and third angle projections. In order to identify the projection technique, use bottom right drawing space of the sheet to denote the projection symbol (Fig. 2.1).

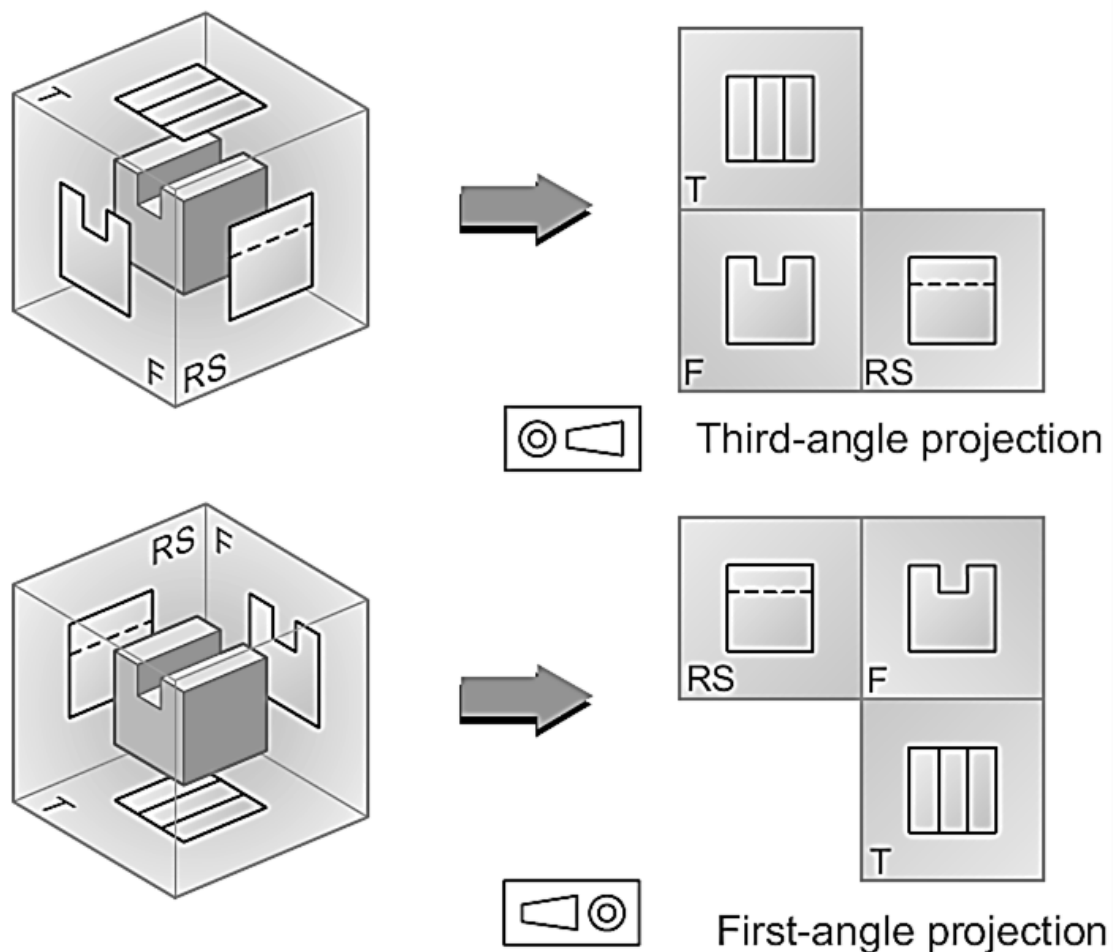


Fig. 2.1

Follow the worksheets on the subsequent pages and bring your understanding of various views to test.

Top View Test

Below are pictures of 20 objects. The top view of each of these is given in the section at the bottom of the page. Select the top view of each object and place its number in the circle given with each picture.

Keep in mind:

1. The same view may be the correct answer for more than one problem.
2. Views are to be used only in the given positions.

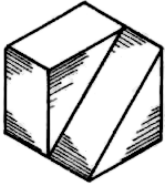
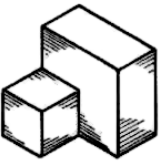
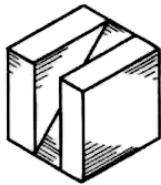
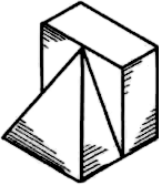
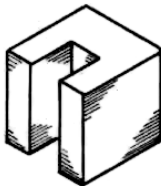
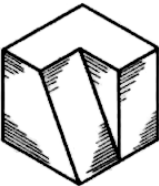
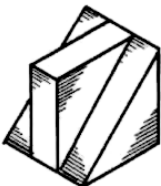
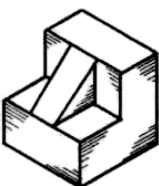
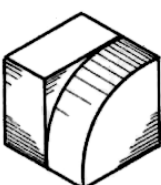
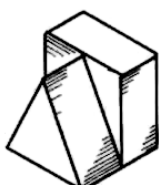
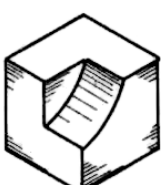
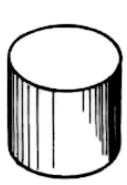
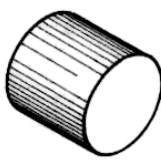
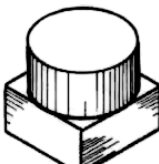
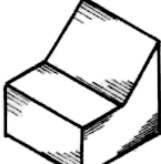
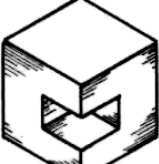
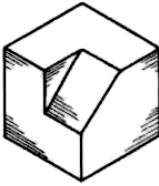
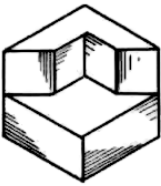
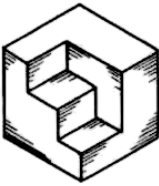
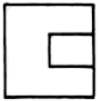
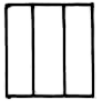
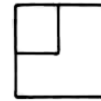
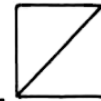
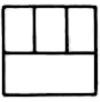
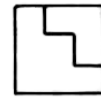
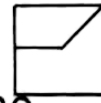
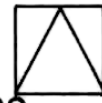
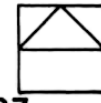
1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24

Front View Test:

Below are pictures of 20 objects. The front view of each of these is given in the section at the bottom of the page. Select the front view of each object and place its number in the circle given with each picture.

Keep in mind:

1. The same view may be the correct answer for more than one problem.
2. Views are to be used only in the given positions.

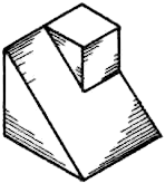
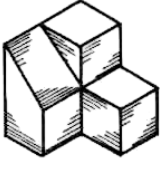
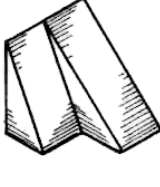
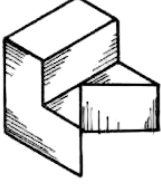
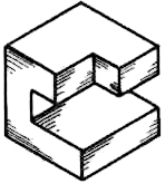
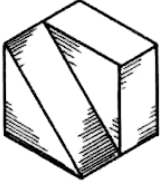
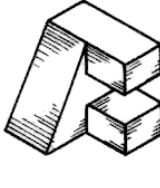
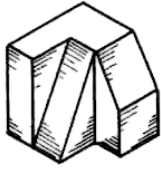
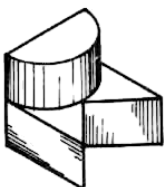
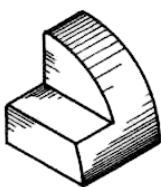
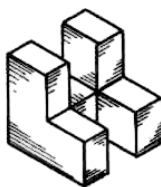
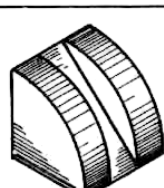
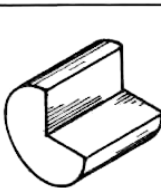
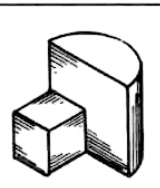
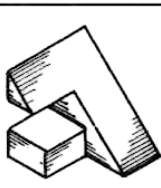
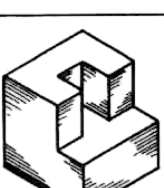
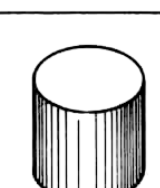
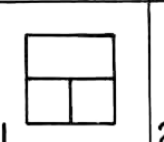
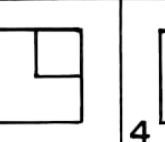
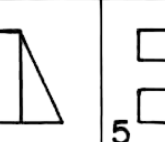
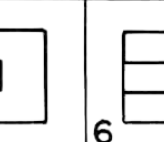
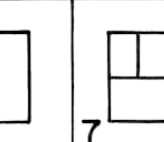
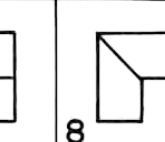
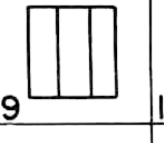
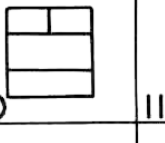
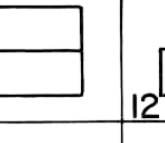
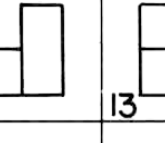
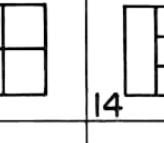
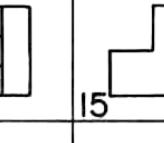
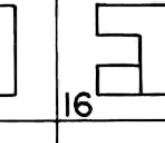
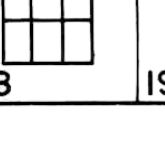
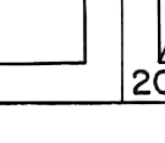
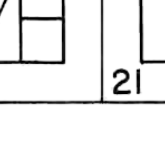
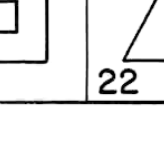
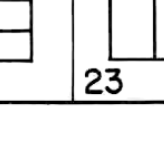
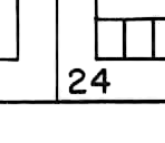
							
							
							
							
							
							
							
							

Right side View Test

Below are pictures of 20 objects. The Right-side view of each of these is given in the section at the bottom of the page. Select the Right-side view of each object and place its number in the circle given with each picture.

Keep in mind:

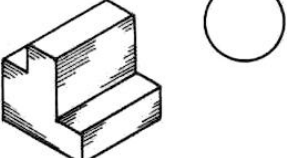
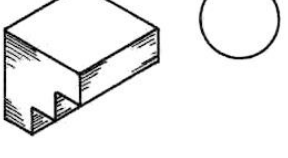
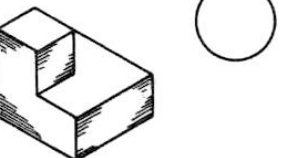
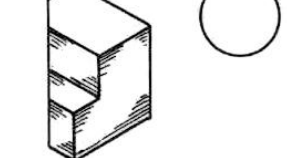
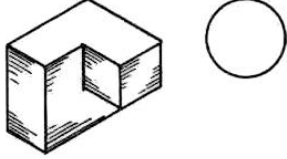
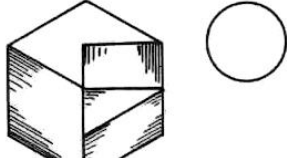
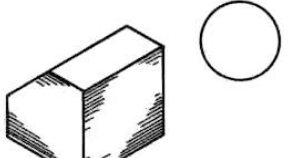
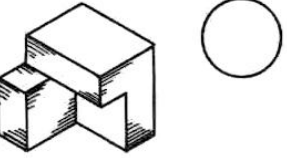
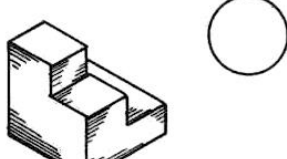
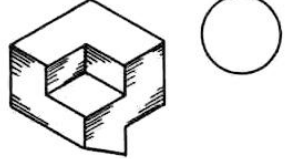
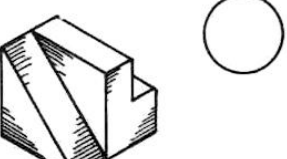
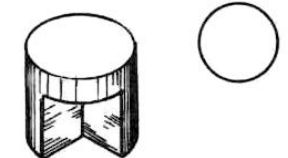
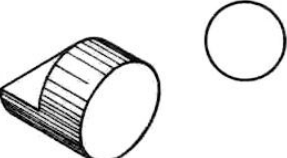
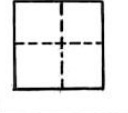
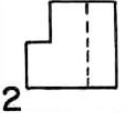
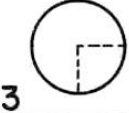
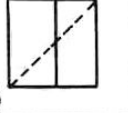
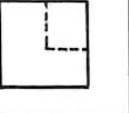
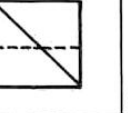
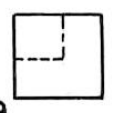
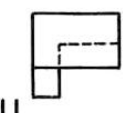
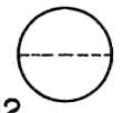
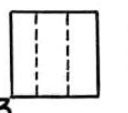
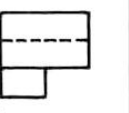
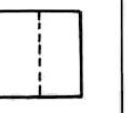
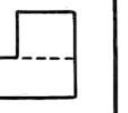
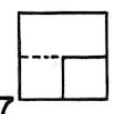
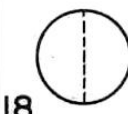
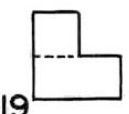
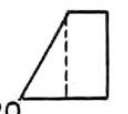
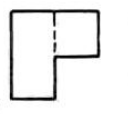
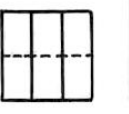
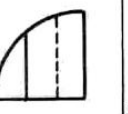
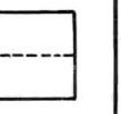
1. The same view may be the correct answer for more than one problem.
2. Views are to be used only in the given positions.

Hidden line test

Below are the pictures of 20 objects. The top, front, or right side of each of these objects having hidden lines is given in the section at the bottom of the page. Select a view and place its number in the circle with each picture. Keep in mind:

1. The same view may be correct answer for more than one problem.
2. Views are to be use only in the position given.

Exercise 2 (Setting up): Fixing and parting the sheet

Start by affixing your sheet according to the introduction, and divide the sheet into 4 quadrants. The four quadrants will each be used for a distinct view of a first angle and a third angle projection.

Exercise 3: Suitable Multiview drawing

Construct a Multiview drawing of the part shown in Fig. 2.2. Use any suitable projection technique to specify the features. All units are in inches.

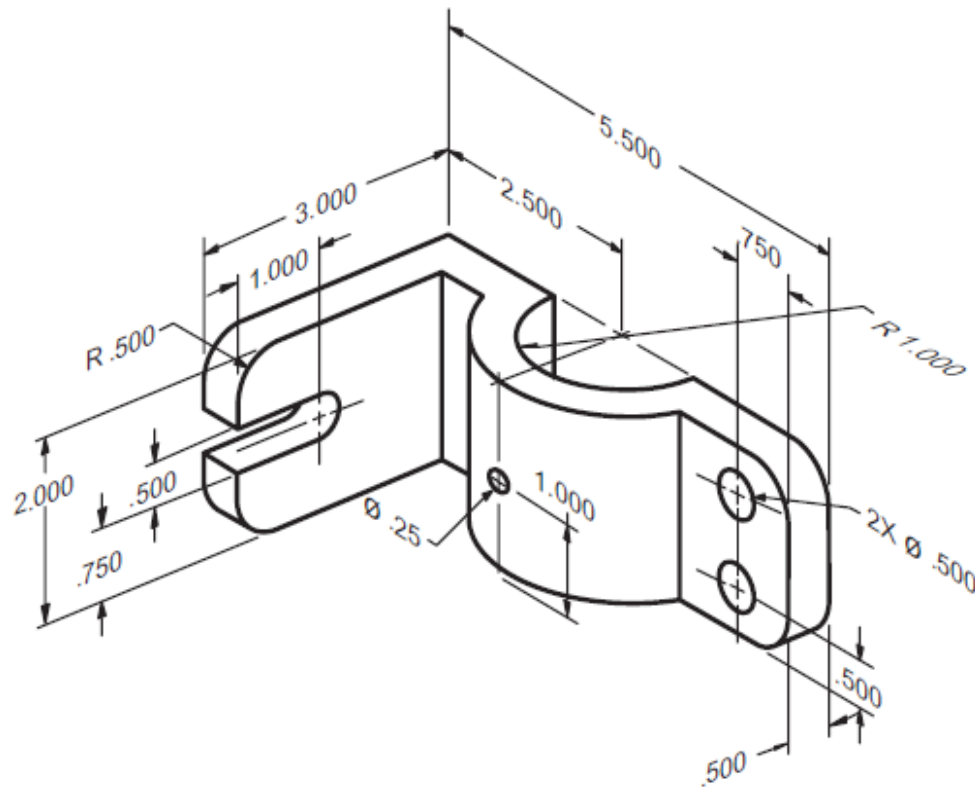


Fig. 2.2

Exercise 4: First angle projection from given pictorial view

Draw the first angle orthographic projection from the given pictorial view in Fig. 2.3. Where the line of sight denoted by X is the front view. All units in millimeters.

Exercise 5: Third angle projection from given pictorial view

Repeat Exercise 2 but draw third angle projection of the object shown in Fig. 2.3.

Hint: Multiples views of Fig. 2.3 can be shown in Fig. 2.4 but they are not aligned in their designated places as they should be.

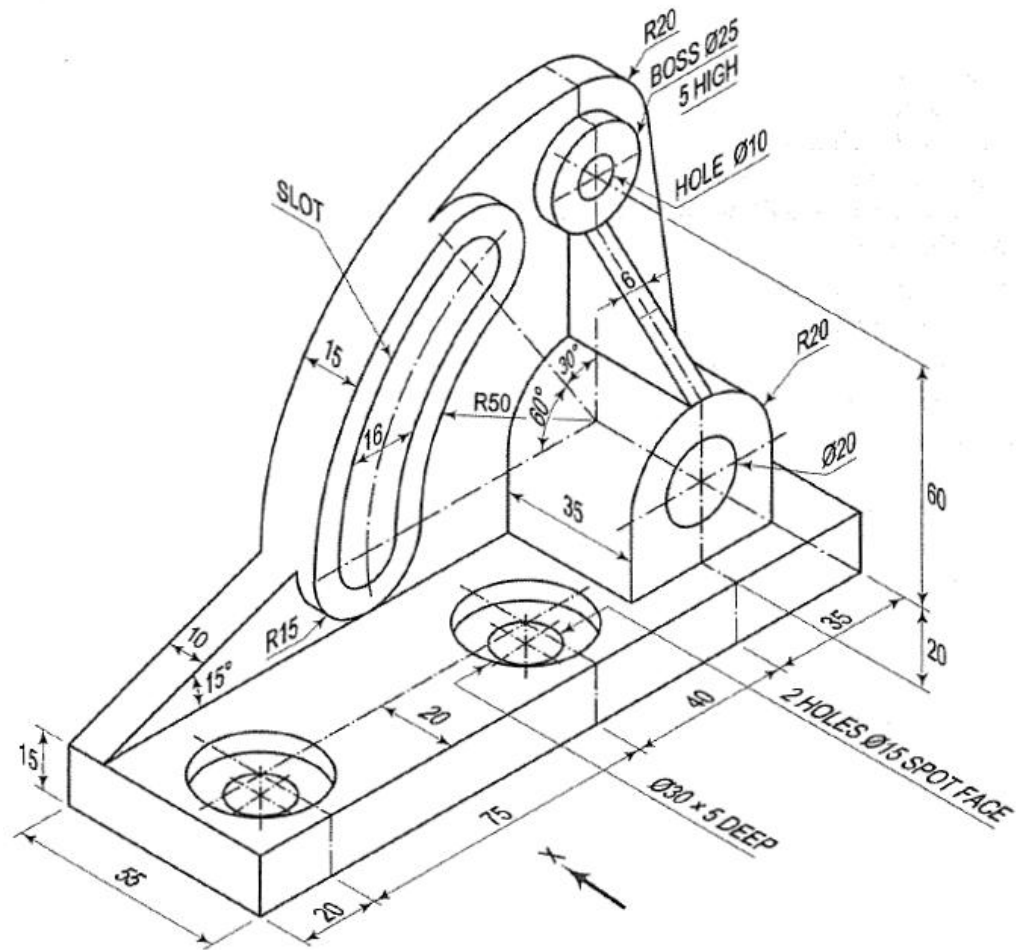


Fig 2.3

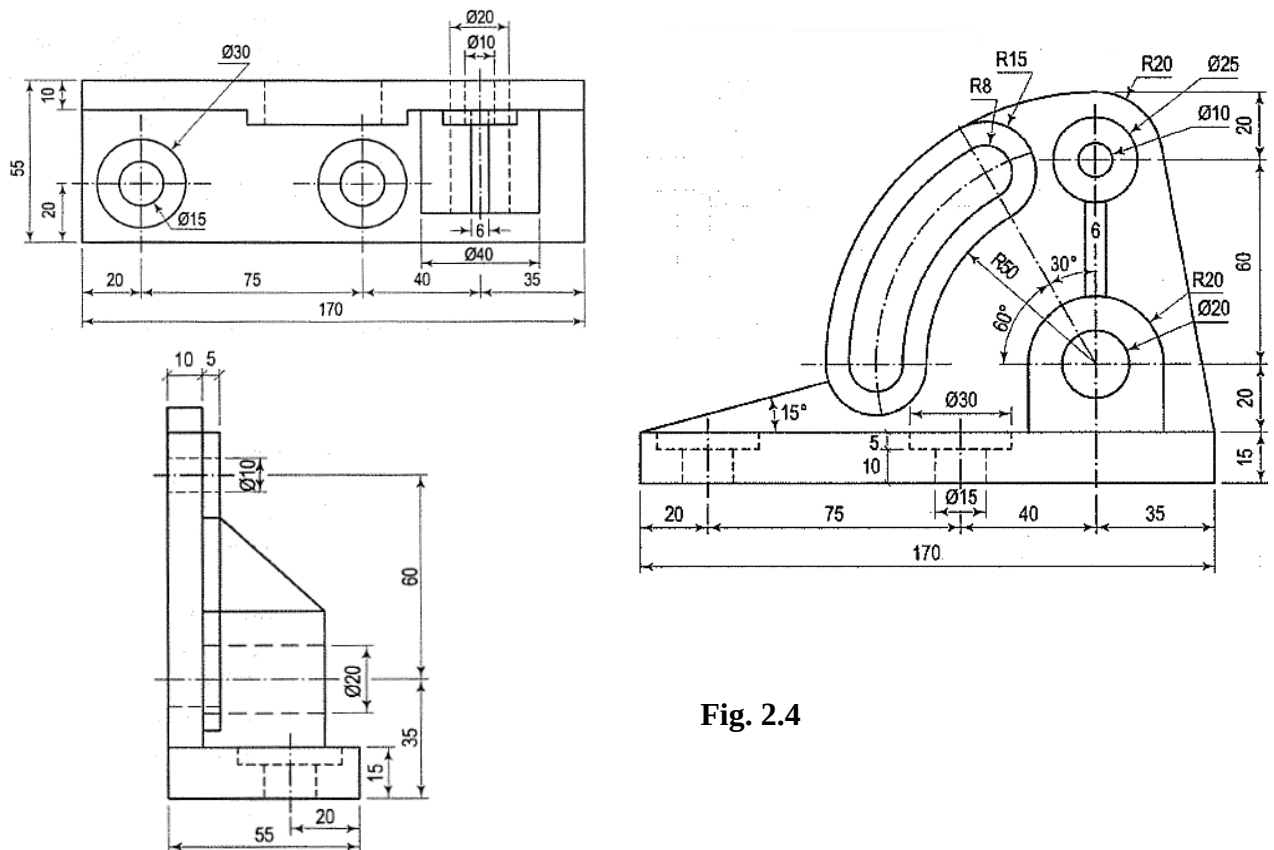


Fig. 2.4

Exercise 6 (Reading): Working sheets

Read the working drawing on the next page and answer the following questions. Write your answer in the same box as the question

What is the name of the part? What views are provided for the part? Which projection technique is used?

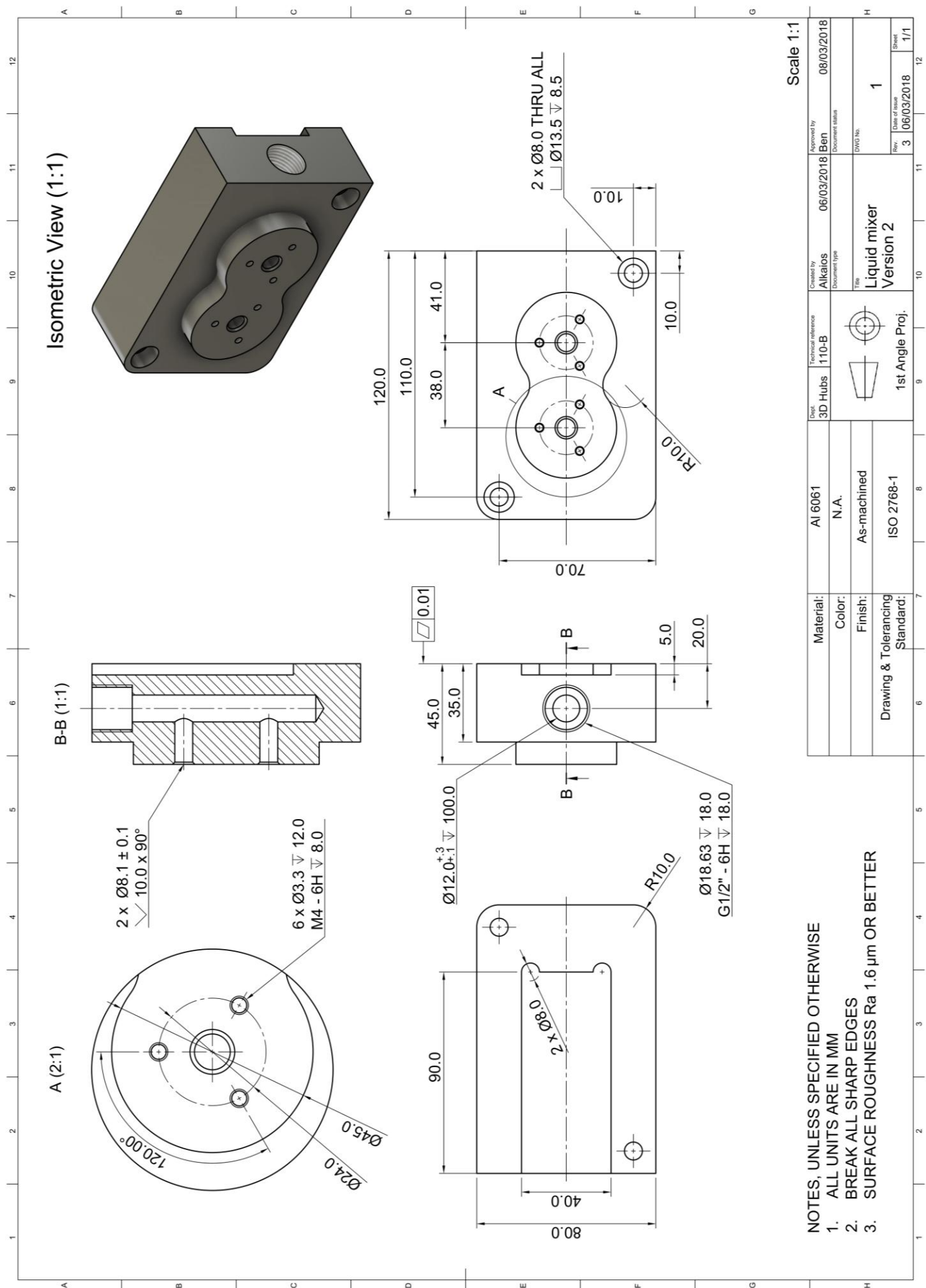
Does the drawing follow any tolerance standard? Are there any overrides on the tolerance? If yes, in which views are they present?

Which production method is to be used for this part? Can you guess from the given format?

Are there any detail views? If yes, why were they required?

What is the material, date of approval, finish, scale, and units of the working drawing?

Which symbols are used in the annotations besides dimensioning? What do these symbol mean?



NED University of Engineering & Technology
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<u>Equipment Identification</u> Sensory skill to <i>identify</i> equipment and/or its component for a lab work.	Not able to identify the equipment.	--	--	--	Able to identify equipment as well as its components.
<u>Equipment Use</u> Sensory skills to <i>demonstrate</i> the use of the equipment for the lab work.	Doesn't demonstrate the use of equipment.	Slightly demonstrates the use of equipment.	Somewhat demonstrates the use of equipment.	Moderately demonstrates the use of equipment.	Fully demonstrates the use of equipment.
<u>Procedural Skills</u> <i>Displays</i> skills to act upon sequence of steps in lab work.	Not able to either learn or perform lab work procedure.	Able to slightly understand lab work procedure and perform lab work.	Able to somewhat understand lab work procedure and perform lab work.	Able to moderately understand lab work procedure and perform lab work.	Able to fully understand lab work procedure and perform lab work.
<u>Response</u> Ability to <i>imitate</i> the lab work on his/her own.	Not able to imitate the lab work.	Able to slightly imitate the lab work.	Able to somewhat imitate the lab work.	Able to moderately imitate the lab work.	Able to fully imitate the lab work.
<u>Observation's Use</u> <i>Displays</i> skills to use the observations from lab work for experimental verifications and illustrations.	Not able to use the observations from lab work for experimental verifications and illustrations.	Slightly able to use the observations from lab work for experimental verifications and illustrations.	Somewhat able to use the observations from lab work for experimental verifications and illustrations.	Moderately able to use the observations from lab work for experimental verifications and illustrations.	Fully able to use the observations from lab work for experimental verifications and illustrations.
<u>Confidence in Completion</u> Ability to <i>complete with confidence</i> the drawing task	Cannot complete the task	Completes with frequent hesitation	Completes with some confidence	Completes with growing confidence	Completes confidently and independently
<u>Work Efficiency</u> Ability to <i>improve efficiency</i> while completing tasks	Shows no efficiency	Efficiency very low	Efficiency slowly improving	Efficiency generally good	Efficiency high and consistent
<u>Safety Adherence</u> Adherence to <i>safety</i> procedures.	Doesn't adhere to safety procedures.	Slightly adheres to safety procedures.	Somewhat adheres to safety procedures.	Moderately adheres to safety procedures.	Fully adheres to safety procedures.
<u>Equipment Handling</u> <i>Equipment care</i> during the use.	Doesn't handle equipment with required care.	Rarely handles equipment with required care.	Occasionally handles equipment with	Often handles equipment with required care.	Handles equipment with required care.

Laboratory Session No. _____

Date: _____

Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	

Practical No. 3

Objective:

To construct isometric views from Multiview drawings.

Instruments:

Drawing sheet (recommended ISO A3, imperial sizes can be used), masking tape/thumb pins, pencils (Grade 2H, HB, 2B), eraser, sharpener, fine-grit sandpaper for removable-lead pencils, T-square, set-squares, protractor, compass, divider, and flexible curve/French curves.

Exercise 1 (Practice): Surface matching test

Use your knowledge of Multiview and match the surfaces with the relevant view.

EXAMPLE

1

2

3

4

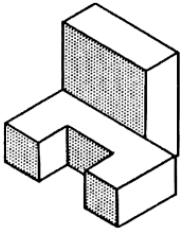
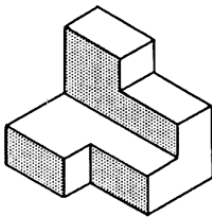
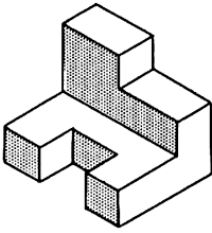
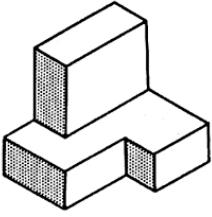
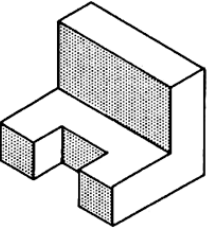
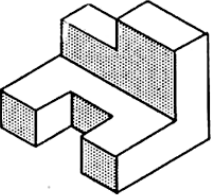
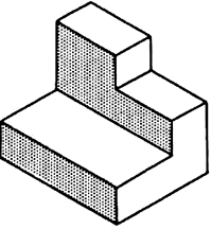
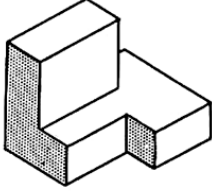
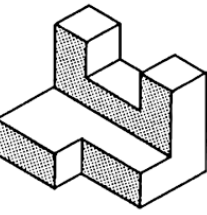
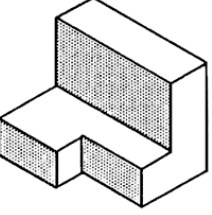
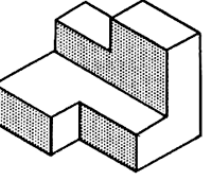
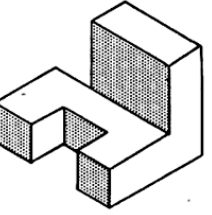
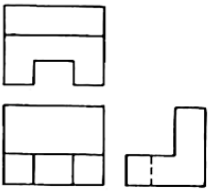
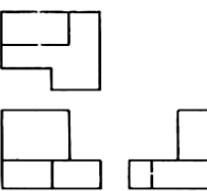
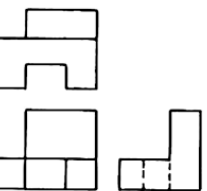
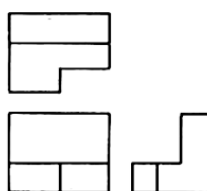
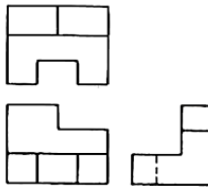
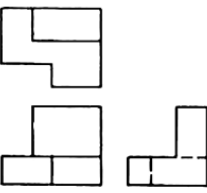
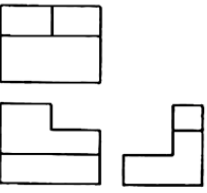
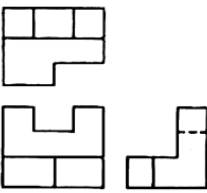
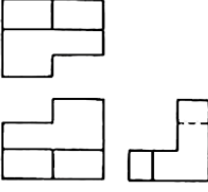
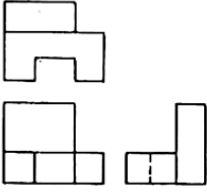
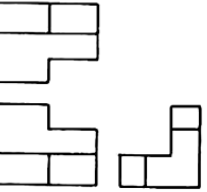
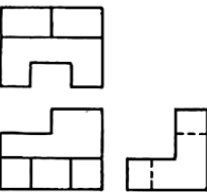
5

6

7

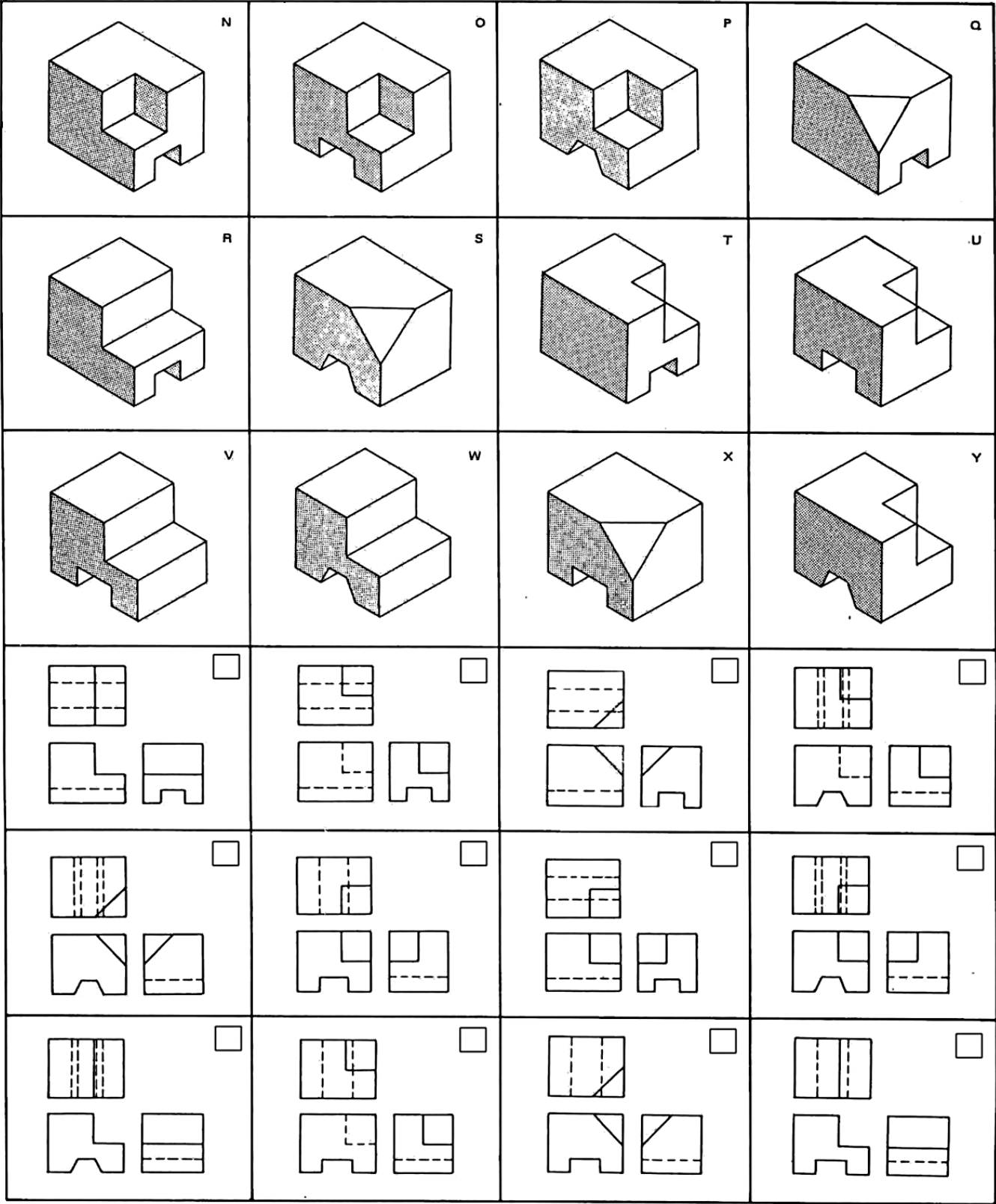
Exercise 2 (Practice): Matching isometric views

Match the given isometric views with the orthographic views given at the bottom of the page. Put the letter given along the isometric views in the small box given with the respective ortho views.

 A	 B	 C	 D
 E	 F	 G	 H
 J	 K	 L	 M
 ☐	 ☐	 ☐	 ☐
 ☐	 ☐	 ☐	 ☐
 ☐	 ☐	 ☐	 ☐

Continued on next page.

Continued.



Exercise 3 (Practice): Missing view test

A circle indicates which view is missing in each of the 20 drawings given below. All missing views are numbered and given in the section at the bottom of the page. Place the number of the missing view in the circle given with each drawing. Keep in mind:

1. The same view may be the correct answer for more than one problem.
2. Views are to be used only in the position given.

Exercise 4 (Setting up): Fixing and preparing the sheet

Start by affixing your sheet. Divide one of your sheets into two equal halves. Prepare your sheet by identifying the center point and draw principal axis in isometric orientation (separated by 120°). Use your set-squares. Recall your understanding of isometric views to draw primitive shapes such as circles and polygons in isometric view.

Exercise 5: Isometric view of straight-edge part

Construct the isometric view from the given projections in Fig. 3.1. Use each half for each isometric drawing. All dimensions in millimeters.

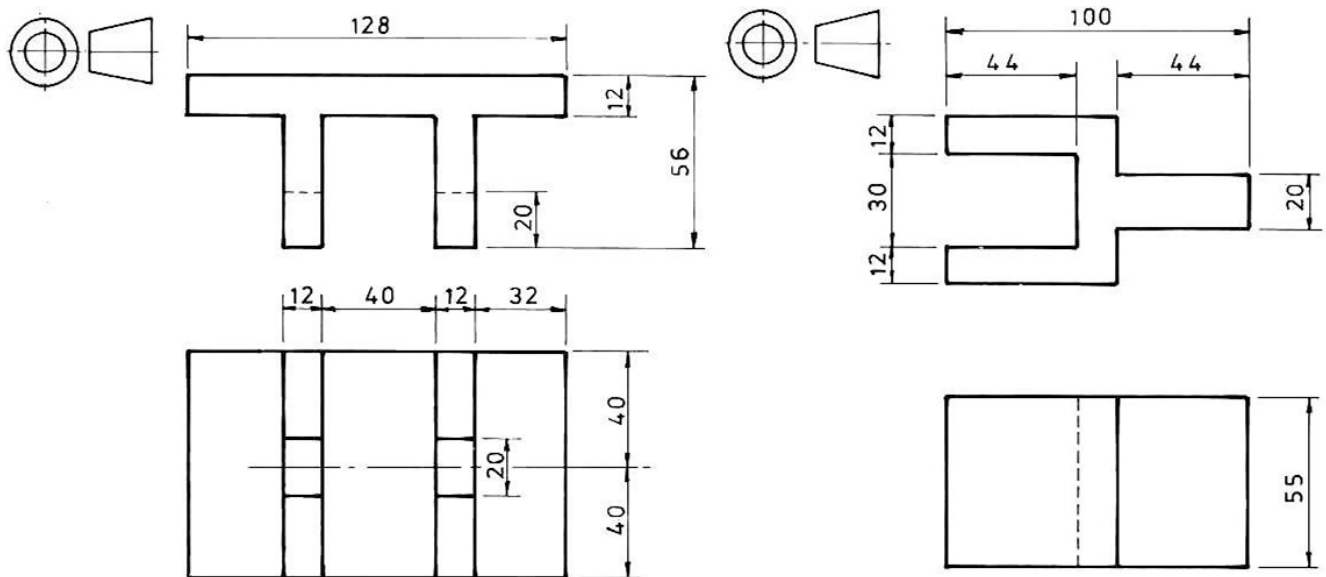


Fig. 3.1

Exercise 6: Isometric view with Multiview Projections

Construct the isometric view from the given projections in Fig. 3.2. All units are in millimeters.

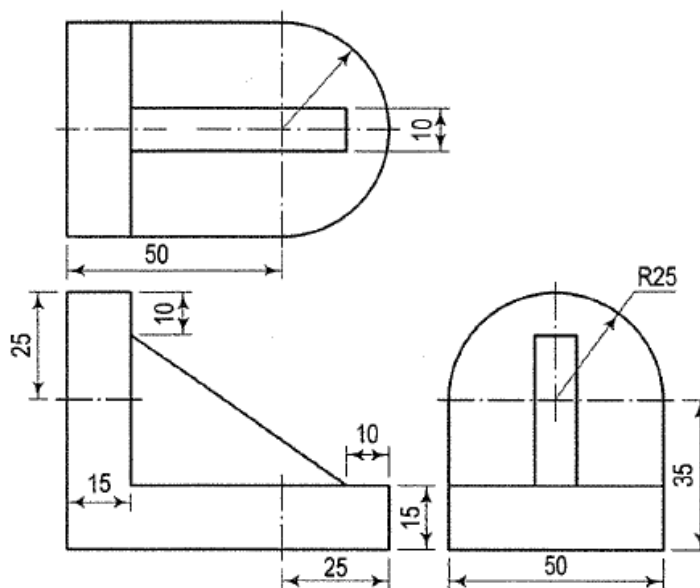


Fig. 3.2

Exercise 7 (Reading): Working sheets

Read the working drawing on the next page and answer the following questions. Write your answer in the same box as the question

What is the name of the part? Is isometric view provided in the sheet? Is the exploded view in isometric orientation? Is the symbol of projection relevant to this drawing?

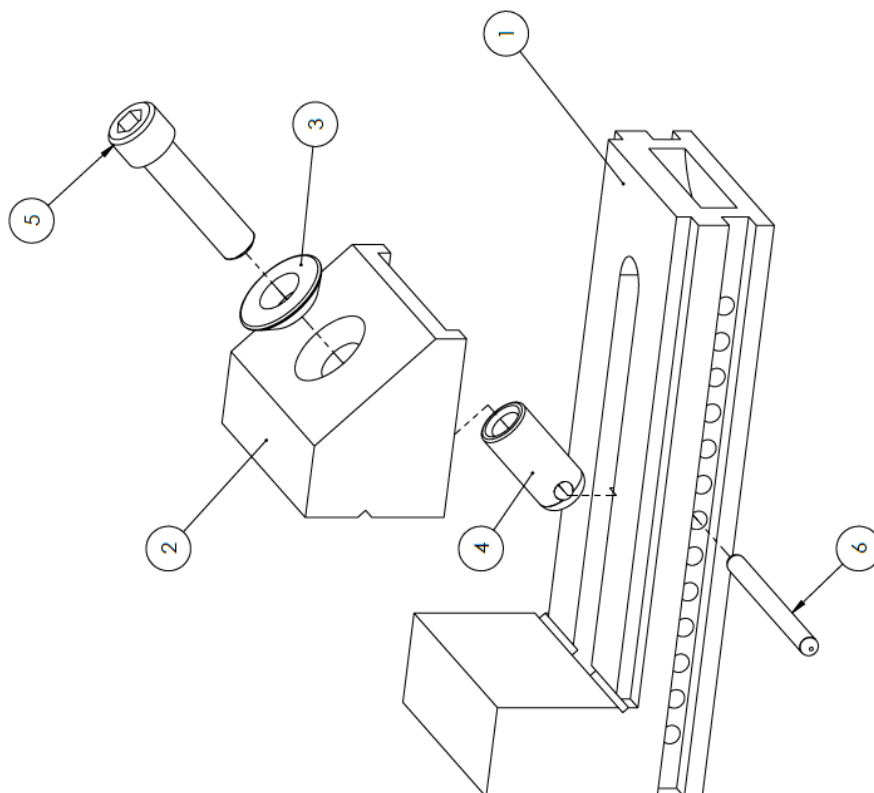
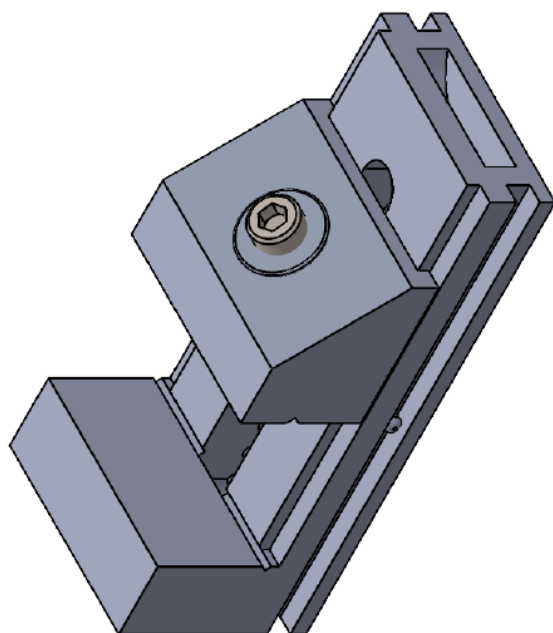
Identify the name of the designing company? What is the drawing number of the sheet, the scale, and the sheet size?

Are there any features which were hidden from the isometric assembled drawing? Would an assembled view with wireframe denotation be adequate?

Which part would be completely concealed in the assembled view?

What is the expected weight of the assembly after manufacturing?

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	50001	Vise Base	1
2	50002	Movable Jaw	1
3	38725	Curved Washer	1
4	48732	Swivel Connector	1
5	5862482	SHCS 0.5-13x2x2	1
6	38726	Vise Pin	1



DRAWN		NAME	DATE	Sub Tooling Co. Grinding Vise TITLE:
CHECKED		LMIL	11/08	
ENG APPR.				
MFG APPR.				
Q.A.		C.B.	11/13	
		SIZE DWG. NO. B 15N53081L REV A		
		SCALE: 1:1 (WEIGHT: 7000) SHEET 1 OF 1		

NED University of Engineering & Technology
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Psychomotor Domain Assessment Rubric-Level P4					
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<u>Equipment Identification</u> Sensory skill to <i>identify</i> equipment and/or its component for a lab work.	Not able to identify the equipment.	--	--	--	Able to identify equipment as well as its components.
<u>Equipment Use</u> Sensory skills to <i>demonstrate</i> the use of the equipment for the lab work.	Doesn't demonstrate the use of equipment.	Slightly demonstrates the use of equipment.	Somewhat demonstrates the use of equipment.	Moderately demonstrates the use of equipment.	Fully demonstrates the use of equipment.
<u>Procedural Skills</u> <i>Displays</i> skills to act upon sequence of steps in lab work.	Not able to either learn or perform lab work procedure.	Able to slightly understand lab work procedure and perform lab work.	Able to somewhat understand lab work procedure and perform lab work.	Able to moderately understand lab work procedure and perform lab work.	Able to fully understand lab work procedure and perform lab work.
<u>Response</u> Ability to <i>imitate</i> the lab work on his/her own.	Not able to imitate the lab work.	Able to slightly imitate the lab work.	Able to somewhat imitate the lab work.	Able to moderately imitate the lab work.	Able to fully imitate the lab work.
<u>Observation's Use</u> <i>Displays</i> skills to use the observations from lab work for experimental verifications and illustrations.	Not able to use the observations from lab work for experimental verifications and illustrations.	Slightly able to use the observations from lab work for experimental verifications and illustrations.	Somewhat able to use the observations from lab work for experimental verifications and illustrations.	Moderately able to use the observations from lab work for experimental verifications and illustrations.	Fully able to use the observations from lab work for experimental verifications and illustrations.
<u>Confidence in Completion</u> Ability to <i>complete with confidence</i> the drawing task	Cannot complete the task	Completes with frequent hesitation	Completes with some confidence	Completes with growing confidence	Completes confidently and independently
<u>Work Efficiency</u> Ability to <i>improve efficiency</i> while completing tasks	Shows no efficiency	Efficiency very low	Efficiency slowly improving	Efficiency generally good	Efficiency high and consistent
<u>Safety Adherence</u> Adherence to <i>safety</i> procedures.	Doesn't adhere to safety procedures.	Slightly adheres to safety procedures.	Somewhat adheres to safety procedures.	Moderately adheres to safety procedures.	Fully adheres to safety procedures.
<u>Equipment Handling</u> <i>Equipment care</i> during the use.	Doesn't handle equipment with required care.	Rarely handles equipment with required care.	Occasionally handles equipment with required care.	Often handles equipment with required care.	Handles equipment with required care.

Laboratory Session No. _____

Date: _____

Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	

Practical No. 4

Objective:

To construct an assembly drawing from an exploded pictorial view.

Instruments:

Drawing sheet (recommended ISO A3, imperial sizes can be used), masking tape/thumb pins, pencils (Grade 2H, HB, 2B), eraser, sharpener, fine-grit sandpaper for removable-lead pencils, T-square, set-squares, protractor, compass, divider, and flexible curve/French curves.

Exercise 1: Fixing and preparing the sheet

Start by affixing your sheet, and divide the sheet into 4 quadrants. The four quadrants will each be used for a distinct view of a third-angle projection. Recall your understanding of isometric views and orthographic projection techniques. You will need as many sheets as there are parts in the assembly of Fig. 4.1 (parts in Fig. 4.2)

Exercise 2: Third-angle projection of the assembly

Construct the third-angle orthographic projection (Front & Side view) of the assembled isometric view.

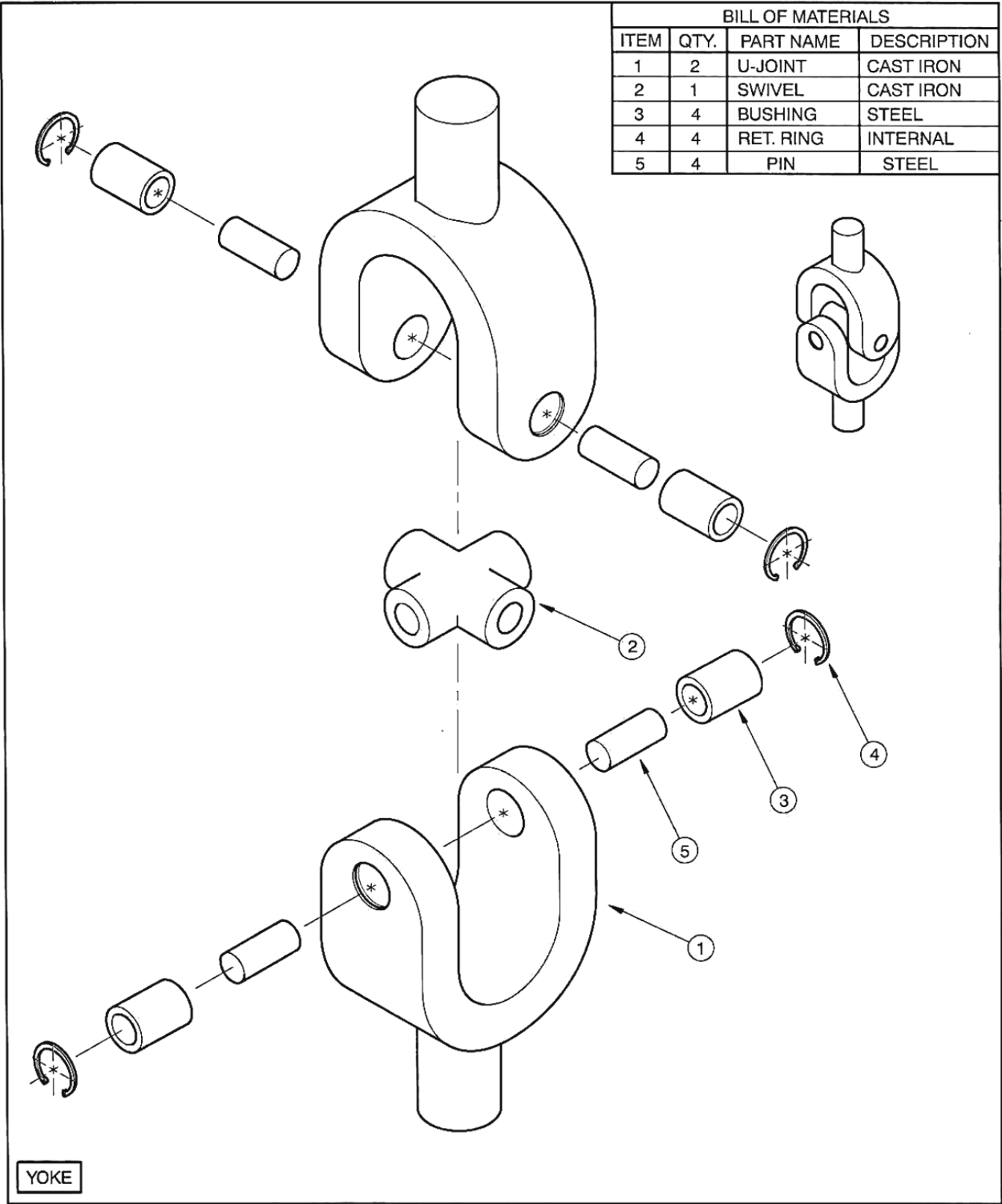


Fig. 4.1

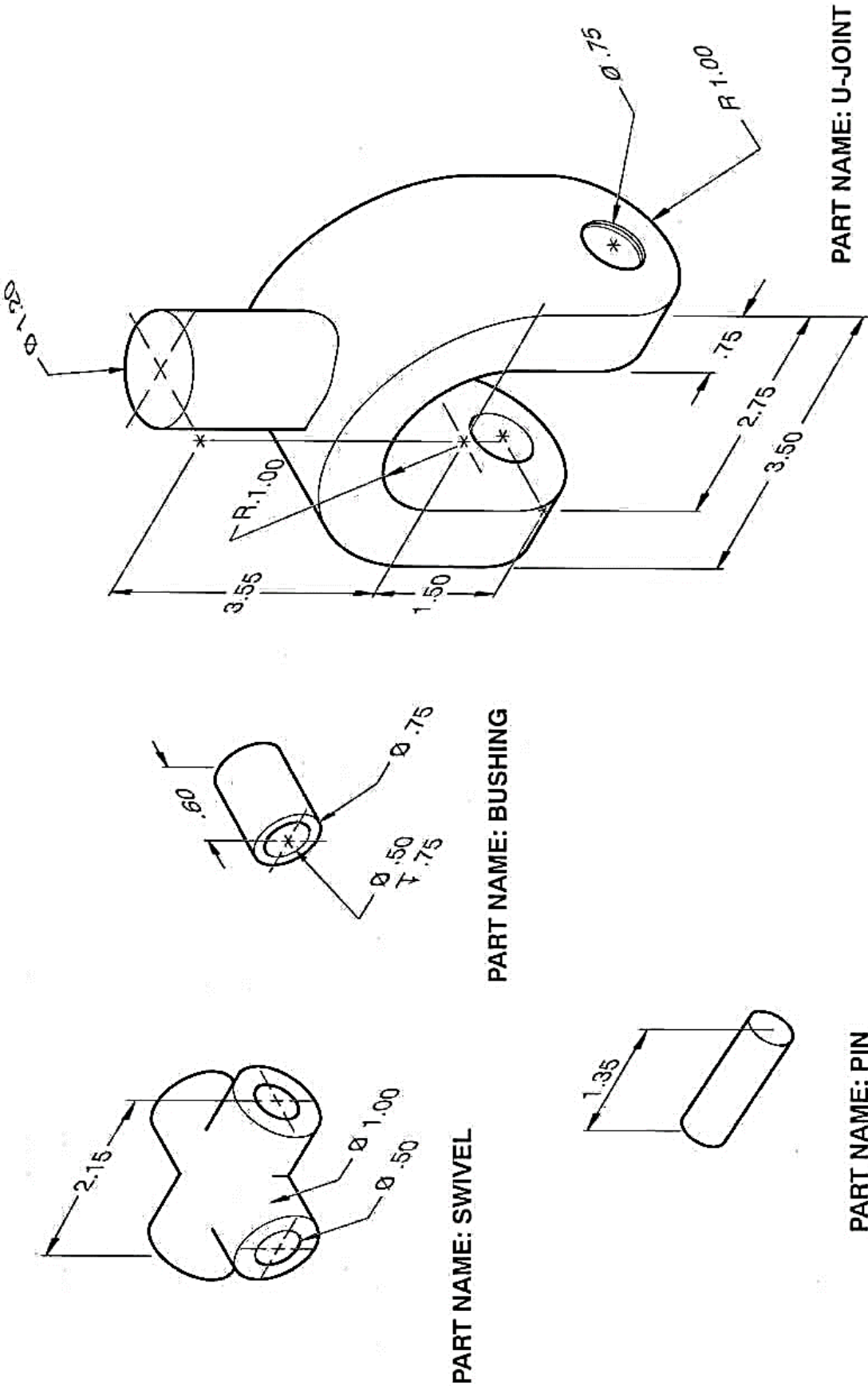


Fig. 4.2

Exercise 5 (Reading): Working sheets

Read the working drawing on the next page and answer the following questions. Write your answer in the same box as the question

What is a BOM in the assembly drawing? Create a BOM for the assembly drawing given. Denote the part number, part description, required quantity, and material.

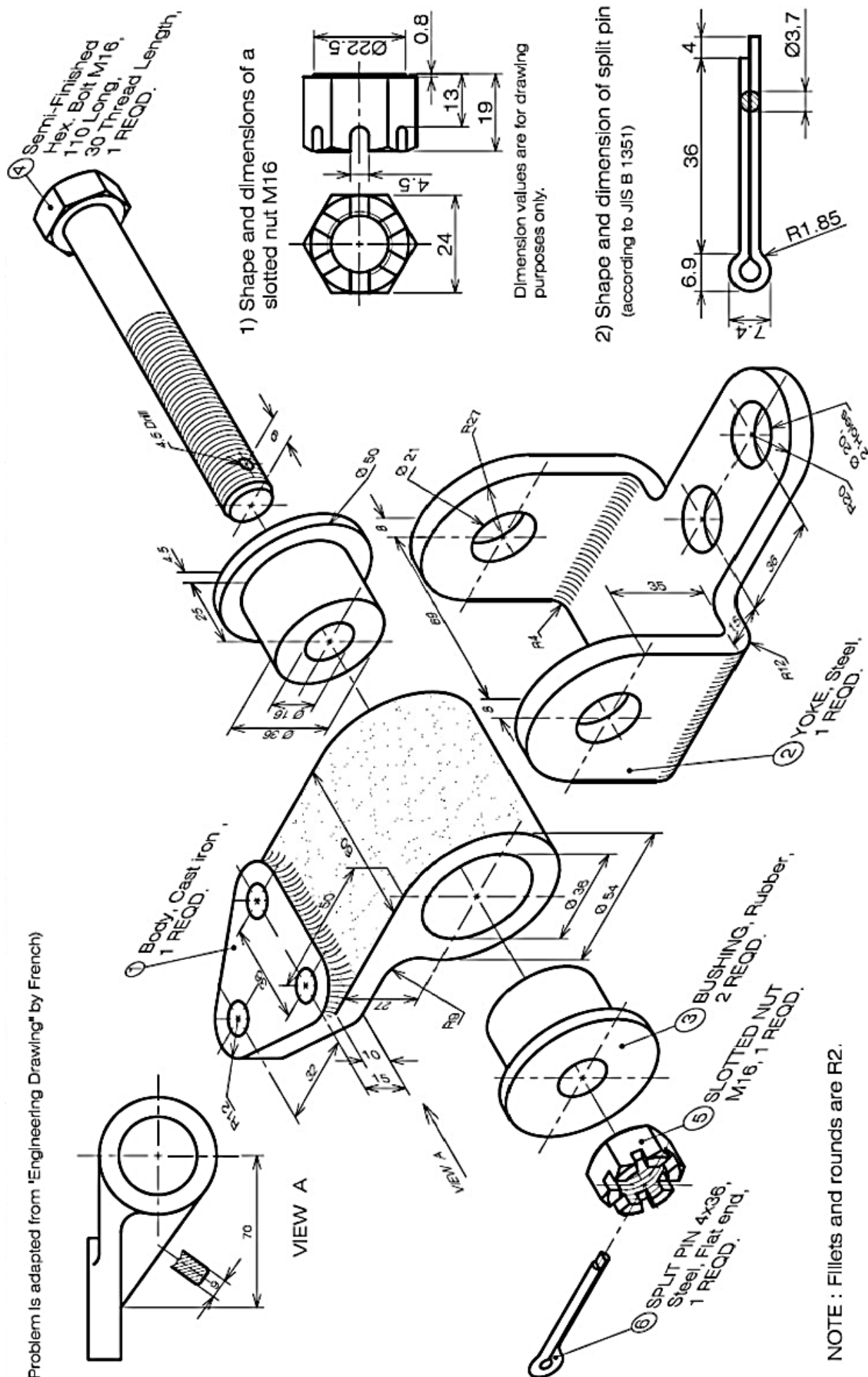
Would there be any clearance between the body and the yoke when bushings are in place? What would be the clearance if the bushing flange thickness is 4 mm?

How many detail views are present in the drawing? Why were they required?

Are there any standard parts used? What standards are cited in the annotations of the drawing?

What is the thread length of the fastener involved in the assembly? Can the thread length be changed? Why?

Are all parts assembled with a tight fit? Is there any clearance fit involved? Hint: Split pin.



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<u>Equipment Use</u> Sensory skills to <i>demonstrate</i> the use of the equipment for the lab work.	Doesn't demonstrate the use of equipment.	Slightly demonstrates the use of equipment.	Somewhat demonstrates the use of equipment.	Moderately demonstrates the use of equipment.	Fully demonstrates the use of equipment.
<u>Procedural Skills</u> <i>Displays</i> skills to act upon sequence of steps in lab work.	Not able to either learn or perform lab work procedure.	Able to slightly understand lab work procedure and perform lab work.	Able to somewhat understand lab work procedure and perform lab work.	Able to moderately understand lab work procedure and perform lab work.	Able to fully understand lab work procedure and perform lab work.
<u>Response</u> Ability to <i>imitate</i> the lab work on his/her own.	Not able to imitate the lab work.	Able to slightly imitate the lab work.	Able to somewhat imitate the lab work.	Able to moderately imitate the lab work.	Able to fully imitate the lab work.
<u>Observation's Use</u> <i>Displays</i> skills to use the observations from lab work for experimental verifications and illustrations.	Not able to use the observations from lab work for experimental verifications and illustrations.	Slightly able to use the observations from lab work for experimental verifications and illustrations.	Somewhat able to use the observations from lab work for experimental verifications and illustrations.	Moderately able to use the observations from lab work for experimental verifications and illustrations.	Fully able to use the observations from lab work for experimental verifications and illustrations.
<u>Confidence in Completion</u> Ability to <i>complete with confidence</i> the drawing task	Cannot complete the task	Completes with frequent hesitation	Completes with some confidence	Completes with growing confidence	Completes confidently and independently
<u>Work Efficiency</u> Ability to <i>improve efficiency</i> while completing tasks	Shows no efficiency	Efficiency very low	Efficiency slowly improving	Efficiency generally good	Efficiency high and consistent
<u>Safety Adherence</u> Adherence to <i>safety</i> procedures.	Doesn't adhere to safety procedures.	Slightly adheres to safety procedures.	Somewhat adheres to safety procedures.	Moderately adheres to safety procedures.	Fully adheres to safety procedures.
<u>Equipment Handling</u> <i>Equipment care</i> during the use.	Doesn't handle equipment with required care.	Rarely handles equipment with required care.	Occasionally handles equipment with	Often handles equipment with required care.	Handles equipment with required care.

Laboratory Session No. _____

Date: _____

Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	

Practical No. 5

Objective: Introduction to AutoCAD Graphical User Interface (GUI)

- 1 In this experiment, the objective is to become familiar with the **AutoCAD user interface** and to practice some of its important **features**. The first step is to open AutoCAD from the Windows operating system.

- 2 Once you click on acad.dwt, a new drawing template will open. The first thing to notice in the top-left corner of the window is the **Application Menu tab**, marked with a large red “A.” Clicking this tab opens a drop-down menu that provides access to your most recently created drawings, along with standard program options, as shown in Fig. 5.1. Although the Application Menu tab may appear slightly different in older versions, it is always located in the top-left corner of the screen.

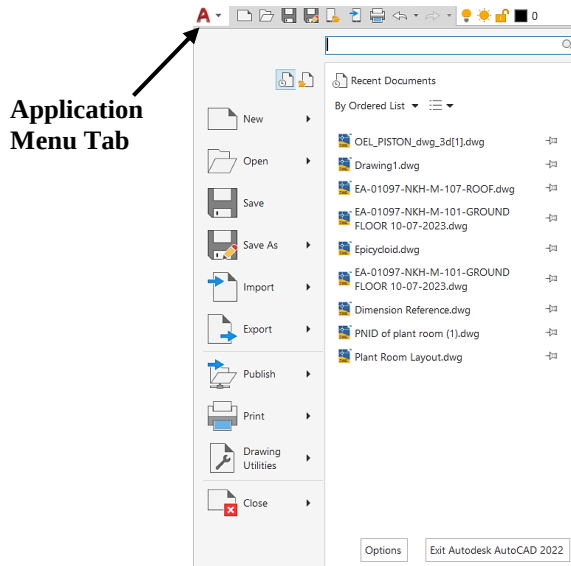


Fig. 5.1

- 3 Immediately to the right of the Application Menu tab is the **Quick Access Toolbar (QAT)**. As shown in Fig. 5.2, the QAT provides shortcut links to the most commonly used commands, such as New, Open, Save, Save As, Open from Web and Mobile, Save to Web & Mobile, Plot & Undo.

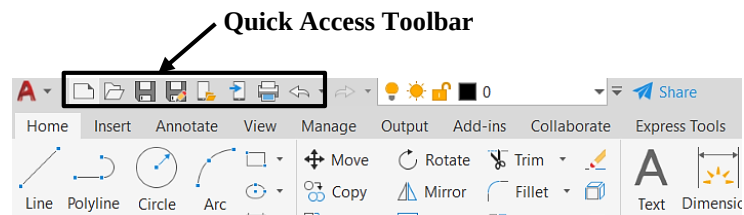
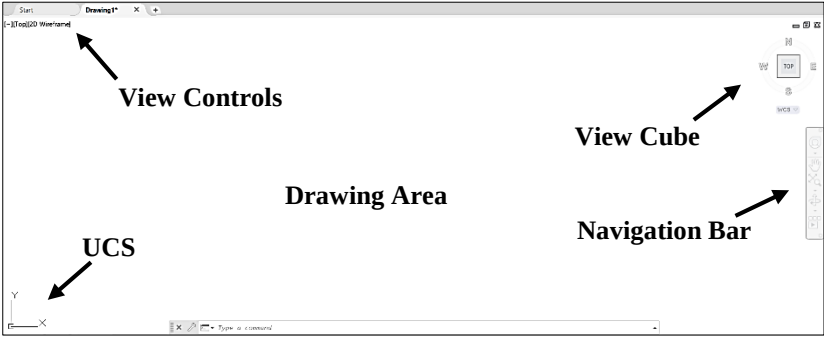
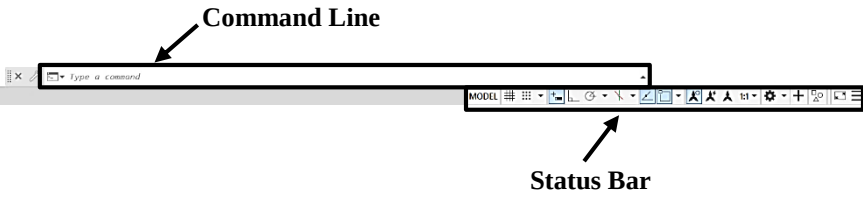


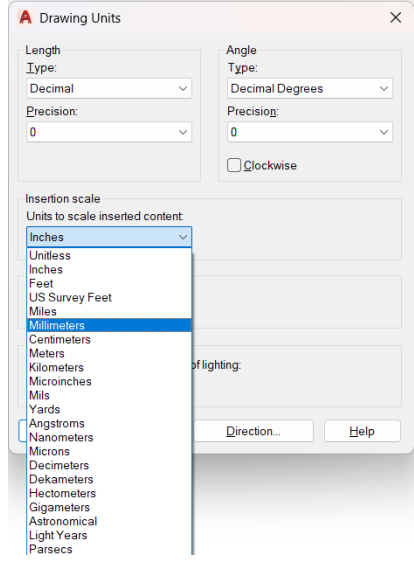
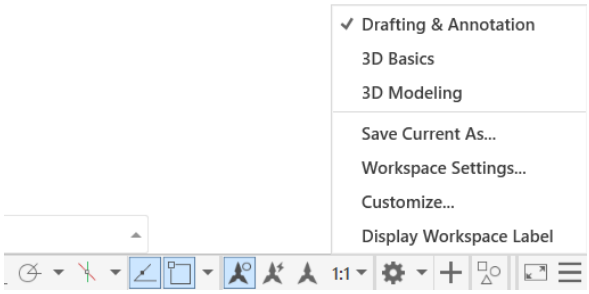
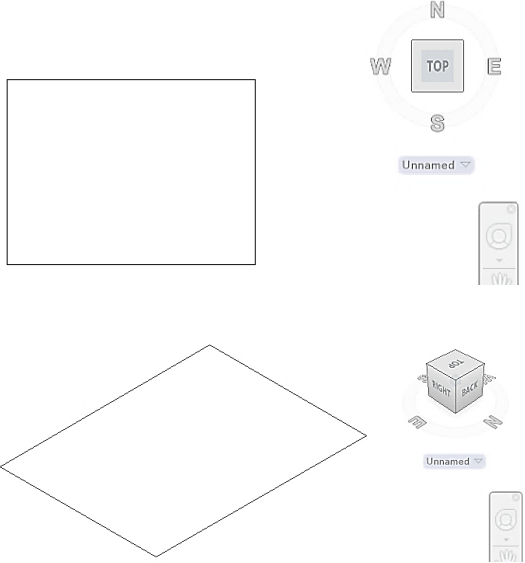
Fig. 5.2

- 4 Beneath the Quick Access Toolbar lies the **Ribbon Area**. As shown in Fig. 5.3, the Ribbon contains most of the essential AutoCAD tools and commands, including Draw, Modify, Annotation, Layers, and many more.



Fig. 5.3

5 Below the Ribbon Area is the Drawing Editor, which contains several important elements, each serving a different purpose, as shown in Fig. 5.4. In the top-left corner of the editor is the View Controls, used to change the view and visual style of the drawing. In the top-right corner is the View Cube, which allows you to rotate or adjust the orientation of the drawing space. Just below the View Cube is the Navigation Bar, which provides tools such as Pan, Zoom Extents, Orbit, and others. In the bottom-left corner is the XY axis, known as the User Coordinate System (UCS). The remaining central space is the Drawing Area, where the actual creation and editing of drawings take place.	 Fig. 5.4
6 At the bottom of the Drawing Area is the Command Line Console, commonly referred to as the Command Line. This is where you can enter commands, view prompts, and track system feedback while working in AutoCAD. Beneath the Command Line, in the bottom-right corner, lies the Status Bar as shown in Fig. 5.5. The Status Bar provides quick access to various drawing aids and settings, including: • GRIDMODE (F7) • SNAPMODE (F9) • DYNMODE • ORTHOMODE (F8) • Polar Tracking (F10) • ISODRAFT • Object Snap Tracking – AUTOSNAP (F11) • Object Snap – OSNAP (F3) • Workspace Switching These tools allow you to control precision, drafting aids, and workspace settings directly from the bottom of the screen.	 Fig. 5.5

7	Now we will learn some useful AutoCAD features.	
8	Before starting any drawing, it is important to ensure that the units are set to millimeters (mm). To do this: • Use the command “Units” and press Enter. • The Drawing Units dialog box will appear as shown in Fig. 5.6. • Under Length heading, set type to Decimal and the precision to 0, 0.0, or as required by the dimensions. Same for Angle. • From the Insertion Scale drop-down menu, select Millimeters. • Click OK to apply the settings.	 Fig. 5.6
9	For creating 2D drawings, go to the Status Bar and click on Workspace Switching as shown in Fig 5.7. From the available options, select Drafting & Annotation.	 Fig. 5.7
10	The View Cube allows you to rotate and adjust the orientation of the drawing space. By clicking on any face, edge, or corner of the cube, you can quickly switch to that specific view or orientation of the drawing as shown in Fig. 5.8.	 Fig. 5.8

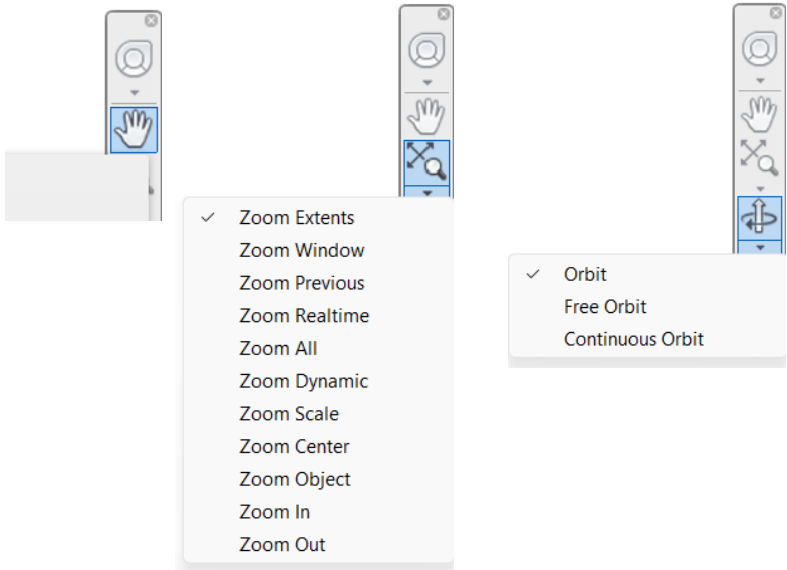
11	On the Navigation Bar, several tools are available to help navigate the drawing space as shown in Fig. 5.9: • Pan – Used to move the view of the drawing without changing its scale or orientation. It can be accessed either by clicking the Pan tool on the Navigation Bar or by typing the “PAN” command in the Command Line. • Zoom – Provides multiple options such as Zoom Window, Zoom Previous, Zoom Realtime, and others, allowing you to adjust the view scale as needed. It can be accessed either by clicking the Zoom tool or by typing the “Zoom” command. • Orbit – Offers options like Orbit, Free Orbit, and Continuous Orbit. Its main function is to rotate the drawing in any direction for better visualization. It can be accessed either by clicking the Orbit tool or by typing the “Orbit” command.	
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Fig. 5.9

12

On the Status Bar, several tools are available as shown in Fig. 5.10:

- **Grid Mode** – ON or OFF the grid line, press F7 or use command “Grid”.
- **Snap Mode** – enable the cursor only on the grid corners. To use this feature, press F9 or use command “Snapmode”.
- **Dynamic Mode** – controls the Dynamic Input feature, which lets you enter commands, dimensions, and coordinates directly at the cursor instead of only through the Command Line.
- **OrthoMode** – restricts cursor movement to horizontal or vertical directions only. Press F8 to use this feature.
- **Polar Tracking** – helps you draw lines or move objects at specific preset angles. Press F10 to use this feature.
- **Iso Drafting** – drawing mode that allows you to create isometric drawings.

It's a 2D representation that looks like 3D. When Isodraft mode is turned ON, the drawing area is divided into three isometric planes:

- ✓ Left Isoplane
- ✓ Right Isoplane
- ✓ Top Isoplane,

Object Snap – allows you to precisely select exact points on objects when creating or editing geometry. Instead of guessing, OSNAP “snaps” the cursor to key points on an object. Some common OSNAP modes include:

- ✓ **Endpoint** – snaps to the end of a line, arc, or polyline.
- ✓ **Midpoint** – snaps to the middle of a line or arc.
- ✓ **Center** – snaps to the center of a circle, arc, or ellipse.
- ✓ **Intersection** – snaps where two objects cross.
- ✓ **Perpendicular** – snaps to a point at 90° to another object.

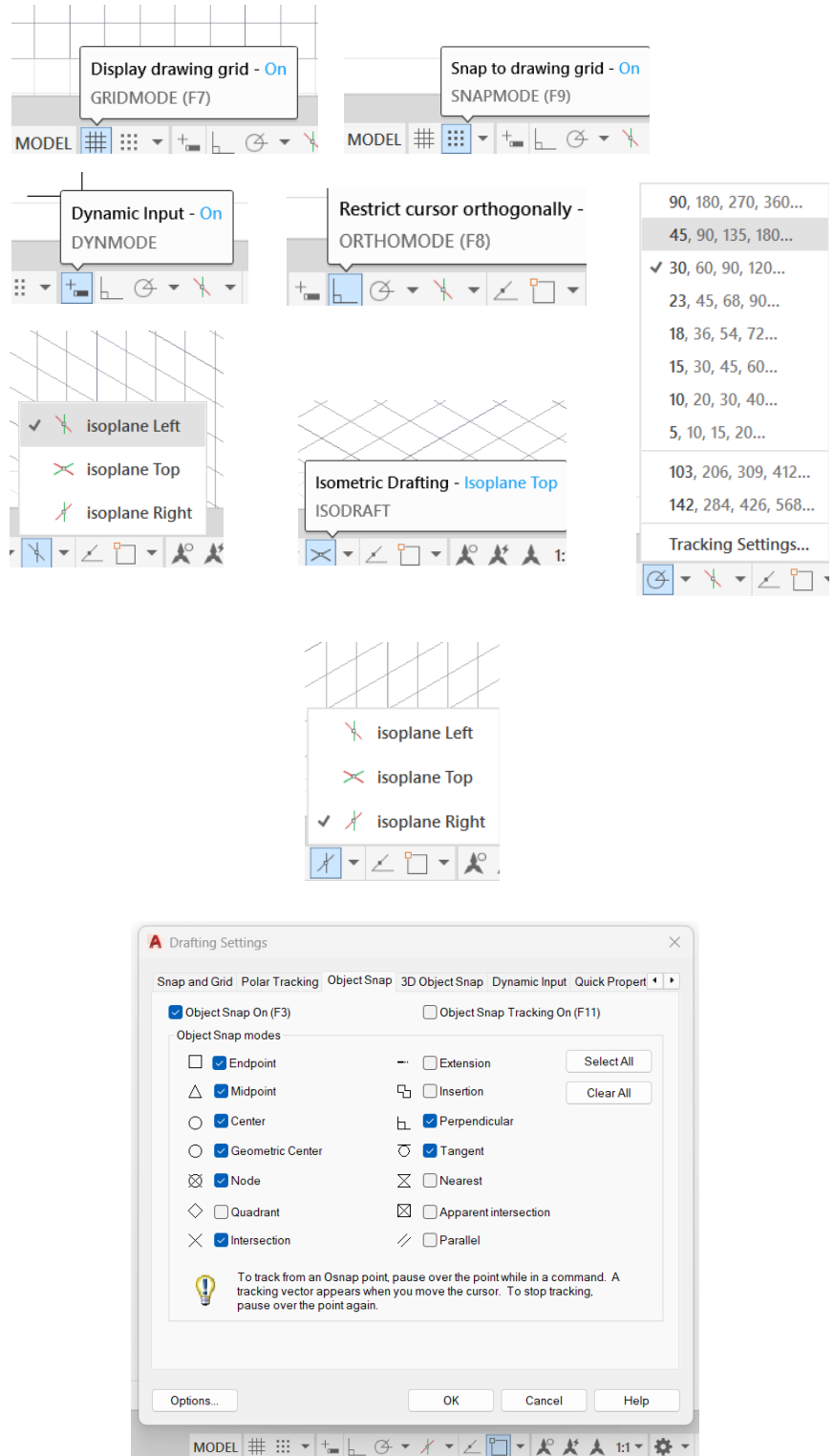
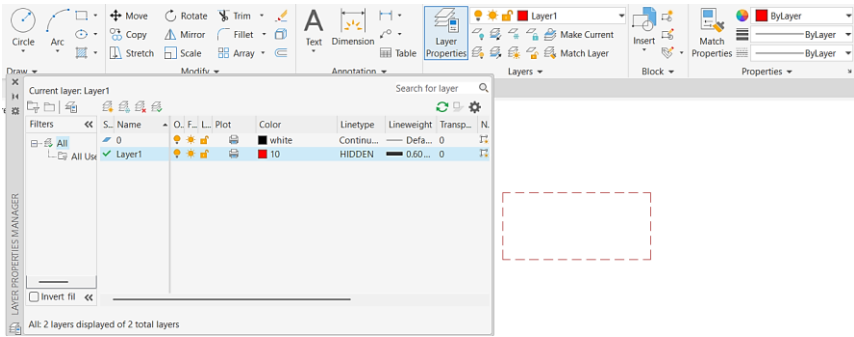
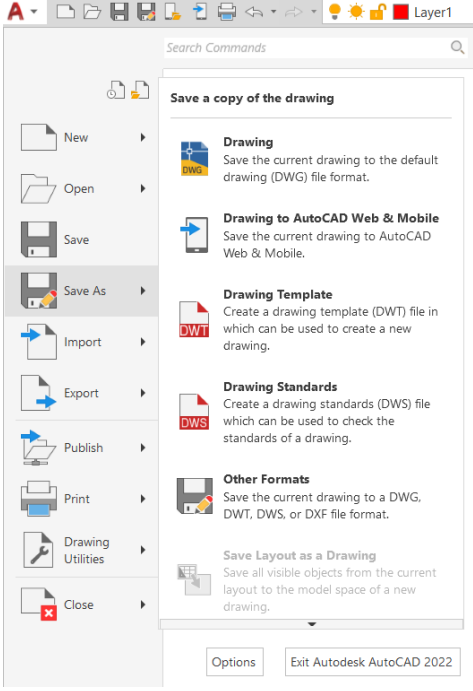


Fig. 5.10

13	Layers are used to organize different elements of a drawing, making it easier to manage and control objects efficiently. To use the Layer function: - Click on Layer Properties. A Layer Properties Manager dialog box will open as shown in Fig. 5.11. - Click New Layer to create a layer. A new row will be added in the dialog box. - You can then turn layers ON/OFF, lock/unlock, freeze/thaw, and change their color. - You can also modify the line type and line weight of each layer. - After selecting a specific layer, draw any object. It will adopt the properties of the chosen layer.	 Fig. 5.11
14	To save your work in AutoCAD: - Click the Application Menu and choose Save or Save As as shown in Fig. 5.12. Save – saves the drawing with the current file name and location. Save As – allows you to save the drawing with a different name, format, or location. File name – Enter a suitable name for your drawing. Save as type – Choose the drawing file format (default is .DWG). - Click Save. - You can also quickly save your drawing by pressing Ctrl + S on the keyboard.	 Fig. 5.12

15

The “Plot” command in AutoCAD is used to print or export your drawing.

- Type PLOT in the Command Line and press Enter.
- In the Plot dialog box as shown in Fig. 5.13, set the following options:
 - ✓ **Printer/Plotter** – Choose your output device (printer, plotter, or “DWG to PDF” for saving as a PDF).
 - ✓ **Paper Size** – Select the desired sheet size (e.g., A4, A3, etc.).
 - ✓ **Plot Area** → Choose what to print: Display, Extents, Window, or Layout.
 - ✓ **Drawing Orientation** – Choose Portrait or Landscape.
 - ✓ Click **Preview** to check how the drawing will appear when printed. If satisfied, click OK to plot/print the drawing.
 - ✓ You can press Ctrl + P to open the Plot dialog box quickly.

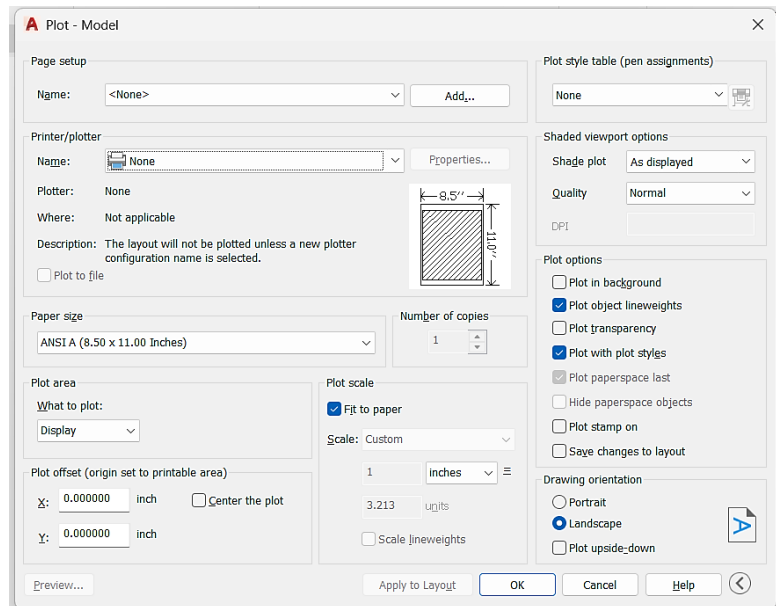


Fig. 5.13

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Course Code and Title: **ME-118 Engineering Drawing**

Software Use Rubric (P3)					
Skill Sets	Extent of Achievement				
	0	1	2	3	4
Identification and usage of correct software menus	Does not use software menus at all	Rarely uses software menus correctly	Occasionally uses software menus correctly	Often uses software menus correctly	Uses software menus correctly and effectively
Skills to use basic options	Not able to use any basic options	Uses basic options with detailed supervision	Uses basic options with some guidance	Uses basic options without guidance	Exhibits all basic options effectively
Skills to use advanced/associated features	Not able to use any advanced/associated features	Uses advanced / associated features with detailed supervision	Uses advanced / associated features with some guidance	Uses advanced / associated features without guidance	Exhibits all advanced / associated features effectively
Performing the assigned task using software	Cannot use the software for assigned task	Uses the software with below-average expertise	Uses the software with average expertise	Uses the software with above-average expertise	Uses the software with proficient expertise
Extraction of required output from software	Does not extract required output at all	Rarely extract required output from software	Occasionally extract required output from software	Often extract required output from software	Effectively extract required output from software

Laboratory Session No. _____

Date: _____

Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	

Practical No. 6

Objective: To create 2D drawings using basic geometrical construction commands in AutoCAD

1

In this experiment, we will learn how to create 2D template drawings using AutoCAD commands. As the first step, set the drawing units to millimeters by using the UNITS command, as explained earlier.

2

Line and Polyline Command: To draw a line or polyline, you can either click on the Line or Polyline tool in the Ribbon menu or type the respective command in the Command Line. For a line, type "LINE (or simply L)" and press Enter, then specify the start and end points in the drawing area. For a polyline, type "PLINE (or simply PL)" and press Enter, then follow the Command Line prompts to create connected segments as shown in Fig. 6.1.

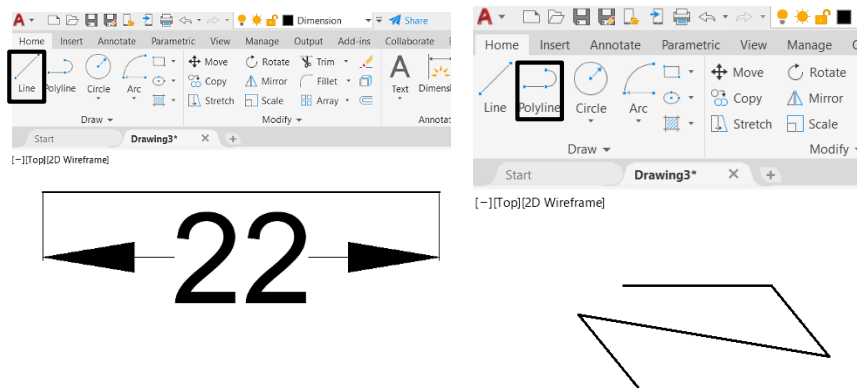


Fig. 6.1

3

Circle Command: To draw a circle, either click on the Circle tool in the Ribbon menu or use the command "CIRCLE (or simply C)". Then, follow the prompts to create the circle using your preferred option (e.g., center-radius, center-diameter, 2-point, or 3-point) as shown in Fig. 6.2.

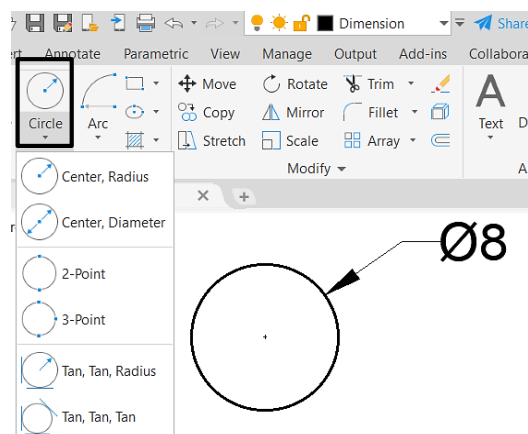


Fig. 6.2

4

Arc Command: To draw an arc, either click on the Arc tool in the Ribbon menu or use the command "ARC (or simply A)". Then, follow the Command Line prompts to specify the required points (start, center, end, etc.) for creating the arc as shown in Fig. 6.3.

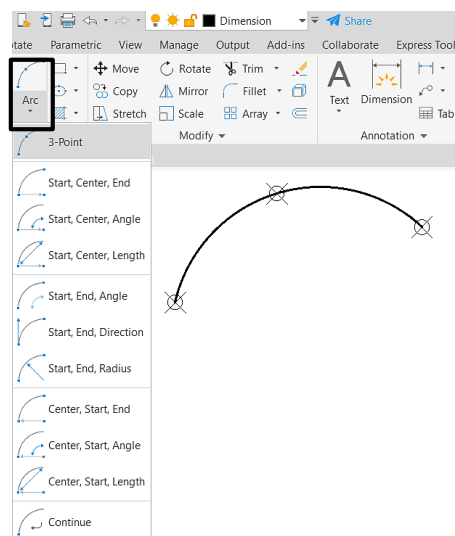
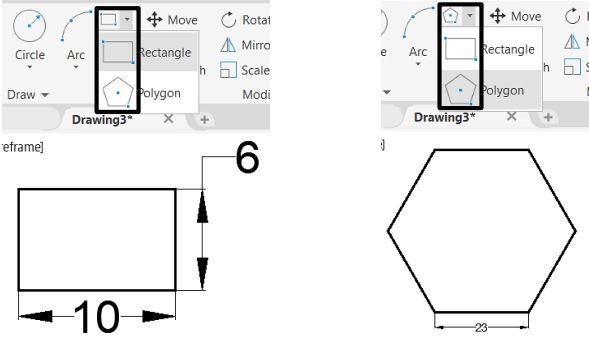
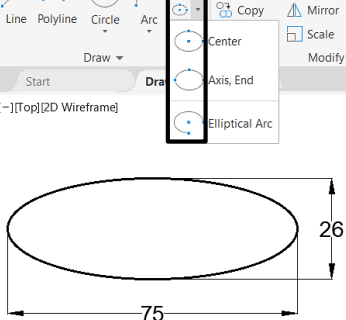
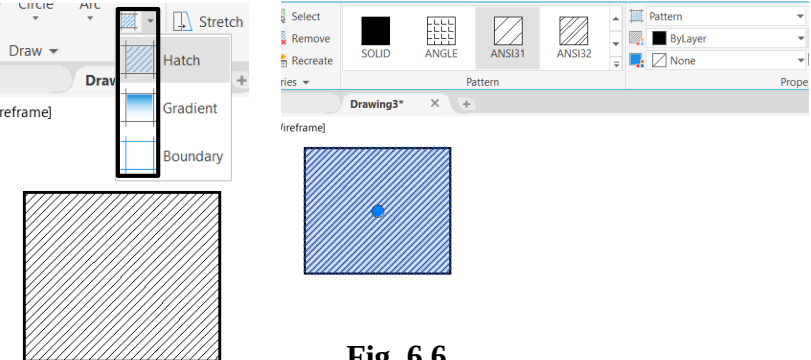
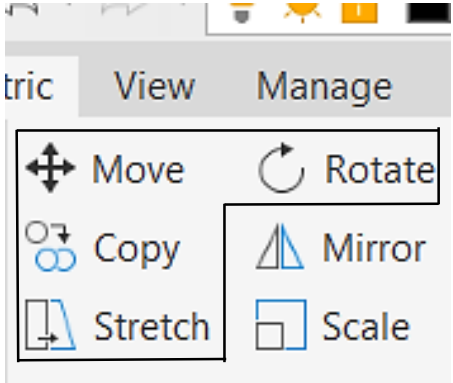
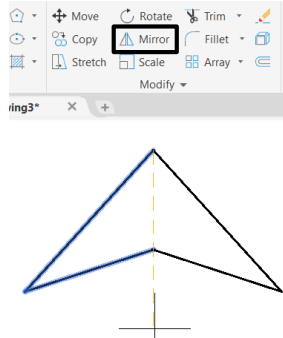
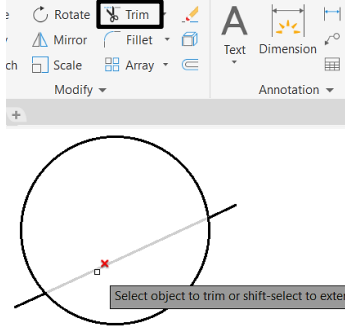
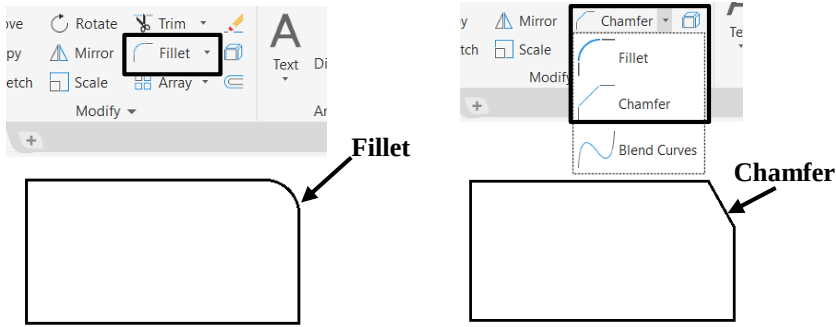


Fig. 6.3

5	Rectangle and Polygon Commands: To draw a rectangle, click on the Rectangle tool in the Ribbon menu or use the command “RECTANGLE (or simply REC)”, then specify the two opposite corners. To draw a polygon, click on the Polygon tool or type “POLYGON”, enter the number of sides, and follow the prompts to inscribe or circumscribe the shape as shown in Fig. 6.4.	 Fig. 6.4
6	Ellipse Command: To draw an ellipse, click on the Ellipse tool in the Ribbon menu or use the command “ELLIPSE”. Then specify the endpoints of one axis and the distance for the other axis as shown in Fig. 6.5.	 Fig. 6.5
7	Hatch Command: To apply hatching, click on the Hatch tool in the Ribbon menu or use the command “HATCH (or simply H)”. Then select the boundary area and choose the desired hatch pattern, scale, and angle as shown in Fig. 6.6.	 Fig. 6.6
8	Move, Copy, Stretch, and Rotate Commands: To modify objects, use the following commands from the Ribbon menu as shown in Fig. 6.7 or use the command “MOVE” to shift an object from one location to another, “COPY” to create duplicates of an object, “STRETCH” to alter the shape or length of an object by moving selected portions, and “ROTATE” to turn an object around a specified base point by a given angle.	 Fig. 6.7

9	Mirror Command: The Mirror command can be accessed from the Ribbon menu, or by using the command “MIRROR (or simply MI)”. It is used to create a mirrored copy of selected objects by specifying a mirror line as shown in Fig. 6.8. You can then choose whether to erase or keep the original objects.	 Fig. 6.8
10	Trim Command: The TRIM command can be accessed from the Ribbon menu or by using the command “TRIM (or simply TR)”. It is used to remove or cut off unwanted parts of objects where they intersect with other objects as shown in Fig. 6.9.	 Fig. 6.9
11	Fillet and Chamfer Commands: The FILLET command can be accessed from the Ribbon menu or by using the command “FILLET (or simply F)”. It is used to create a rounded corner between two objects. The CHAMFER command, accessed from the Ribbon or by using the command “CHAMFER (or simply CHA)”, is used to create a beveled edge between two objects by specifying distances or angles as shown in Fig. 6.10.	 Fig. 6.10

Exercise 1:

Draw the 2D mechanical part in AutoCAD as shown in the Fig. 6.11, including circular ends, hexagonal cutouts, handle slot, and connecting fillets. Use appropriate drawing and modify commands, and add dimensions to complete the drawing.

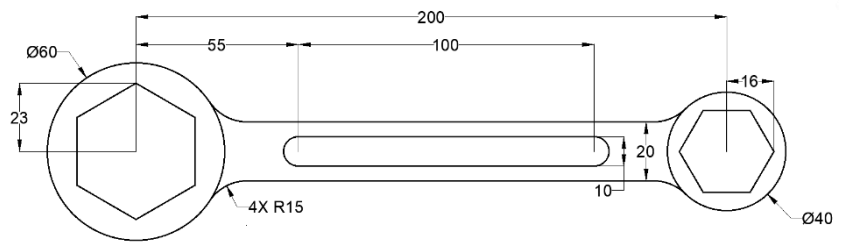


Fig. 6.11

Exercise 2:

Draw the given 2D mechanical part in AutoCAD as shown in Fig. 6.12 using the provided radii and geometry. Ensure that all circles, arcs, and fillets are constructed accurately and maintain proper symmetry.

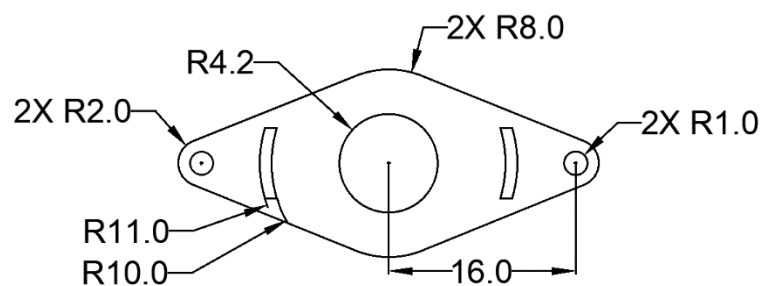


Fig. 6.12

Exercise 3:

Draw the given 2D mechanical part in AutoCAD as shown in Fig. 6.13 using the provided radii and geometry. Ensure that all circles, arcs, and fillets are constructed accurately and maintain proper symmetry.

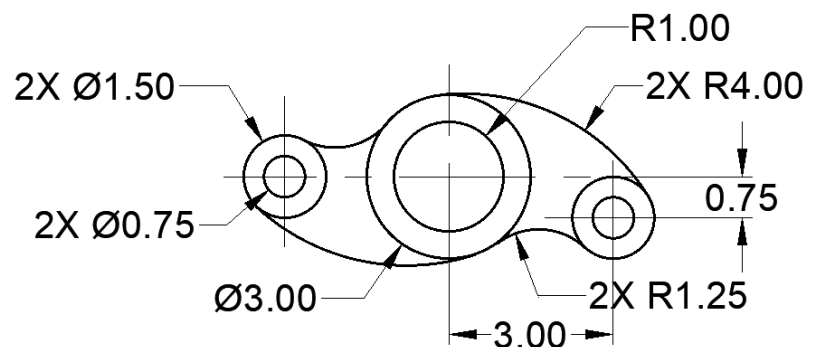


Fig. 6.13

Exercise 4:

Draw the given 2D mechanical part in AutoCAD as shown in Fig. 6.14 using the provided radii and geometry. Ensure that all circles and arcs are constructed accurately and maintain proper symmetry.

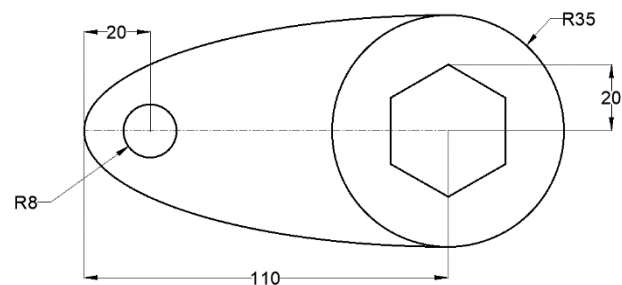


Fig. 6.14

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Laboratory Session No. _____

Date: _____

Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	

Practical No. 7

Objective: To develop 2D drawings using Array function in AutoCAD

1 In this experiment, we will learn how to create 2D drawings using Array Function.
As the first step, set the drawing units to millimeters by using the UNITS command, as explained earlier.

2 **Rectangular Array Command:**
The Rectangular Array command creates multiple copies of selected objects arranged in rows and columns at specified distances. It can be accessed from the Ribbon menu or by using the command “ARRAYRECT (or AR)” as shown in Fig. 7.1.

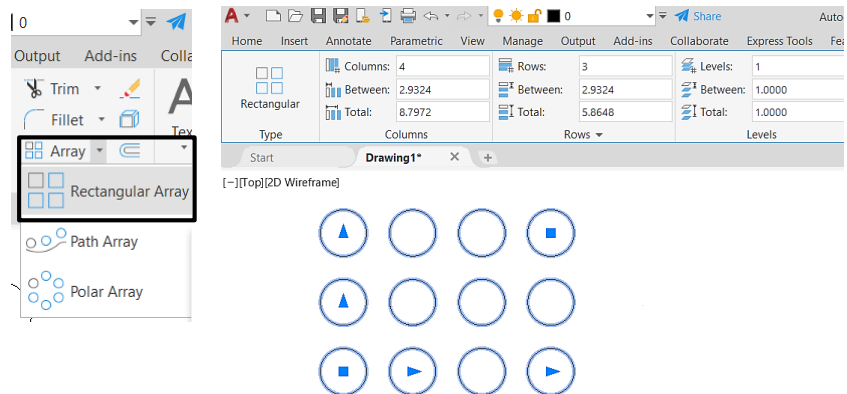


Fig. 7.1

3 **Polar Array Command:**
The POLAR ARRAY command is used to create multiple copies of objects in a circular pattern around a center point. To use it, go to the Array tool in the Ribbon (Modify panel) and select Polar Array, or use the command “ARRAYPOLAR (or simply ARRAY and choose Polar)”. Then, specify the object, center point, number of items, and angle to fill as shown in Fig. 7.2.

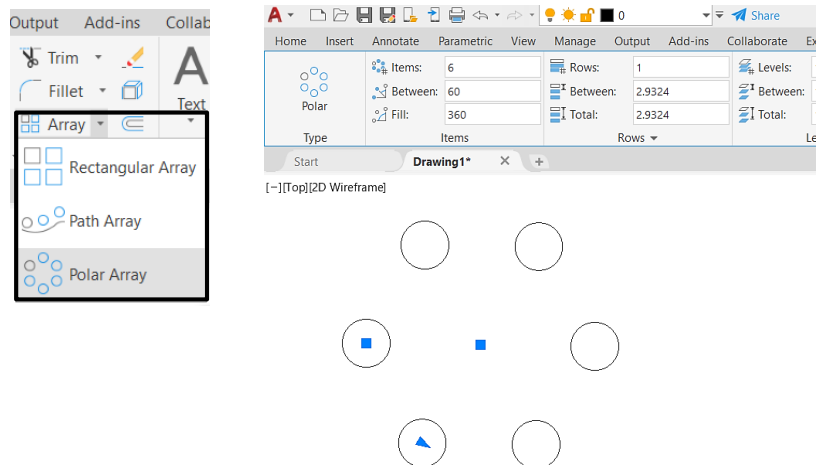


Fig. 7.2

4 **Path Array Command:**
The PATH ARRAY command is used to create multiple copies of objects along a specified path such as a line, polyline, arc, circle, or spline. To use it, go to the Array tool in the Ribbon (Modify panel) and select Path Array, or use the command “ARRAYPATH”. Then, select the object to array, choose the path, and adjust the number of items, spacing, and alignment options as required as shown in Fig. 7.3.

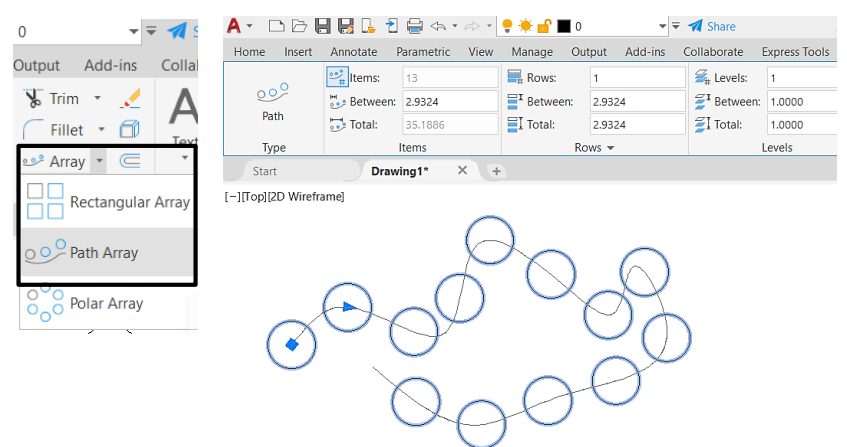


Fig. 7.3

5

Exercise 1:

Recreate the given figure in AutoCAD, as shown in Fig. 7.4, by applying the array command.

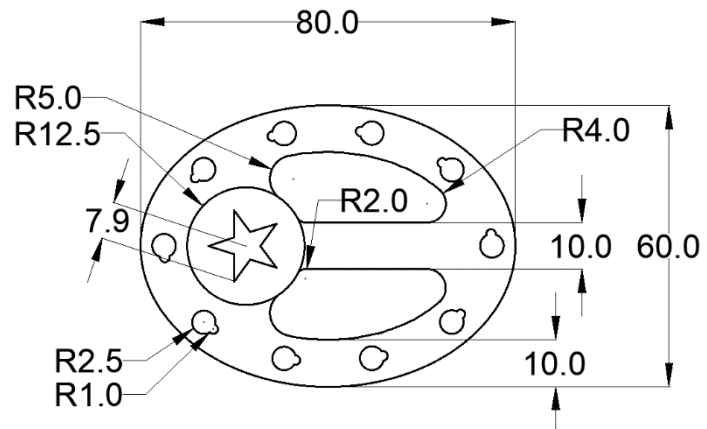


Fig. 7.4

6

Exercise 2:

Recreate the given figure in AutoCAD, as shown in Fig. 7.5, by applying the array command.

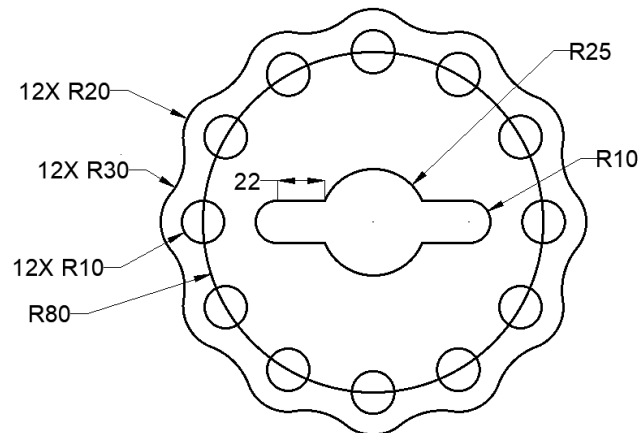


Fig. 7.5

7

Exercise 3:

Recreate the given figure in AutoCAD, as shown in Fig. 7.6, by applying the array command.

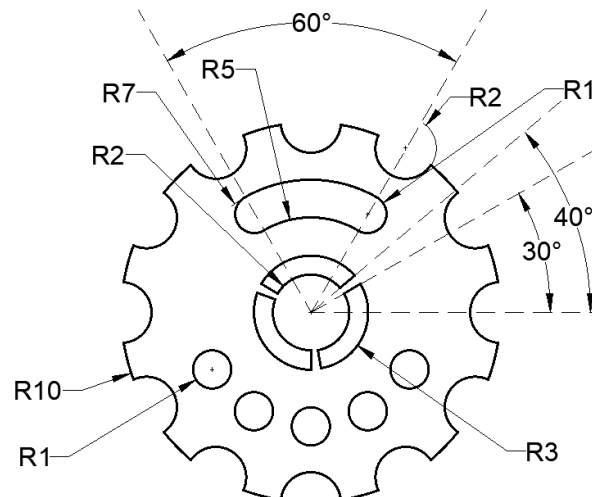


Fig. 7.6

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Course Code and Title: **ME-118 Engineering Drawing**

Software Use Rubric (P3)					
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Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	

Practical No. 8

Objective: Applications of Engineering Curves (Involute and Epicycloid) in 2D Drawing with AutoCAD

1 In this experiment, we will learn how to draw an **involute & Epicycloid** in AutoCAD.
As a first step, change the drawing units to millimeters by using the UNITS command, as explained earlier.

2 First, draw a circle with a radius of 20 mm.
Next, construct a tangent to the circle using the arc length formula $C = 2\pi r$, as shown in Fig. 8.1.

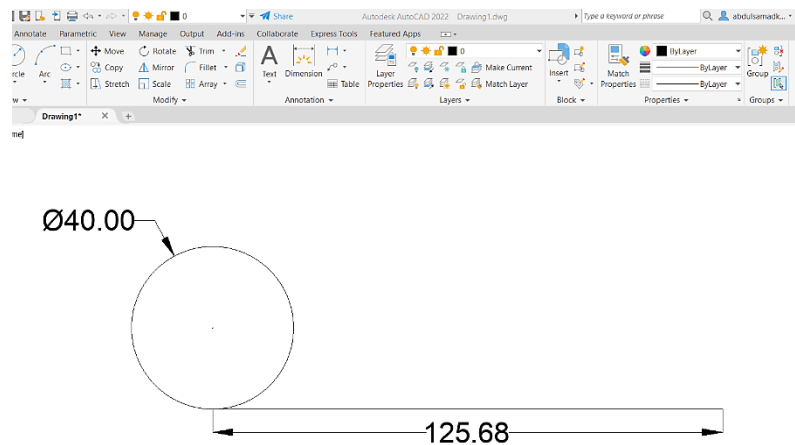


Fig. 8.1

3 Now, divide the tangent into 12 equal parts using the command "Divide".
To make the division points more visible, use the command "DDPTYPE" and select an appropriate point style and size for clear display as shown in Fig. 8.2.

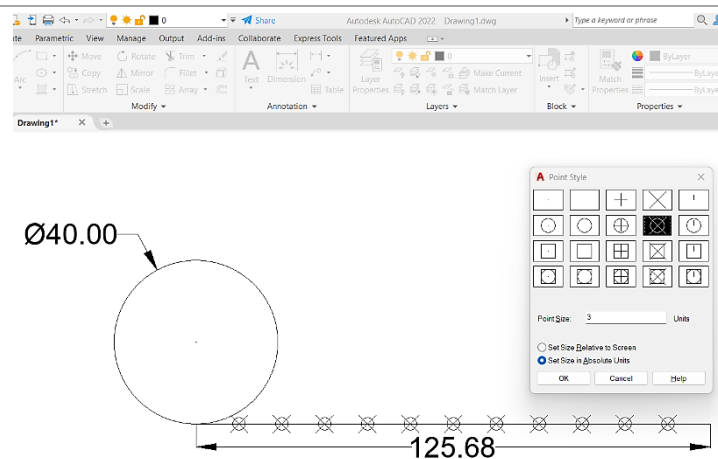


Fig. 8.2

4 Use the command "Arraypolar" to create 12 equidistant tangents around the circle as shown in Fig. 8.3.
After that, apply the command "Explode" to the entire array so that each tangent becomes an individual object, allowing you to select points on the tangents individually.

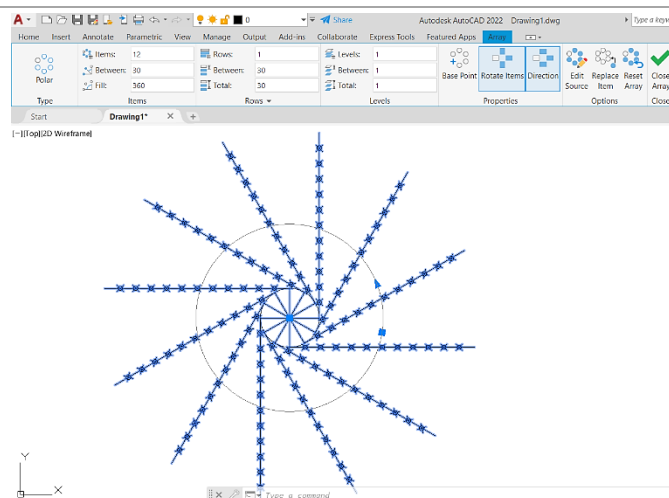
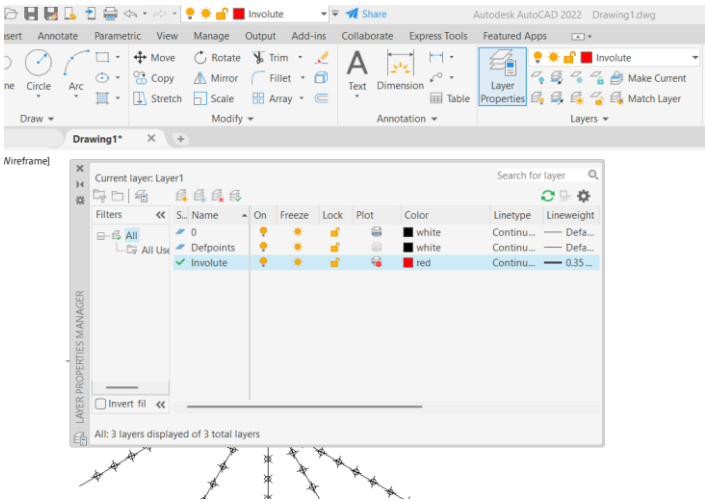
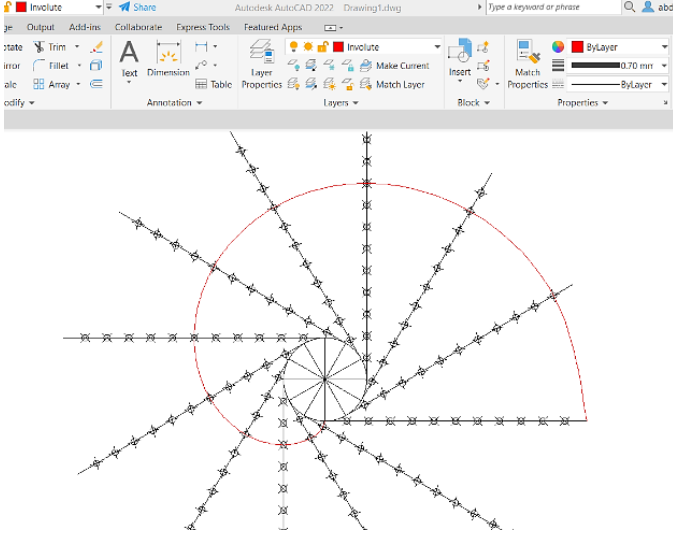
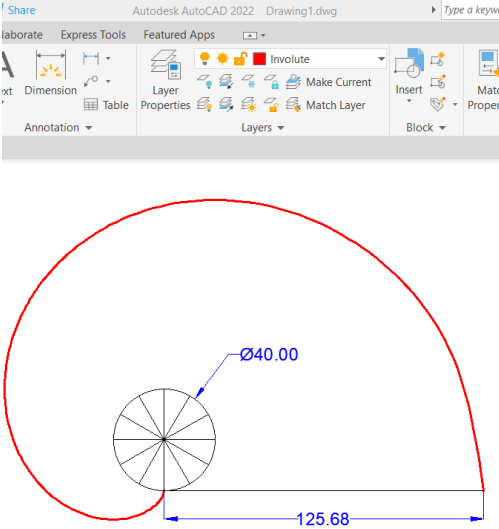
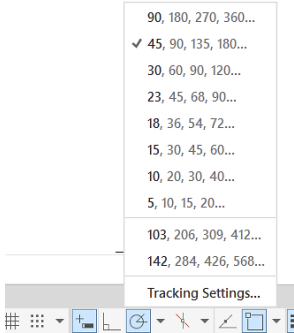
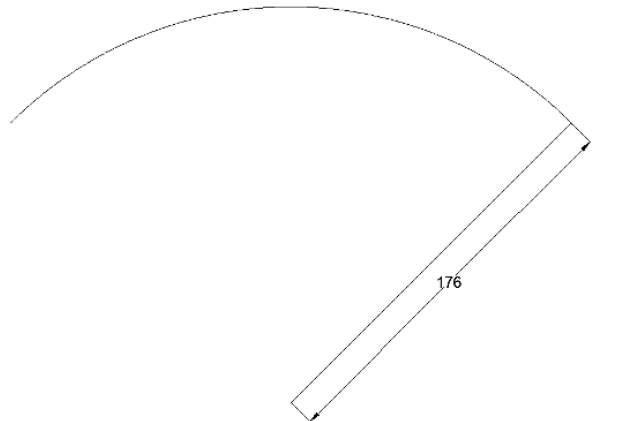
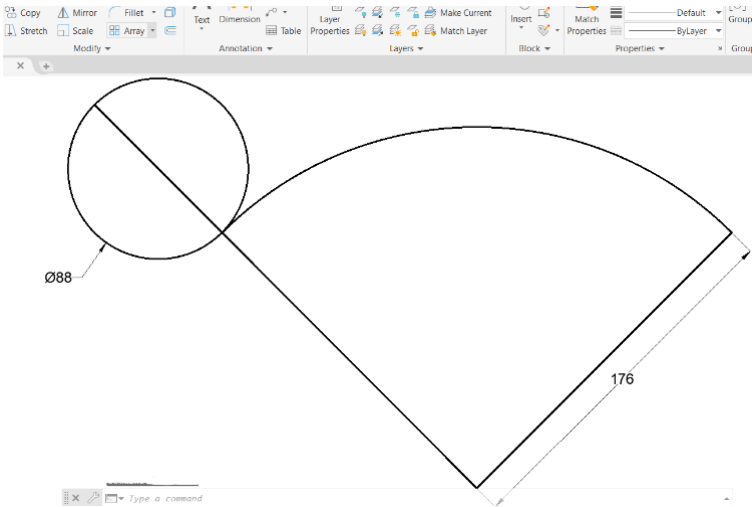
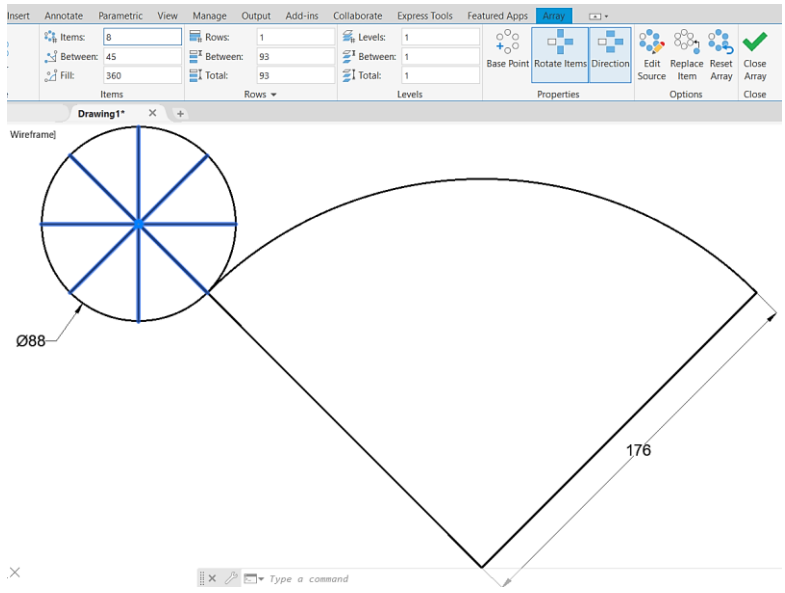
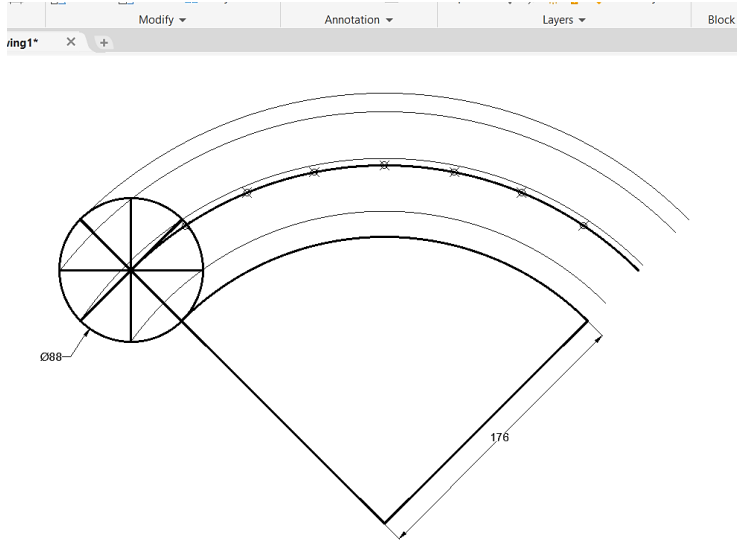
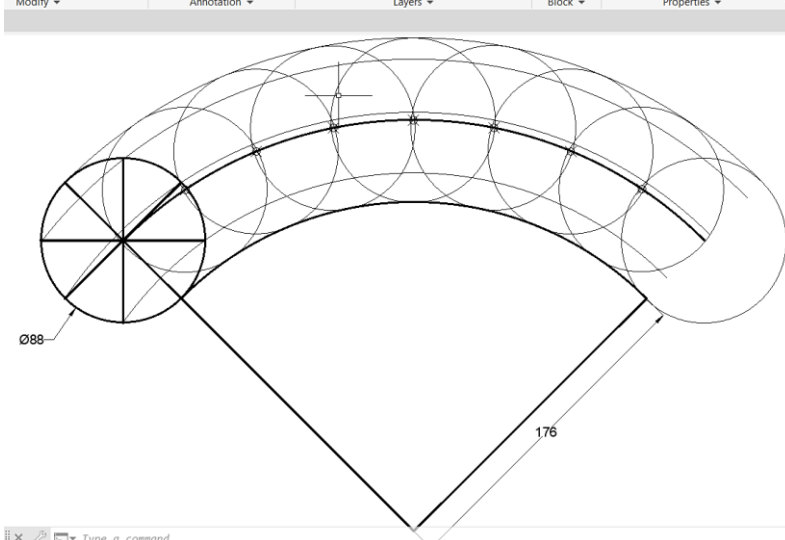
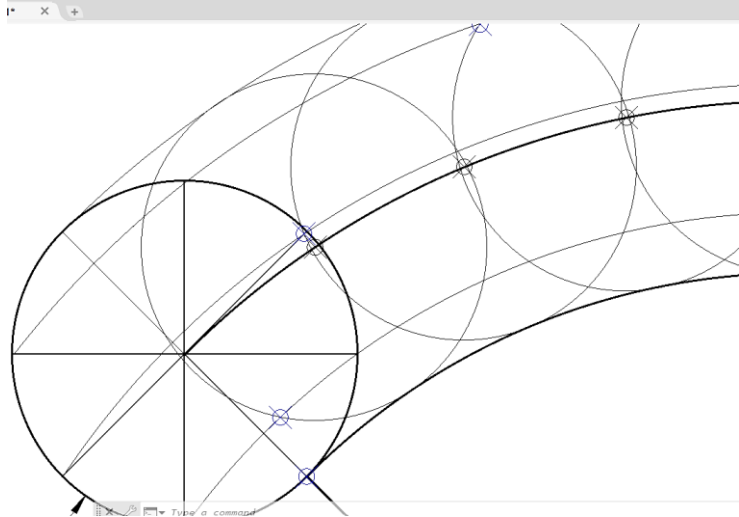
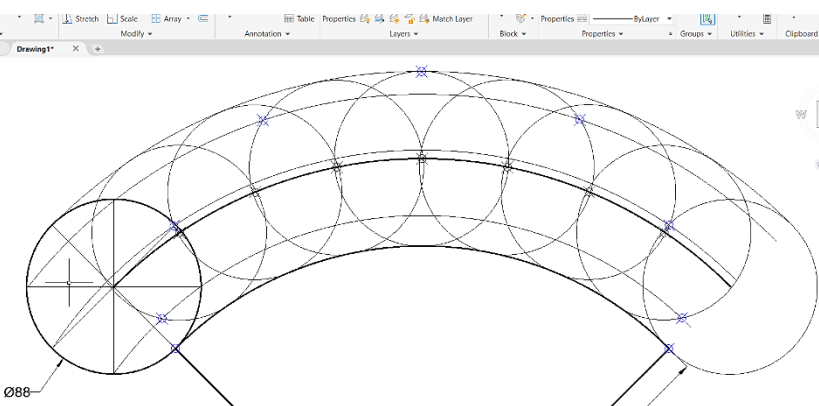
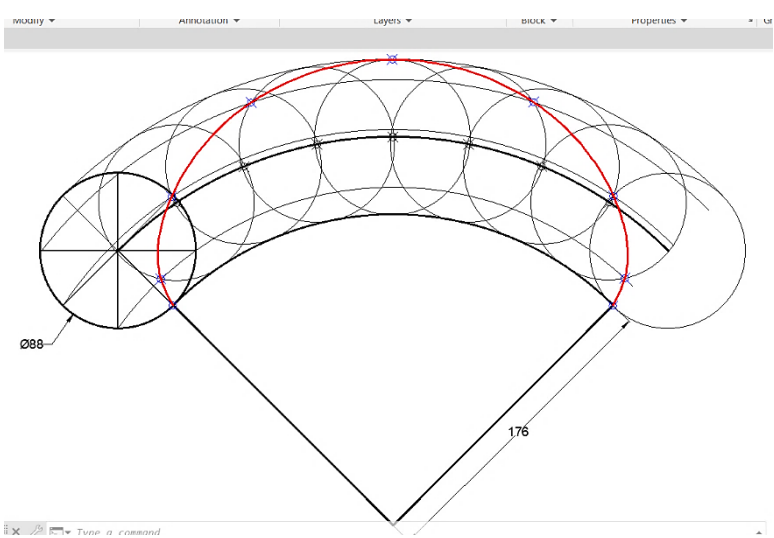


Fig. 8.3

5	Now, we will begin drawing the involute curve. For better clarity and organization, create a new layer using the Layer Properties option and name it “Involute”, as shown in Fig. 8.4.	 The screenshot shows the AutoCAD 2022 interface with the Layer Properties Manager open. A new layer named 'Involute' has been created, highlighted in blue. It is set to be 'On', 'Unfrozen', and 'Plotted'. The color is set to 'red' and the lineweight is '0.35'. The current layer is 'Layer1'. Fig. 8.4
6	To draw the involute curve, use the “Spline” command and press Enter. Starting from the lowest point on the circle, sequentially join the division points located on the array tangents. Continue this process by following the concept of the involute curve until all points are smoothly connected through the spline as shown in Fig. 8.5.	 The screenshot shows the AutoCAD 2022 interface with a red circle and its involute curve. The involute curve is drawn using the Spline command, connecting the division points on the array tangents. The curve is shown in red, matching the 'Involute' layer settings. Fig. 8.5
7	Remove the unnecessary lines and dimension the drawing as shown in Fig. 8.6.	 The screenshot shows the final drawing of the involute curve. The unnecessary lines have been removed, and the drawing is dimensioned. The circle has a diameter of $\varnothing 40.00$ and the involute curve has a length of 125.68. Fig. 8.6

8	We will now learn how to draw an Epicycloid in AutoCAD. As a first step, change the drawing units to millimeters by using the UNITS command, as explained earlier.	
9	First set the polar tracking to 45, 90, 135, 180 etc. as shown in Fig 8.7.	
10	Draw a line of 176 mm at an angle of 45°. Use “Center, Start, End” from the Ribbon bar and draw the curve at an angle of 90°. Then join the two open ends as shown in Fig. 8.8.	
11	The included angle can be found by the formula $\theta = \left(\frac{r}{R}\right) 360^\circ$. Draw the generating circle of diameter 88 mm outside the circumference of larger circle as shown in Fig. 8.9.	

12	Use the command “Arraypolar” to divide the generating circle into 8 equal parts as shown in Fig. 8.10.	 Fig. 8.10
13	From the center of the generating circle, draw an arc using the “Center, Start, End” option from the Ribbon menu. Next, divide this arc into 8 equal segments. Using each division point on the generating circle, draw additional arcs with the same method as shown in Fig. 8.11.	 Fig. 8.11
14	Copy the generating circle with center on the division points using the command “Copy” as shown in Fig. 8.12.	 Fig. 8.12

15	Use the “Multiple Points” option from the Ribbon menu to place points at the intersections of the circles and their corresponding arcs, as illustrated in Fig. 8.13.	 Fig. 8.13
16	The points will look like this as shown in Fig. 8.14.	 Fig. 8.14
17	Use the “Spline” command to join the plotted points and generate the Epicycloid curve as shown in Fig. 8.15.	 Fig. 8.15

Exercise 1:

Draw a **General Cycloid** in AutoCAD as shown in Fig. 8.16 using the following specifications:

- A generating circle of diameter 88 mm.
- A directing line of length 276 mm.
- Use the method of rolling the generating circle along the directing line.
- Mark the division points on the circle and line, then plot the curve by joining the points to form the cycloid.

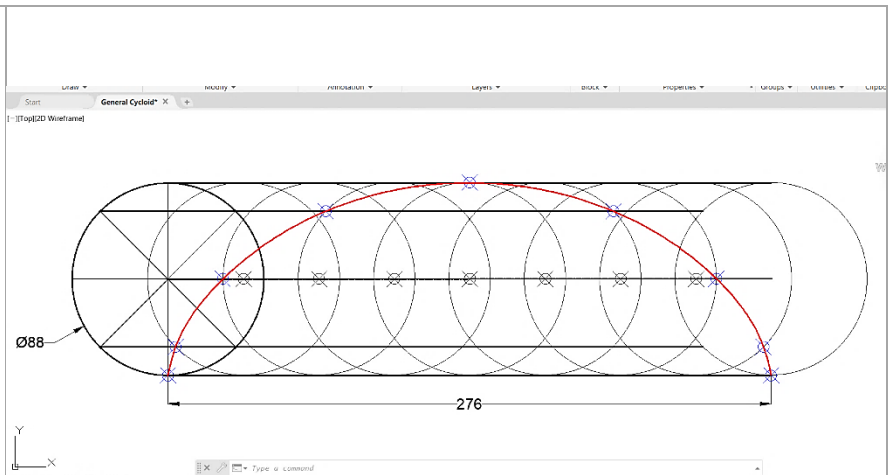


Fig. 8.16

NED University of Engineering & Technology
Department of Mechanical Engineering
Course Code and Title: **ME-118 Engineering Drawing**

Software Use Rubric (P3)					
Skill Sets	Extent of Achievement				
	0	1	2	3	4
Identification and usage of correct software menus	Does not use software menus at all	Rarely uses software menus correctly	Occasionally uses software menus correctly	Often uses software menus correctly	Uses software menus correctly and effectively
Skills to use basic options	Not able to use any basic options	Uses basic options with detailed supervision	Uses basic options with some guidance	Uses basic options without guidance	Exhibits all basic options effectively
Skills to use advanced/associated features	Not able to use any advanced/associated features	Uses advanced / associated features with detailed supervision	Uses advanced / associated features with some guidance	Uses advanced / associated features without guidance	Exhibits all advanced / associated features effectively
Performing the assigned task using software	Cannot use the software for assigned task	Uses the software with below-average expertise	Uses the software with average expertise	Uses the software with above-average expertise	Uses the software with proficient expertise
Extraction of required output from software	Does not extract required output at all	Rarely extract required output from software	Occasionally extract required output from software	Often extract required output from software	Effectively extract required output from software

Laboratory Session No. _____

Date: _____

Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	

Practical No. 9

Objective: To develop and visualize parametric 3D models using basic geometrical construction commands in AutoCAD

- 1 In this experiment, we will learn how to draw **3D drawings** using AutoCAD commands.
As a first step, change the drawing units to millimeters by using the UNITS command, as explained earlier.

- 2 **Isometric Drawing:**
Isometric drawing is a type of 3D representation used in technical drafting where an object is drawn with its axes typically set at 120° to each other, allowing all three dimensions (length, width, and height) to be shown in a single view as depicted in Fig. 9.1.

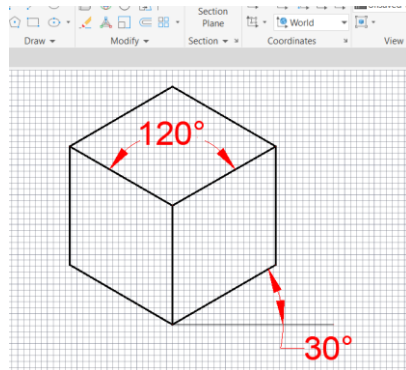


Fig. 9.1

- 3 For creating 3D drawings, go to the Status Bar and click on Workspace Switching as shown in Fig 9.2. From the available options, select **3D modeling**.

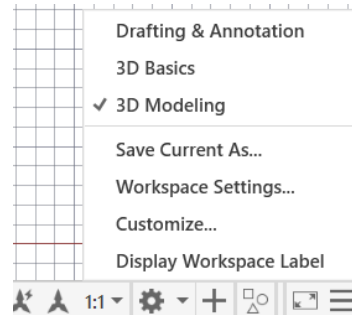


Fig. 9.2

- 4 To change the 2D axis into 3D, select **SW Isometric** from the Ribbon menu, and the axis will switch to a 3D orientation as shown in Fig. 9.3.

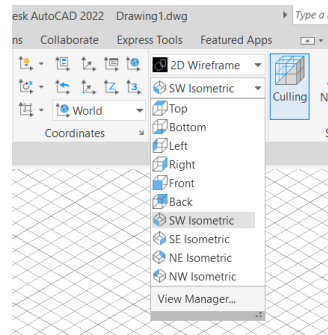


Fig. 9.3

- 5 **Box, Cylinder, Cone Command etc.:** The BOX, CYLINDER, and CONE commands in AutoCAD are used to create basic 3D solid models. They can be accessed either from the Ribbon menu or by typing BOX, CYLINDER, or CONE in the command line, making them essential tools for constructing 3D shapes in AutoCAD as shown in Fig. 9.4.

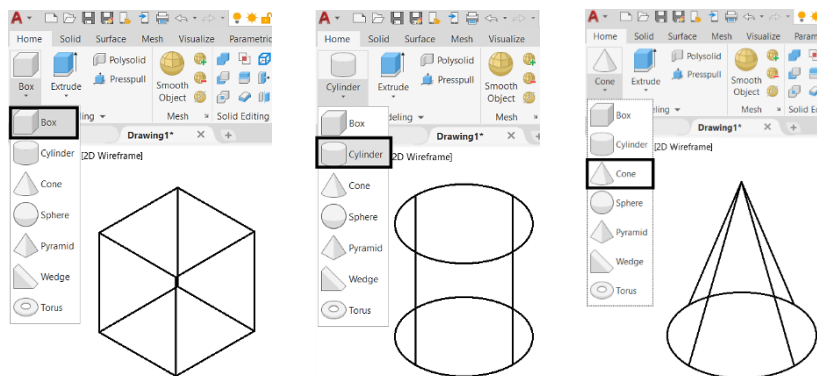
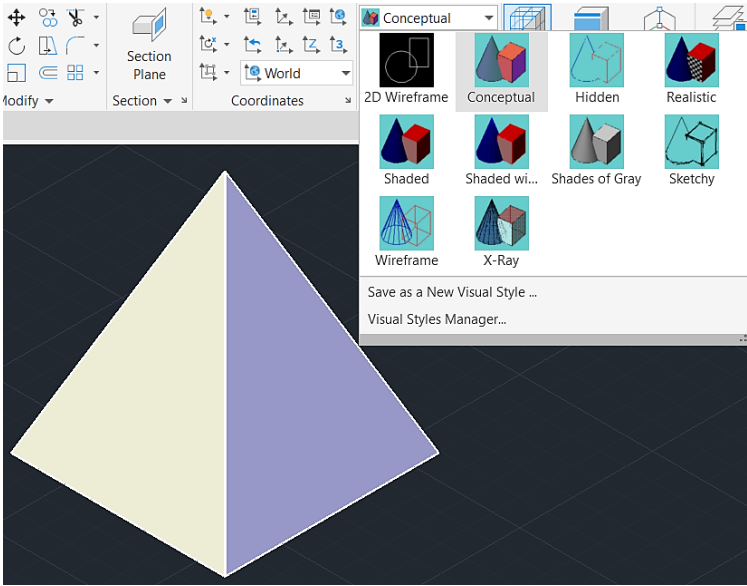
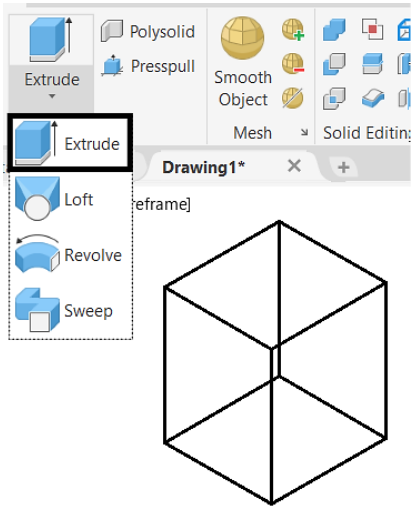
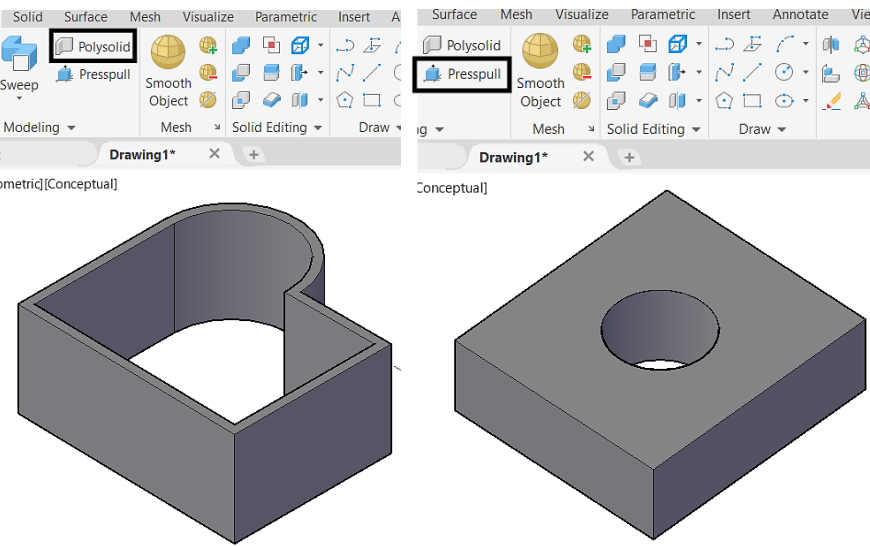
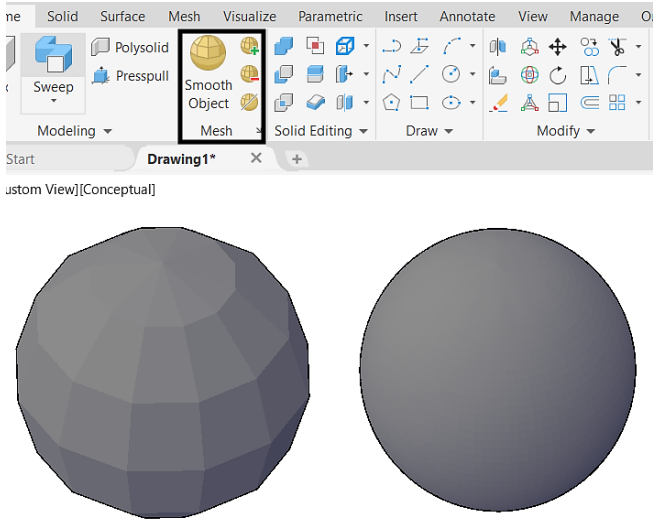
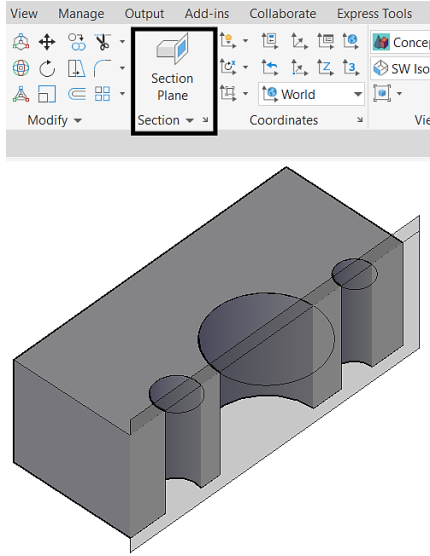
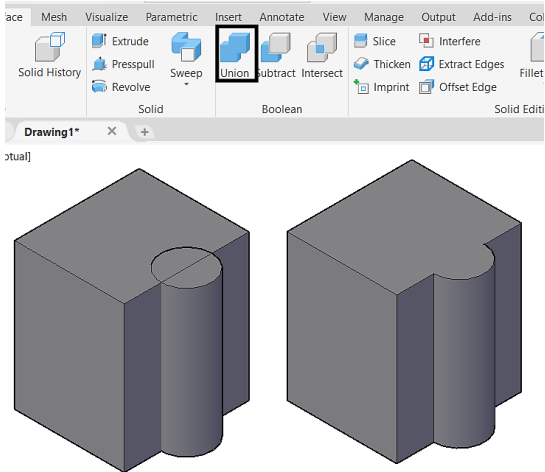
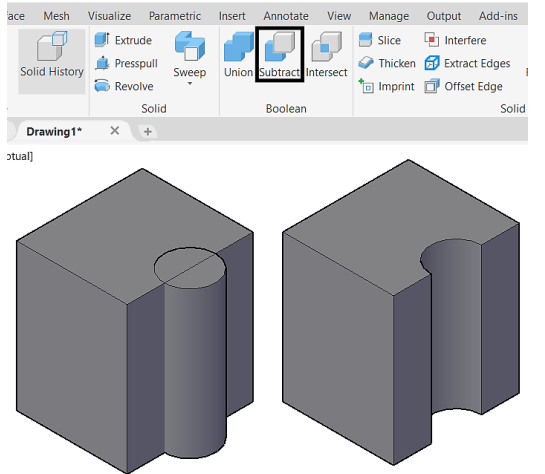
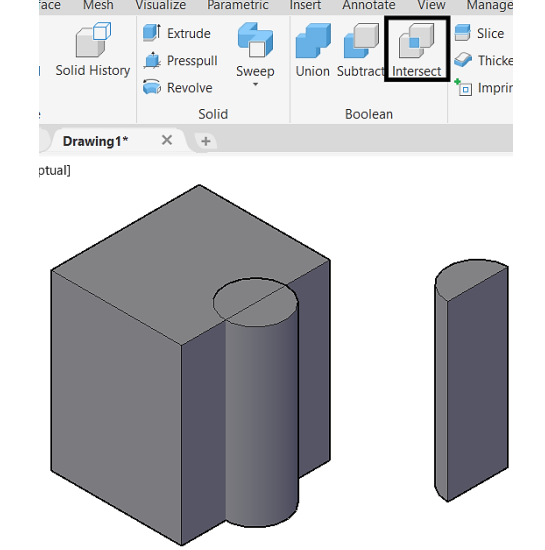
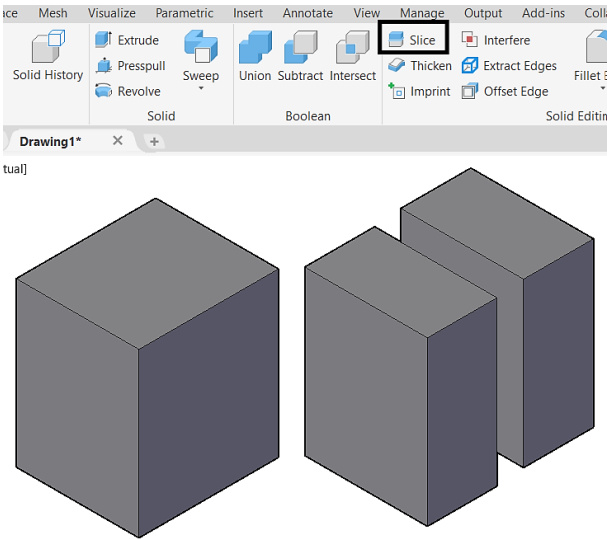
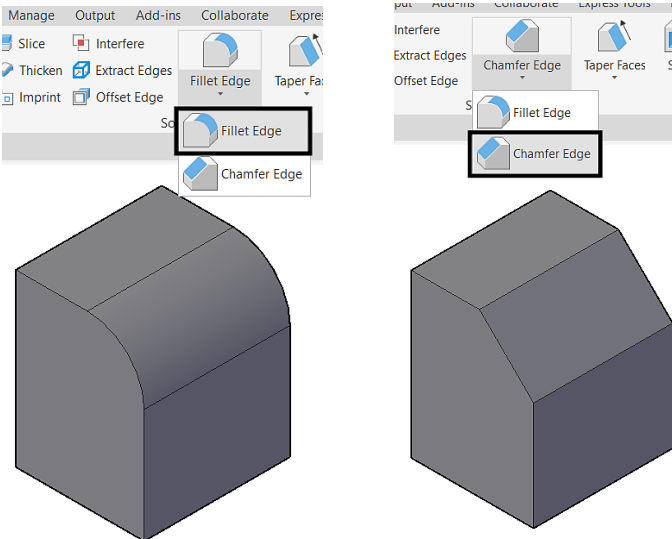
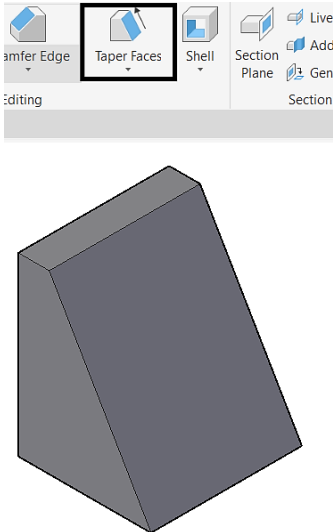


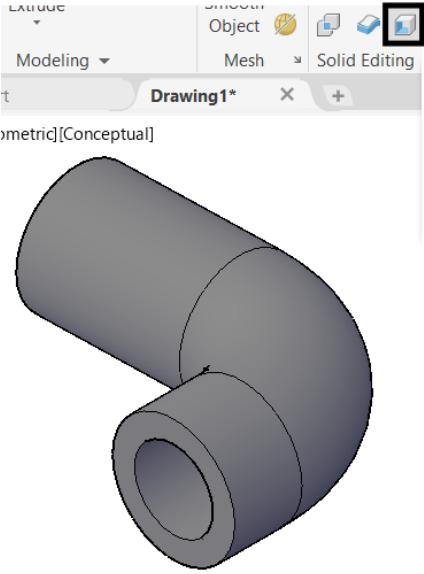
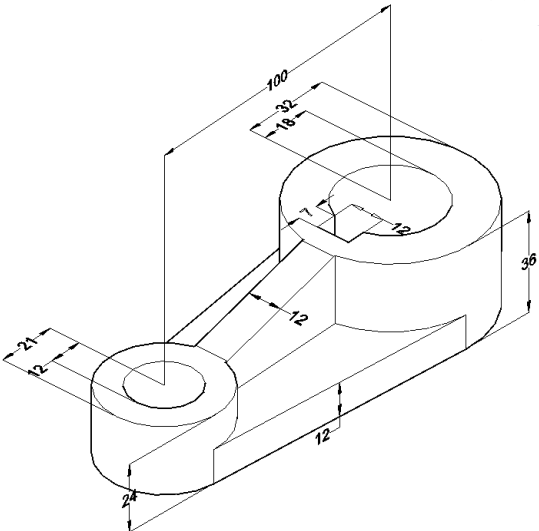
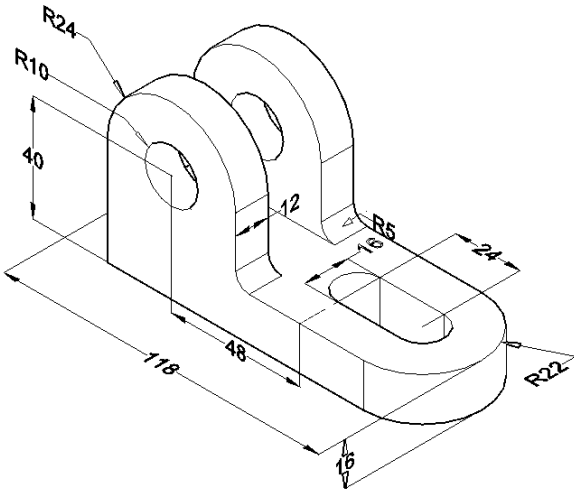
Fig. 9.4

6	Drawing Visual Styles: In AutoCAD, Visual Styles control how 3D models are displayed on the screen. They change the appearance of edges, shading, and lighting to make visualization easier. Common styles include: • 2D Wireframe – shows only edges without surfaces. • 3D Wireframe – displays 3D objects as transparent with all edges visible. • Conceptual – applies shaded faces with smooth edges for quick visualization. • Realistic – displays materials, textures, and lighting effects. You can access them from the Visual Styles panel on the Ribbon or by using the command “VISUALSTYLES” as shown in Fig. 9.5.	 Fig. 9.5
7	EXTRUDE command: The EXTRUDE command is used to turn 2D objects into 3D solids by giving them height or extending them along a path. It is accessed by using the command “EXTRUDE (or EXT)” as shown in Fig. 9.6.	 Fig. 9.6

8	POLYSOLID & PRESSPULL commands: By using the command “POLYSOLID”, you can draw 3D walls with specified height and thickness, similar to drawing a polyline but in 3D. By using the command “PRESSPULL” you can extrude, offset, or cut a closed 2D boundary by simply clicking inside it as shown in Fig. 9.7.	 The screenshot shows the AutoCAD ribbon with the 'Solid' tab selected. The 'Polysolid' and 'Presspull' icons are highlighted with red boxes. Below the ribbon, two 3D models are shown: a U-shaped wall on the left and a rectangular block with a circular hole on the right. Fig. 9.7
9	SMOOTH command: SMOOTH command is used to convert a polyline or mesh object into a smoother, more refined shape by adding curves or smoothing out edges. By using the command “SMOOTH”, you can increase the smoothness level of objects like meshes or polylines, which is especially useful for creating organic or free-form 3D models as shown in Fig. 9.8.	 The screenshot shows the AutoCAD ribbon with the 'Mesh' tab selected. The 'Smooth Object' icon is highlighted with a red box. Below the ribbon, two spheres are shown: one with a visible mesh on the left and one smooth on the right. Fig. 9.8
10	SECTION command: The SECTION command in AutoCAD is used to create a 2D cross-section of a 3D solid or surface. By using the command “SECTION”, you can define a cutting plane across the object, and AutoCAD will generate the intersecting profile as a 2D region or polyline. This is particularly useful for engineering drawings where internal details of 3D models need to be shown as depicted in Fig. 9.9.	 The screenshot shows the AutoCAD ribbon with the 'Section' tab selected. The 'Section Plane' icon is highlighted with a red box. Below the ribbon, a 3D model of a mechanical part is shown with a section plane cutting through it, revealing internal details. Fig. 9.9

11	UNION command: The UNION command in AutoCAD is used to combine two or more 3D solids into a single solid object. By using the command “UNION”, overlapping or touching solids merge into one continuous body, which is useful for simplifying models and preparing them for further editing or visualization as shown in Fig. 9.10.	 Fig. 9.10
12	SUBTRACT command: This command is used to remove the volume of one 3D solid from another. By using the command “SUBTRACT”, you first select the main solid (from which material will be removed) and then select the solid(s) to subtract. The result is a single object with the selected portions cut out, making it useful for creating holes, slots, or cavities in 3D models as shown in Fig. 9.11.	 Fig. 9.11
13	INTERSECT command: The INTERSECT command in AutoCAD is used to create a 3D solid from the common volume of two or more solids. By using the command “INTERSECT”, only the overlapping portion of the selected objects is retained, as shown in Fig. 9.12.	 Fig. 9.12

14	SLICE command: This command is used to cut a 3D solid into two or more parts. By using the command “SLICE”, you can specify a plane through points, objects, or view direction, and AutoCAD will divide the solid accordingly as shown in Fig. 9.13.	
15	FILLETEDGE & CHAMFEREDGE commands By using the command “FILLETEDGE”, you can create a rounded edge (fillet) on the selected solid edges. By using the command “CHAMFEREDGE”, you can create a beveled edge (chamfer) by specifying distances or angles as shown in Fig. 9.14.	
16	TAPER command: The TAPER command in AutoCAD is used to angle the faces of a 3D solid relative to a base point or axis. By using the command “TAPER”, you can select one or more faces of a solid and apply a taper angle, which is especially useful for designing molds, castings, or parts that require draft angles, as shown in Fig. 9.15.	

17	SHELL command: The SHELL command in AutoCAD is used to hollow out a 3D solid by removing one or more selected faces and applying a uniform wall thickness to the remaining surfaces. By using the command “SHELL”, you can quickly convert a solid object (like a box, cylinder, or sphere) into a hollow object, which is useful for creating models such as containers, casings, or housings as shown in Fig. 9.16.	 The screenshot shows the AutoCAD software interface. The 'Solid Editing' tab is active in the ribbon, and the 'SHELL' command is highlighted. Below the ribbon, a 3D model of a hollowed-out cylindrical part is displayed, demonstrating the result of the SHELL command.
18	Exercise 1: Draw the 3D model of the mechanical part shown in the Fig. 9.17 using AutoCAD. The part consists of two cylindrical bosses of different diameters and heights, connected by a rib. Each boss has a central hole, and the larger boss includes a rectangular keyway cut.	 The isometric drawing shows a mechanical part with two cylindrical bosses. The larger boss on the right has a central hole with a diameter of 32 and a rectangular keyway cut with a width of 18 and a depth of 12. The smaller boss on the left has a central hole with a diameter of 21 and a height of 12. The two bosses are connected by a rib with a width of 12. The overall dimensions of the part are 100 in length, 36 in height, and 24 in width.
19	Exercise 2: Draw the 3D model of the mechanical bracket shown in the Fig. 9.18 using AutoCAD. The part consists of a base plate with rounded ends, two vertical supports with circular holes, and a slot cut on the base.	 The isometric drawing shows a mechanical bracket with a base plate and two vertical supports. The base plate has rounded ends with a radius of R24 and a central slot with a width of 16 and a depth of 12. The two vertical supports have circular holes with a diameter of 10 and a height of 40. The overall dimensions of the part are 118 in length, 48 in width, and 16 in height.

Exercise 3:

Draw the 3D pipe elbow with flanges in AutoCAD as shown in the Fig. 9.19. The model should include the elbow created with a fillet, circular and rectangular flanges with holes, and apply proper dimensions and annotations.

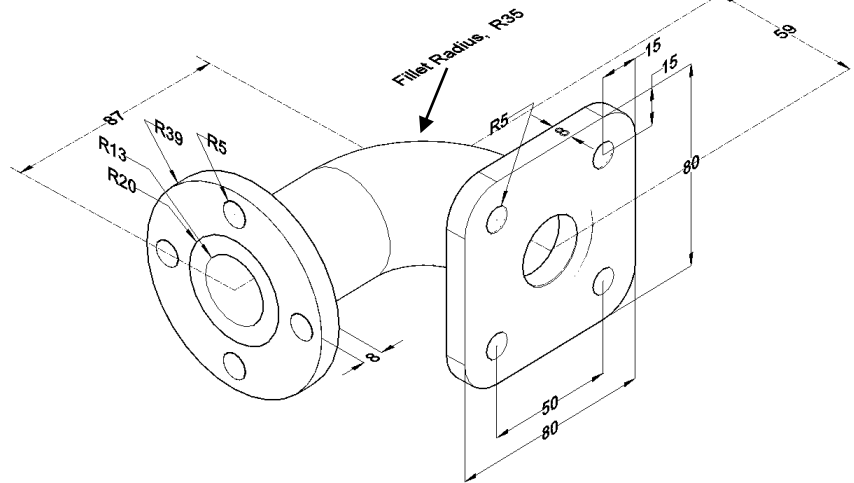


Fig. 9.19

NED University of Engineering & Technology
Department of Mechanical Engineering
Course Code and Title: **ME-118 Engineering Drawing**

Software Use Rubric (P3)					
Skill Sets	Extent of Achievement				
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Skills to use advanced/associated features	Not able to use any advanced/associated features	Uses advanced / associated features with detailed supervision	Uses advanced / associated features with some guidance	Uses advanced / associated features without guidance	Exhibits all advanced / associated features effectively
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Extraction of required output from software	Does not extract required output at all	Rarely extract required output from software	Occasionally extract required output from software	Often extract required output from software	Effectively extract required output from software

Laboratory Session No. _____

Date: _____

Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	

Practical No. 10

Objective: Modifying dimensions and units in AutoCAD on 2D & 3D drawings

1 In this experiment, we will learn how to apply **dimensions to 2D & 3D drawings** using AutoCAD commands.

Before starting any drawing, it is important to ensure that the units are set to **millimeters (mm)**. To do this:

- Use the command “Units” and press Enter.
- The Drawing Units dialog box will appear as shown in Fig. 10.1.
- Under Length heading, set type to Decimal and the precision to 0, 0.0, or as required by the dimensions. Same for Angle.
- From the Insertion Scale drop-down menu, select Millimeters.

Click OK to apply the settings.

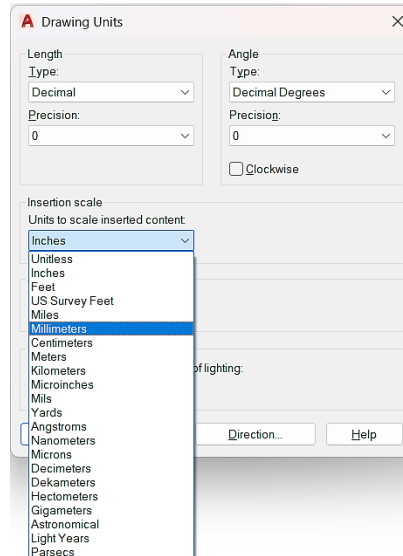


Fig. 10.1

Dimension Style Manager

The Dimension Style Manager in AutoCAD is used to control the appearance of dimensions such as text, arrows, lines, and units. By using the command DIMSTYLE (or typing D in the command line), you can create, modify, and manage dimension styles to ensure consistency as shown in Fig. 10.2.

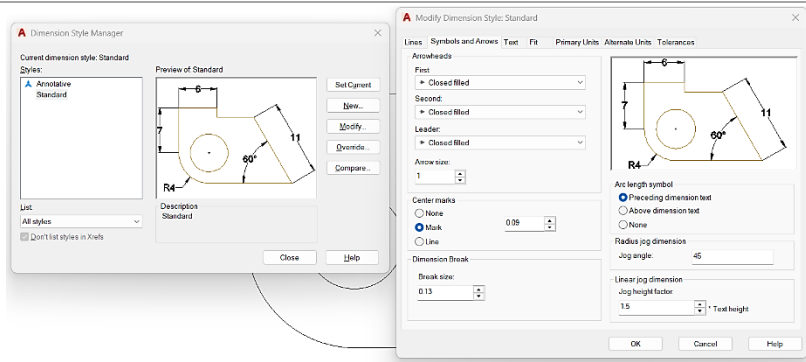


Fig. 10.2

Dimensioning the 2D Drawing

- 4 A simple geometry of a Bar Locking Plate is drawn in AutoCAD, as shown in Fig. 10.3.

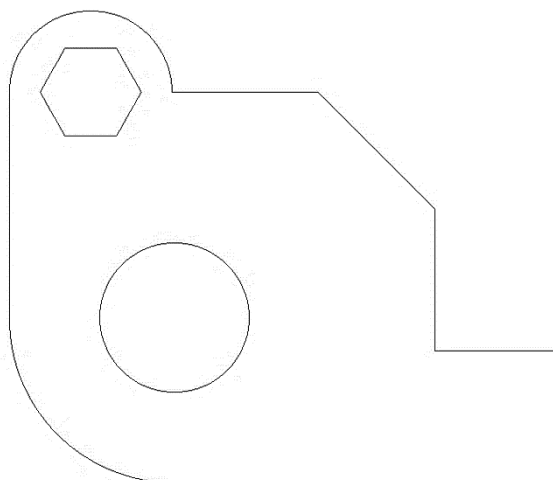
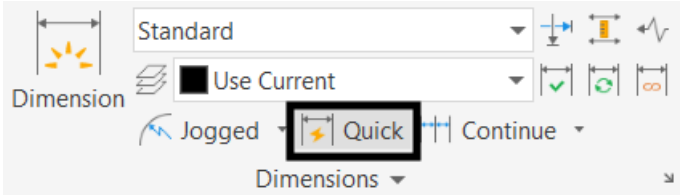
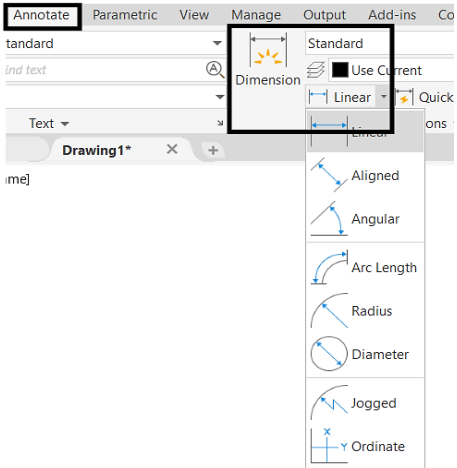
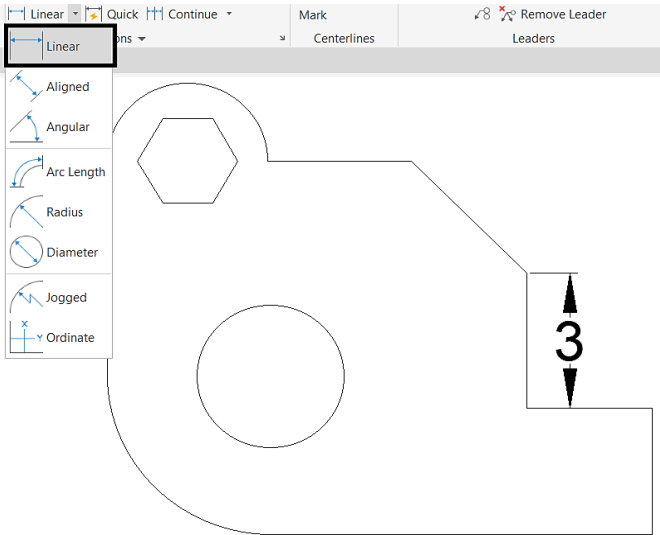
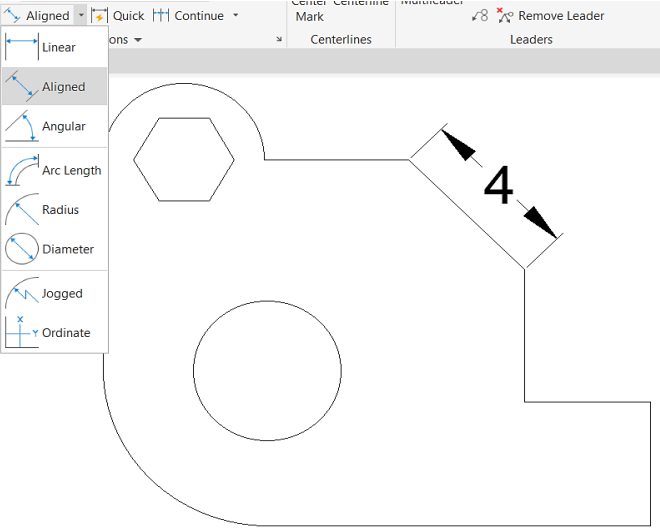
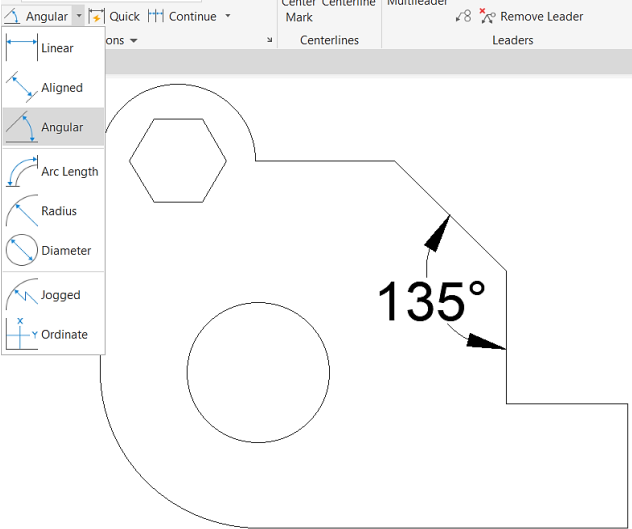
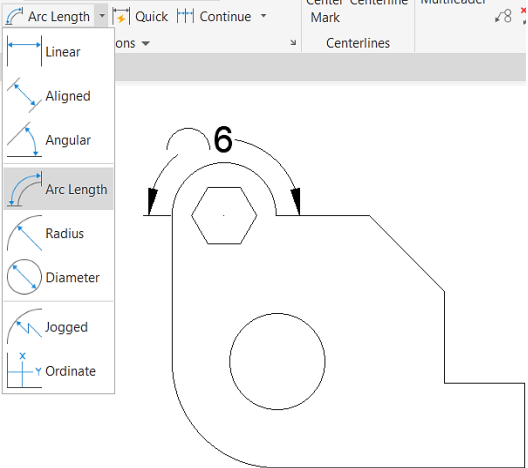
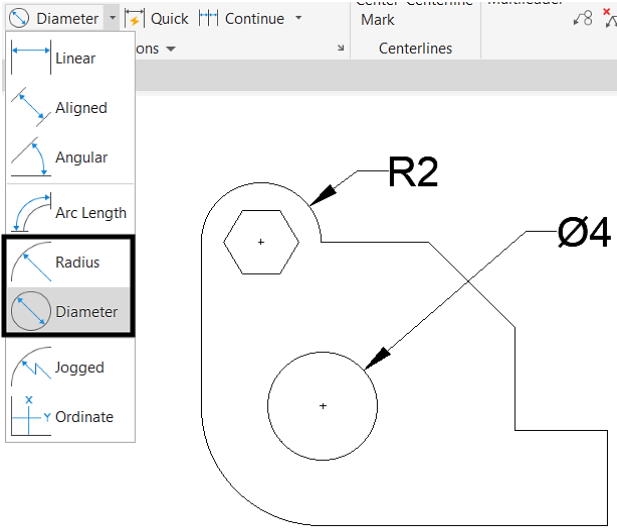
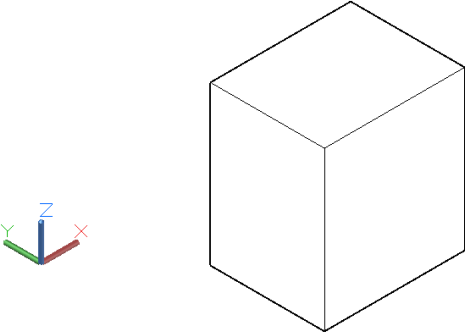
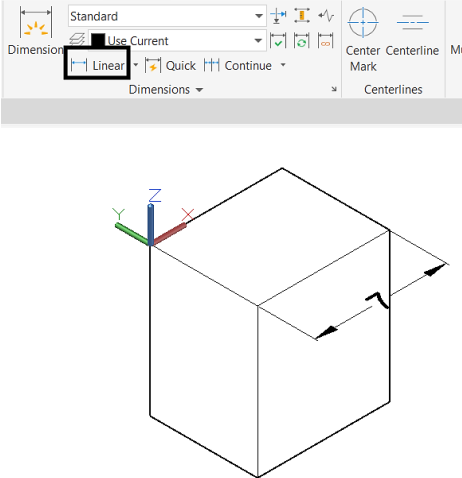
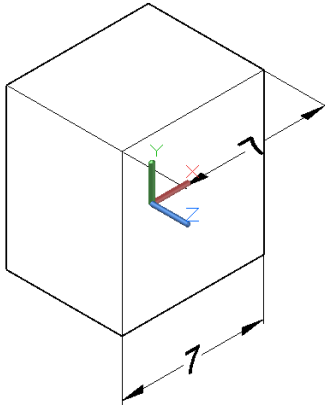
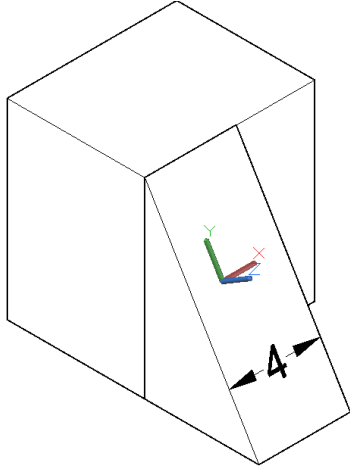


Fig. 10.3

5	Quick Dimension The Quick Dimension tool in AutoCAD allows you to automatically generate multiple dimensions at once. By using the command QDIM, you can select objects, and AutoCAD will create a series of dimensions (linear, angular, or radial) based on the selected geometry as shown in Fig. 10.4.	 The image shows the 'Dimensions' ribbon in AutoCAD. The 'Quick' button, which features a sunburst icon, is highlighted with a black rectangular box. Other buttons visible include 'Dimension', 'Jogged', and 'Continue'. Fig. 10.4
6	Go to Annotate bar in the Ribbon Menu then select multiple dimension option as shown in Fig. 10.5.	 The image shows the 'Annotate' ribbon in AutoCAD. The 'Multiple Dimension' panel is expanded, showing a list of dimension types: 'Standard', 'Linear', 'Quick', 'Aligned', 'Angular', 'Arc Length', 'Radius', 'Diameter', 'Jogged', and 'Ordinate'. The 'Quick' option is highlighted with a black rectangular box. Fig. 10.5
7	Linear Dimension The Linear Dimension in AutoCAD is used to measure and display the distance between two points along the X or Y axis. By using the command DIMLINEAR, you can create horizontal, vertical, or aligned dimensions quickly. This command helps in accurately defining lengths and distances of objects as shown in Fig. 10.6.	 The image shows the 'Linear' dimension tool selected in the 'Dimensions' ribbon. Below the ribbon, a mechanical drawing of a part is shown. A vertical linear dimension is applied to a step in the drawing, with the dimension value '3' displayed. The dimension line is vertical, and the text '3' is placed to the right of the dimension line. Fig. 10.6

8	Aligned Dimension The Aligned Dimension in AutoCAD is used to measure and display the distance between two points along the actual direction of the object, rather than restricting it to horizontal or vertical. By using the command DIMALIGNED, you can place dimensions parallel to inclined or slanted objects for accurate representation as shown in Fig. 10.7.	 The image shows the AutoCAD dimensioning toolbar with the 'Aligned' tool selected. Below the toolbar is a mechanical part drawing. A dimension of '4' is applied to a slanted edge of the part, parallel to its direction.
9	Angular Dimension The Angular Dimension in AutoCAD is used to measure and display the angle between two lines, three points, or the vertices of an arc or circle. By using the command DIMANGULAR, you can create dimensions that show precise angular values on drawings as shown in Fig. 10.8.	 The image shows the AutoCAD dimensioning toolbar with the 'Angular' tool selected. Below the toolbar is the same mechanical part drawing. An angular dimension of '135°' is applied to a vertex of the part, indicating the angle between two intersecting lines.
10	Arc Length Dimension The Arc Length Dimension in AutoCAD is used to display the length of an arc rather than its angle or radius. By using the command DIMARCLENGTH, you can select an arc and place the dimension to clearly show its curved length as shown in Fig. 10.9.	 The image shows the AutoCAD dimensioning toolbar with the 'Arc Length' tool selected. Below the toolbar is the same mechanical part drawing. An arc length dimension of '6' is applied to a small circular arc on the part, following its curvature.

11	Radius & Diameter Dimension The Radius and Diameter Dimension tools in AutoCAD are used to specify the size of circles and arcs. By using the commands DIMRADIUS for radius and DIMDIAMETER for diameter, you can select the circle or arc and place the required dimension in the drawing as shown in Fig. 10.10.	 The image shows the AutoCAD dimensioning ribbon with the 'Diameter' dropdown menu open, highlighting the 'Diameter' option. To the right is a mechanical part drawing featuring a circular hole with a diameter dimension of Ø4 and a rounded corner with a radius dimension of R2.
12	Dimensioning the 3D Drawing A simple geometry of a Box is drawn in AutoCAD, as shown in Fig. 10.11.	 The image displays a 3D coordinate system with X, Y, and Z axes. Next to it is a 3D wireframe model of a rectangular box.
13	In the Ribbon menu, go to the Annotate tab and select the Multiple Dimension option. It is important to ensure the UCS is set correctly, as dimensions are generated according to the XY plane. For example, in Fig. 10.12, since the XY plane is horizontal, the dimensions will also appear horizontally.	 The image shows the AutoCAD ribbon with the 'Multiple Dimension' option selected. Below it is a 3D wireframe model of a rectangular box with horizontal dimension lines applied to its top edges, indicating the current User Coordinate System (UCS) is in the XY plane.

14	As shown in Fig. 10.13, when the UCS is aligned with the front face of the cube/box, the XY axis shifts to the vertical plane, and the dimensions are displayed accordingly in the vertical orientation.	 Fig. 10.13
15	As shown in Fig. 10.14, when the UCS is aligned with the inclined face, the XY axis shifts to the inclined plane, and the dimensions are displayed accordingly in that orientation.	 Fig. 10.14

NED University of Engineering & Technology
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Remarks	
Instructor's Signature with Date:	

Practical No. 11

Objective: To create Orthographic 2D Views from a 3D Model using AutoCAD

1 In this experiment, we will learn how to create **orthographic 2D views** from a 3D model using AutoCAD commands.

Before starting any drawing, it is important to ensure that the units are set to **millimeters (mm)**. To do this:

- Use the command “Units” and press Enter.
- The Drawing Units dialog box will appear as shown in Fig. 11.1.
- Under Length heading, set type to Decimal and the precision to 0, 0.0, or as required by the dimensions. Same for Angle.
- From the Insertion Scale drop-down menu, select Millimeters.

Click OK to apply the settings.

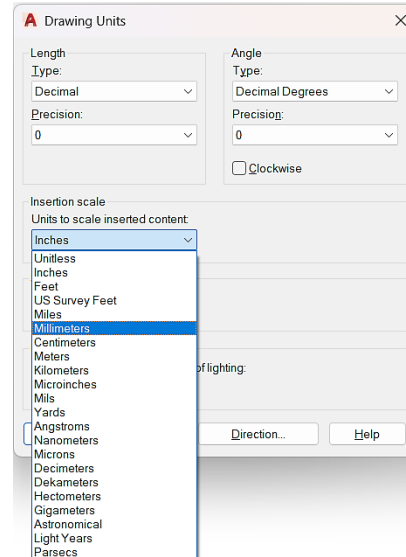


Fig. 11.1

3 Open the 3D drawing in AutoCAD and click on Layout 1 at the bottom as shown in Fig. 11.2.

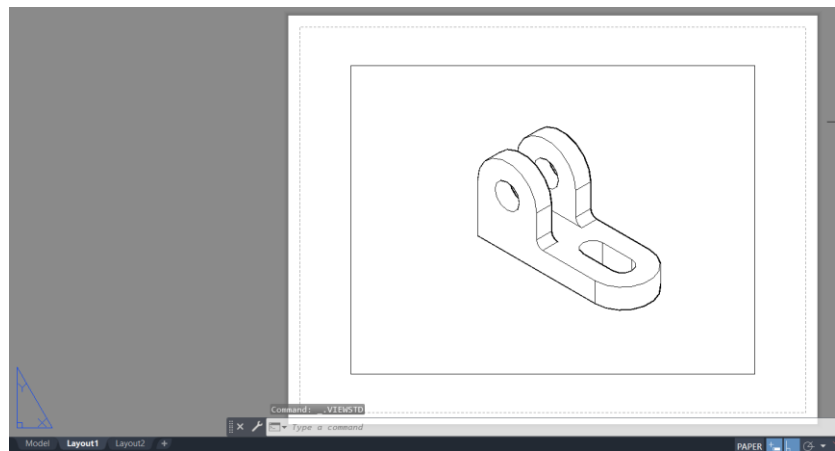


Fig. 11.2

Right click on Layout 1 then select Drafting Standard Setup. A new window will be opened, select the desired angle of projection as shown in Fig. 11.3 then click OK.

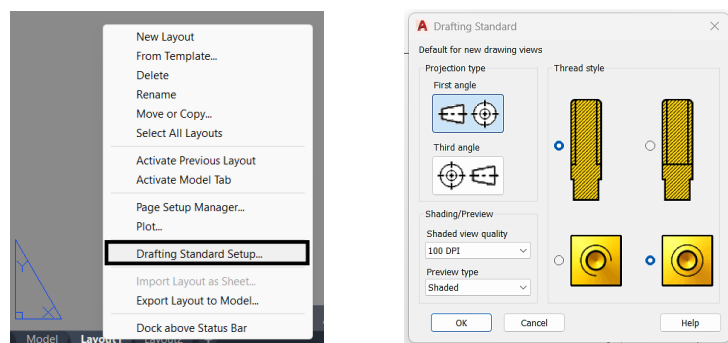
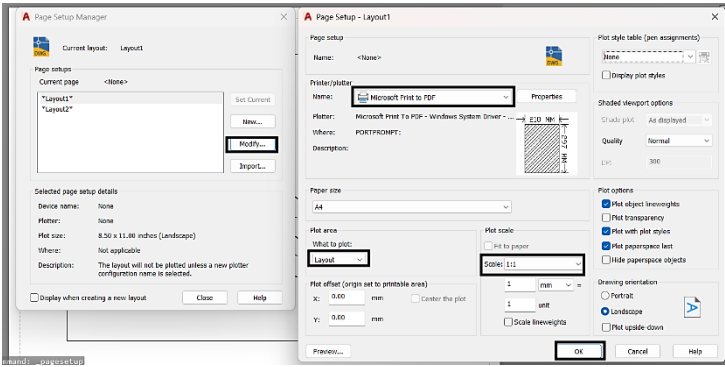
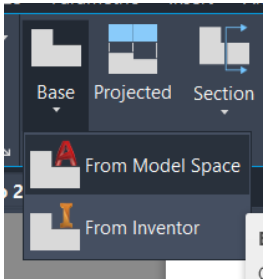
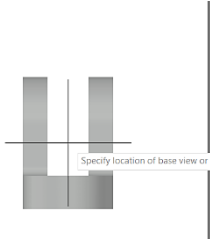
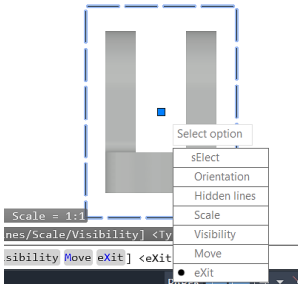
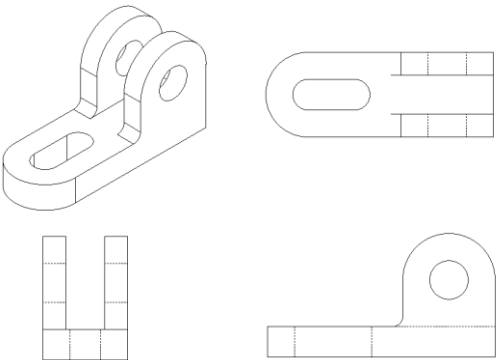


Fig. 11.3

5	Go to Layout tab in the Ribbon menu then select Page setup. A new window will open representing the setup manager. Click on modify option then a new window will open again. Change the required options for e.g., Name, what to plot, Scale as shown in Fig. 11.4 then click OK.	 Fig. 11.4
6	Click on Base option in the Ribbon menu then select From Model Space as shown in Fig. 11.5.	 Fig. 11.5
7	After selecting from Model Space, a 2D view will be formed as shown in Fig. 11.6. this view can be placed at any location by moving the cursor.	 Fig. 11.6
8	After placing the first view, click on Orientation to select the desired view to be placed like front view, side view and so on as shown in Fig. 11.7.	 Fig. 11.7
8	After placing the first view, click on Orientation to select the desired view to be placed like front view, side view and so on as shown in Fig. 11.8.	 Fig. 11.8

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Instructor's Signature with Date:	

Practical No. 12

Objective: To create Piping & Instrumentation Diagram (P&ID) using AutoCAD Plant 3D

1 In this experiment, we will learn how to create **P&ID** using AutoCAD Plant 3D.

2 Open AutoCAD Plant 3D then select New Project. After that, Enter a suitable file name and its path. Set the units to Metric then select ISO as shown in Fig. 12.1.

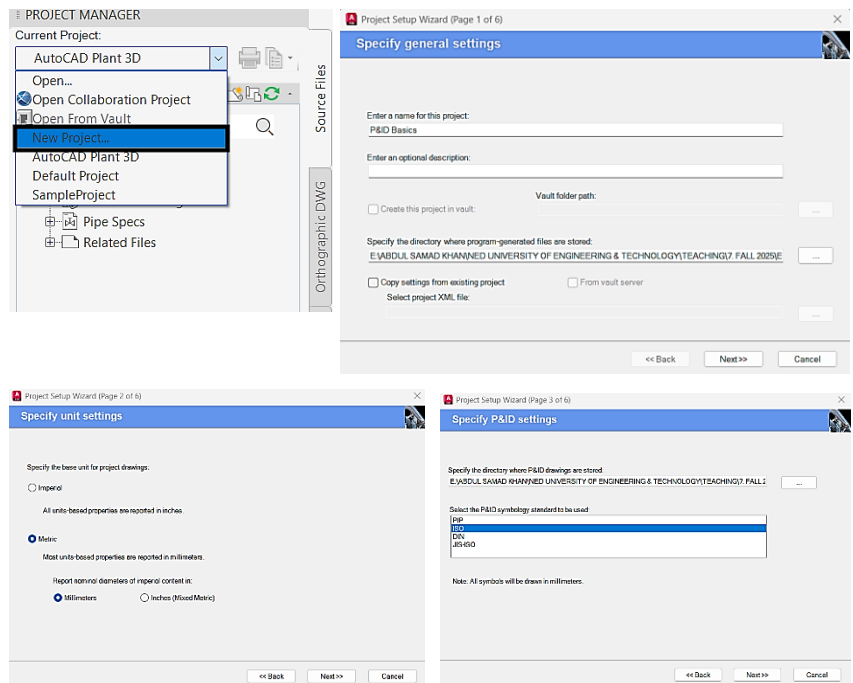


Fig. 12.1

3 Right Click on the Folder created in the previous step then select New Drawing. Enter a File name and its Path. Press OK as shown in Fig. 12.2.

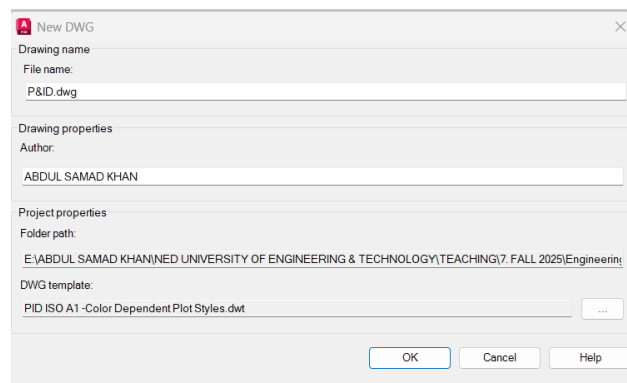


Fig. 12.2

4 Change the Workspace to P&ID ISO as shown in Fig. 12.3.

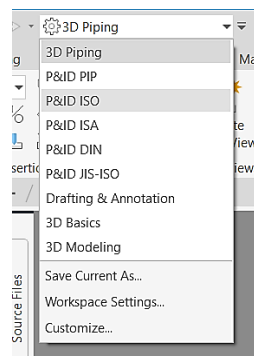


Fig. 12.3

5

A Tool Palettes will be present on the right side of the window as shown in Fig. 12.4.

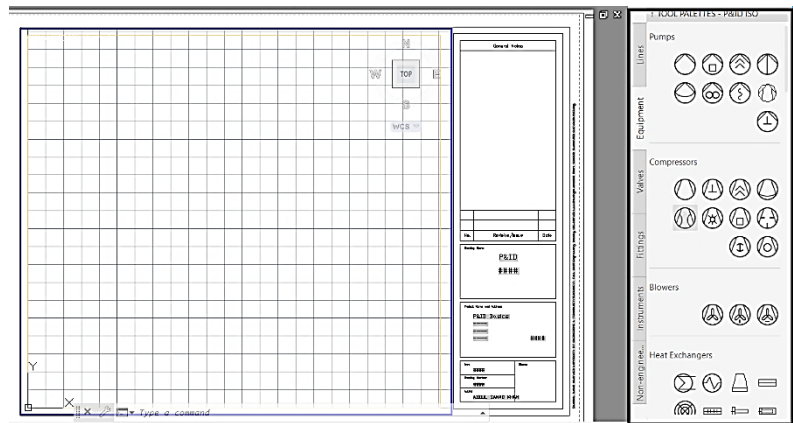


Fig. 12.4

6

Drag and drop the equipment from Tool Palettes. For example, a Positive Displacement Pump is dragged and drop then give it a number. The Tag number will be automatically generated. Check the box of “Place annotation after assigning tag” then press assign. A pump will be created with its annotation as shown in Fig. 12.5.

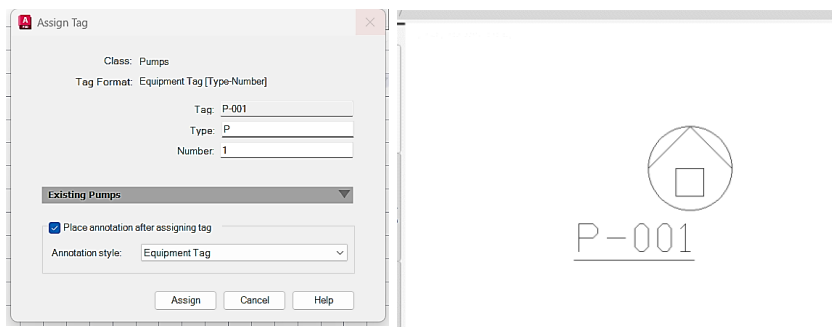


Fig. 12.5

7

Go to Project Manager then select Project Setup. Expand P&ID Class definition, Pumps, Positive displacement pump. Click on Edit symbol. A new window will be opened. Select ISO Flanged Nozzle. Press OK as shown in Fig. 12.6.

In this way, any fitting type can be selected.

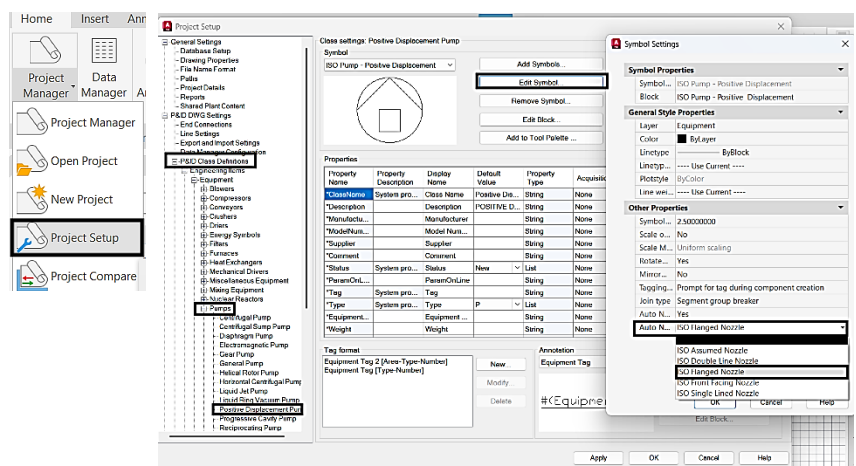


Fig. 12.6

8

Select a line to represent the pipe and create it. The pipe is connected to the pump through a flange, as illustrated in Fig. 12.7.

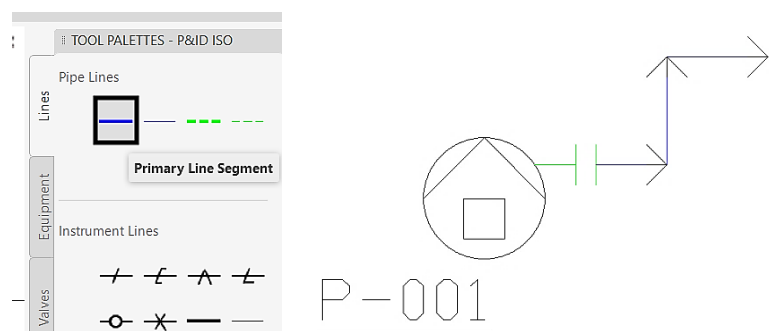
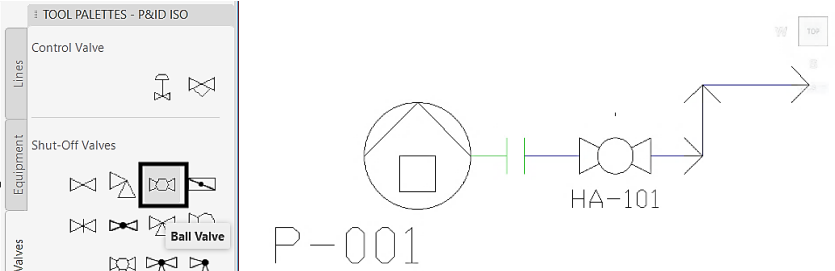
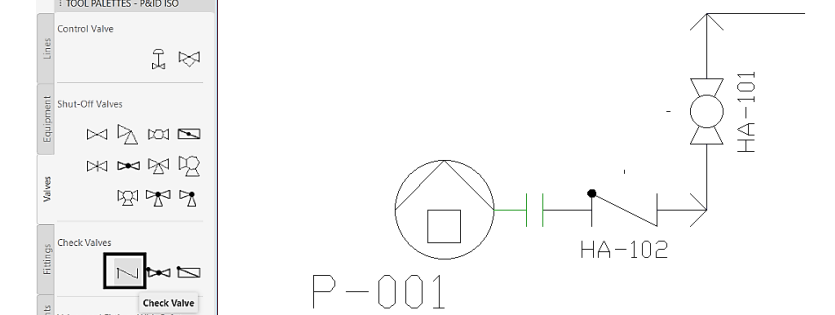
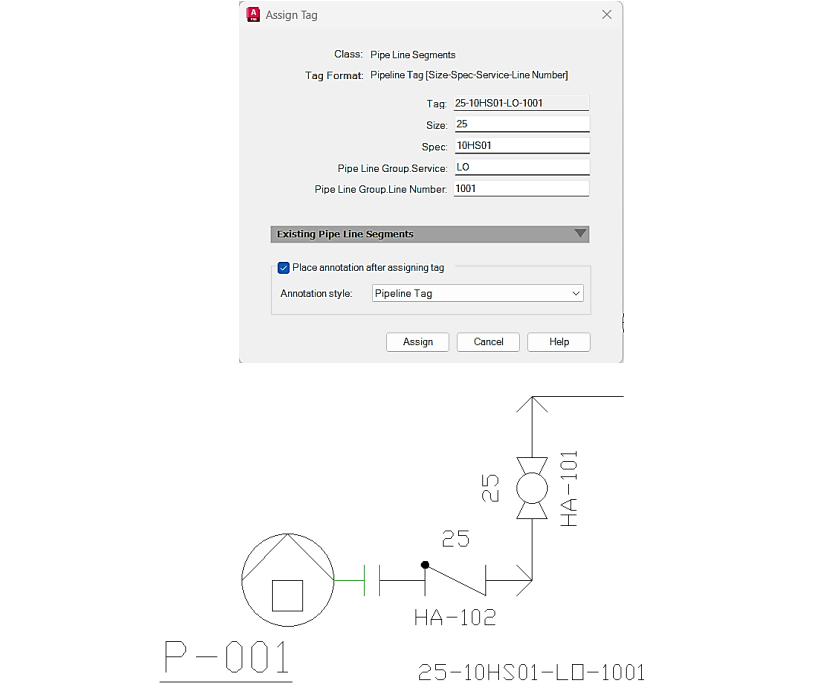


Fig. 12.7

9	Similarly, select a valve for e.g., Ball valve from the Tool Palettes and insert it after the flange as shown in Fig. 12.8.	 Fig. 12.8
10	Select a check valve and installed it in between the pump and the ball valve as shown in Fig. 12.9. <i>Info: The Check valve protects the pump from backflow when the pump stops.</i>	 Fig. 12.9
11	We need to assign the pipe size and tags so select the pipe and go to Assign Task option in the Ribbon Menu. A dialog box will be opened. Select the pipe size 25 (25 mm or 4 inch), specs as Stainless Steel, Pipe line group service as Lube Oil and enter number like 1001 then press Assign. The tagging will be applied on the drawing as shown in Fig. 12.10.	 Fig. 12.10

Exercise 1:

An engineer is designing a piping system to transport a hot fluid from a pump to a heat exchanger for cooling before it is stored in a spherical vessel. The system requires proper flow control, filtration, and connection elements to ensure safe and efficient operation.

The students are required to develop a 2D P&ID diagram in AutoCAD Plant 3D representing this system layout.

Tasks:

12

Select a Three-Way Ball Valve from the Tool Palettes and insert in between perpendicular pipes as shown in Fig. 12.11.

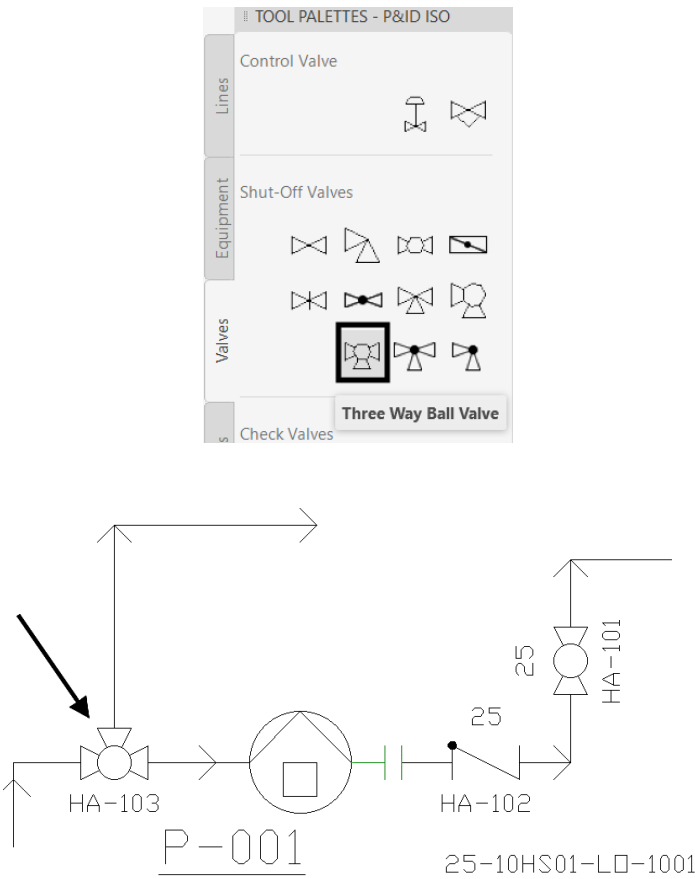


Fig. 12.11

13

Select a Strainer from the Fitting section of the Tool Palettes and install at the suction of the Pump as shown in Fig. 12.12.

Info: A strainer installed upstream of pumps to protect them from damage caused by contaminants.

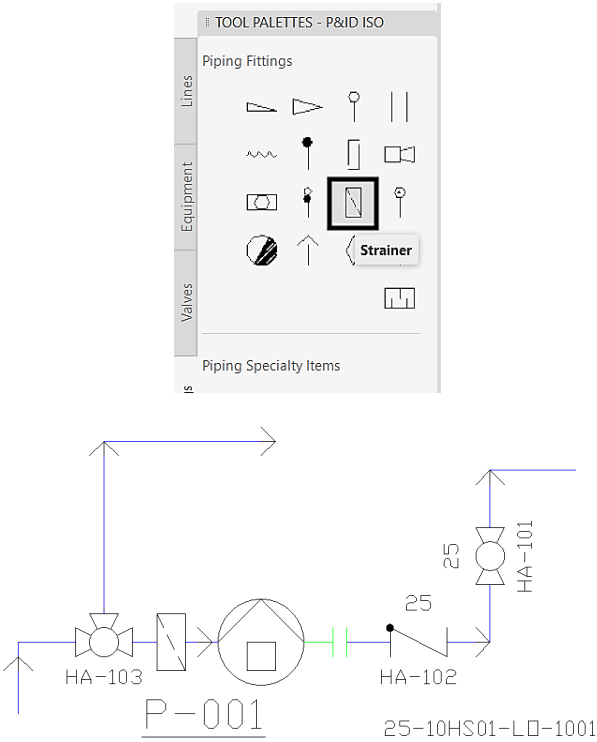


Fig. 12.12

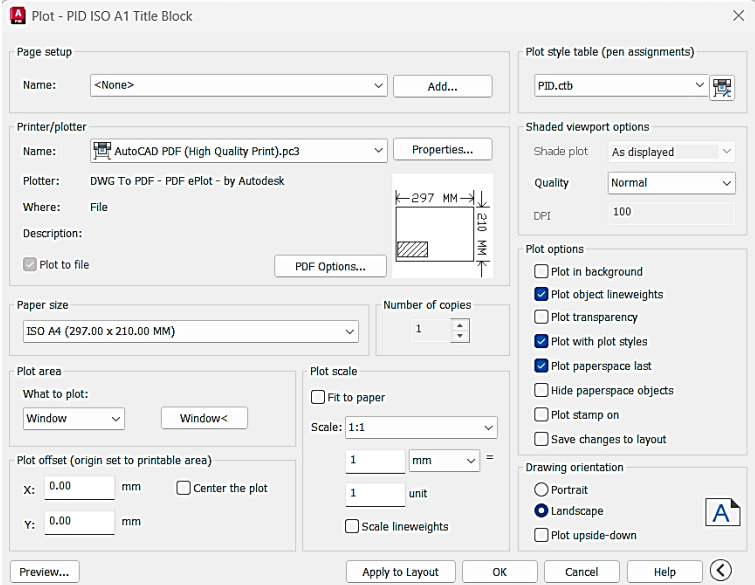
14	The “Plot” command in AutoCAD Plant 3D is used to print or export your drawing. - Type PLOT in the Command Line and press Enter. - In the Plot dialog box as shown in Fig. 12.13, set the following options: ✓ Printer/Plotter – Choose your output device (printer, plotter, or “DWG to PDF” for saving as a PDF). ✓ Paper Size – Select the desired sheet size (e.g., A4, A3, etc.). ✓ Plot Area → Choose what to print: Display, Extents, Window, or Layout. ✓ Drawing Orientation – Choose Portrait or Landscape. ✓ Click Preview to check how the drawing will appear when printed. If satisfied, click OK to plot/print the drawing. ✓ You can press Ctrl + P to open the Plot dialog box quickly.	
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Fig. 12.13

1. Insert a *Centrifugal Pump (P-100)* as the starting equipment of the system. Label it properly.
2. Draw a *pipeline (25 mm dia, Material=100HS01, 201)* from the pump discharge using *Flanged Nozzle* and connect it to a *Strainer* to remove impurities from the *Lube Oil* before entering the Heat Exchanger.
3. Add a *Ball Valve (HA-108)* downstream of the strainer to control the flow through the system.
4. Connect the outlet of the ball valve to a *Shell and Tube Heat Exchanger (W-100)* via piping where the hot fluid will be cooled.
5. From the outlet of the heat exchanger, draw a pipeline leading to a *Spherical Vessel (K-300)* for temporary fluid storage.
6. Place *Gate Valves (HA-109, HA-110)* before the pump and after the heat exchanger to allow isolation during maintenance.
7. Annotate all equipment, valves, and instruments using standard P&ID tags and ensure consistent line types and flow directions.
8. After completing the drawing, take a printout (PDF or A4 size) of the final P&ID diagram showing all equipment, valves, pipelines, and instrument tags clearly.

Exercise 2 (Reading): P&ID

Read the P&ID on the next page and answer the following questions. Write your answer in the same box as the question:

Explain the function of the following components visible in the drawing:

- a) Pump
- b) Strainer
- c) Ball valve
- d) Heat exchanger

Where is the Ball Valve used in the drawing? Answer by looking at the surrounding components.

What is the total capacity of Oil Tanker?

Can you identify where “10000 I.G.” is written? If yes, where is it mentioned, and what does I.G. stand for?

What does  show in the drawing?

When the operator leaves the cabin, what is the first area he will enter?

What is the height of the wall near Fill Point?

What is the distance between the Dispensing Unit and the Electrical Cables?

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ARE YOU READY FOR THIS?

